MA'ADEN معادن

Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **DESIGN CALCULATION**
6000 VILLAGE-VIP ACCOMMODATION
DESIGN OF BUILDING STRUCTURE

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities
Typ.	:	Typical
HCS	:	Hollow Core Slab
Fc'	:	Concrete Compressive Strength
Fy	:	Yield Strength of Reinforcing Steel
ϕ	:	Strength Reduction Factor
SBC	:	Saudi Building Code
SASO	:	Saudi Standards, Metrology and Quality Organization
ACI	:	American Concrete Institute
ASCE	:	American Society of Civil Engineers
PCI	:	Precast/Prestressed Concrete Institute

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1. INTENT

This Document is intended to produce the technical needs in Engineering including particular reference to design justification & Engineering validation works. The work described below is related to Design of VIP Accommodation in Village Area.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The Scope of this document is limited to Complete Analysis & Structural Design Verification of VIP Accommodation only.

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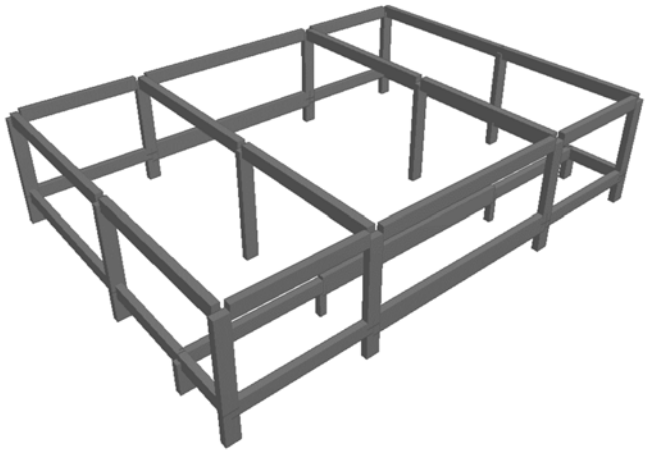
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4. DESIGN METHODOLOGY & ASSUMPTIONS

This Building Is a Single Story Frame Structure with Foundation Being Cast in Site. The Building Shall Have Insulated Block Walls and Cast in Site Framing Beams to support the HCS (Hollow Core Slab)

The Superstructure for this Accommodation Building was modeled using STAAD software. The Analysis of Frames (Cast-In-Situ Columns & Beams) has been performed meeting design requirements & also keeping in view the Architectural requirements. The loads were applied as per SBC/ASCE/ACI standards & analysis was done to generate the Column Reactions at the Base which are assumed to be fixed. The Foundation design has been done by importing Column Base Reactions into Foundation Design Software.



STAAD 3D MODEL

4.1. Codes

[1] SBC-301-2007-Structural Loading & Forces.

[2] SBC-303-2007-Structural Soil & Foundations.

[3] SBC-304-2007-Structural Concrete Structures.

[4] SASO SSA 2 – Steel Bars for the Reinforcement of Concrete.

[5] SASO SSA 224 – Steel Fabric for Reinforcement of Concrete

[6] ASCE-7-2005 Minimum Design Loads for Buildings and other Structures.

[7] ACI 318-08 Building Code Requirements.

[8] PCI –MNL-120-04 –Design Handbook, 6th edition.

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4.2. Geotechnical investigations

Foundations have been designed based on recommendations of the Soil Investigation Report, Doc # MD-216-3000-EG-CV-RPT-1008-A02.

4.3. Material Properties

Compressive Strength of Concrete at 28th days

fc' = 35 MPa (5.00 Ksi) for beams, columns, suspended slabs.

fc' = 35 MPa (5.00 Ksi) for foundations, grade beams, columns.

Yield Strength of Steel

fy = 414 MPa (60 Ksi) for all reinforcing steel

5. DESIGN LOADS

5.1. Dead Load

5.1.1. Roof Dead Loads

- Provision for Electrical Lighting = 0.05 kN/m²

Ceiling

Suspended steel channel system	= 0.10 kN/m²
Acoustical Fiber Tile (16mm)	= 0.05 kN/m²
Sub-total	= 0.2 kN/m²

- Hollow Core Slab (PCI 6th Edition)

(HCS-250 Topped) b/w Grid B & C	= **3.43 KN/m2**
(HCS-150 Topped) b/w Grid A & B, C& D	= **2.52 KN/m2**

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Prestressed Hollowcore Floor

Overall Structural Depth (mm)	Self Wt- KN/M2	No. of Cores in Unit	Spans indicated below allow for Characteristic Service Load (Live Load) Plus Self Weight Plus 1.5Kn/m² for Finishes Characteristic Service Loads Kn/m²							
			0.75	1.50	2.00	2.50	3.00	4.00	5.00	10.00
			Clear Span in metres							
100	2.38	0	5.39	5.19	5.08	4.91	4.72	4.40	4.14	3.30
150	2.52	11	7.40	7.40	7.40	7.40	7.28	6.80	6.40	5.12
150 (Dense	2.94	11	7.40	7.40	7.40	7.35	7.08	6.64	6.27	5.06
200	3.10	11	9.40	9.40	9.40	9.40	9.26	8.69	8.21	6.65
250	3.43	6	10.80	10.80	10.80	10.80	10.80	10.80	10.56	8.60
300	3.88	6	13.50	13.50	13.50	13.50	13.18	12.44	11.81	9.70
350	4.23	6	15.00	15.00	15.00	15.00	14.89	14.08	13.39	11.06
400	4.57	6	16.00	16.00	16.00	16.00	15.97	15.13	14.42	11.98
450	4.92	6	18.00	18.00	18.00	17.70	17.19	16.33	15.58	13.02
500	5.27	6	18.00	18.00	18.00	18.00	18.00	17.13	16.38	13.76

- Built-in Water Proofing System

100mm thk Polystyrene Foam

0.01 kN/m² per 25.4mm thickness x (100/25.4mm) = 0.039 kN/m²

Water proofing membrane

Bituminous, smooth surface = 0.07 kN/m²

90 mm thick average cement foam concrete screed Lightweight

Concrete 0.38 kN/m² per 25mm thickness x (90/25mm) = 1.368 kN/m²

75 mm Concrete screed above HCS = 21 x 0.075 =1.6 kN/m²

Sub-total = **3.1 kN/m²**

Total Load b/w Grid B & C = **6.75 kN/m²**

Total Load b/w Grid A & B, C& D = **5.62 kN/m²**

UDL on External Beams along Grid Line-A ,D = 5.10 x 5.62/2 =**14.33 Kn/m**

UDL on External Beams along Grid Line-B,C = 14.33+(8.55 x 3.65/2)=**30 Kn/m**

Moment on all the External Beams due to Cantilever Portion of slab and Screed

= (0.15x24+ (0.05+0.2)/2x20)*0.6*0.6/2 = **1 knm/m**

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5.1.2. Ground Floor Walls

- 250 thk. Outer CMU

16 x 0.25 x 2.5m clear height = **10.0 kN/m**

5.2. Live Load

5.2.1. Roof Live Loads

Live loads were taken from SEI/ASCE 7-05, Table 4-1.

Ordinary Flat = **0.50 kN/m²**

UDL on External Beams along Grid Line-1 = 0.50 x 7.35/2 =**1.84 Kn/m**

UDL on External Beams along Grid Line-3 = 0.50 x 6.47/2 =**1.62 Kn/m**

UDL on Internal Beams along Grid Line-2 =**3.5 Kn/m**

Moment on all the External Beams due to Live Load on

Cantilever Portion of slab = 1*0.6*0.6/2 = **0.18 kN/m /m**

5.3. Wind Load

Wind loads were automatically generated in STAAD as per ASCE 7-05. The parameters used for the wind load generation are as follows (Refer STRUCTURAL DESIGN CRITERIA NON PROCESS BUILDINGS Doc No MD-216-0000-EG-ST-CRT-0001)

All structures and foundations (except for buried structures) designed and shall be constructed to resist the wind effects determined in accordance with Section-7.2, SBC-301-Analytical Procedure.

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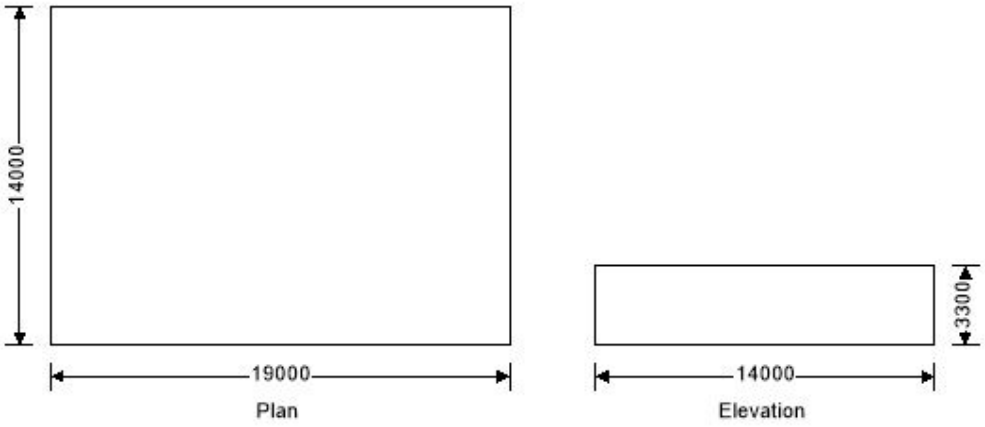
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5.4. Wind Load (Analytical Verification)

WIND LOADING (SBC-301)

Using the directional design method



Building data

Type of roof;	Flat
Length of building;	b = 19000 mm
Width of building;	d = 14000 mm
Height to eaves;	H = 3300 mm
Mean height;	h = 3300 mm

General wind load requirements

Basic wind speed;	V = **49** m/s (or) 177 kmph (Refer Clause-4.2.6, Doc- MD-216-1000-EG-ST- CRT-1001)
Risk category;	II(As per SBC-301, Table 1.6-1)
Velocity pressure exponent coeff;	K_d = 0.85(As per SBC-301, Table 6.4-1)
Exposure category ;	D(As per SBC-301, Table 6.4.2.3)
Enclosure classification ;	Partially enclosed buildings(As per SBC-301, Section-6.2)
Internal pressure coef +ve;	GC_{pi_p} = 0.55(As per SBC-301, Figure-7.2-1)

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Internal pressure coef –ve; GC_{pi_n} = -0.55(As per SBC-301, Figure-7.2-1)

Gust effect factor; G_f = 0.85(As per SBC-301, Section-7.2.7.1)

Importance Factor; 1(As per SBC-301, Section-9.5)

Topography

Topography factor not significant; K_{zt} = 1.0(As per SBC-301, Section-6.4.3.2)

Velocity pressure equation; $q = 0.0473x10^{-3}Kn/m^2 \times K_z \times K_{zt} \times K_d \times V^2 \times I$

V in Km/hr(As per SBC-301, Section-7.2.8)

Velocity pressures table

z (mm)	K_z (SBC-301,Table 7.2-2)	q_z (kN/m^2)
3300	1.03	1.29
3301	1.03	1.29

Pressures and forces

Net pressure; $p = q \times G_f \times C_{pe} - q_i \times GC_{pi}$;

Net force; $F_w = p \times A_{ref}$;

Walls load case 1 - Wind 0, GC_{pi} 0.55, - c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A	3.30	0.80	1.29	0.17	62.72	10.51
B	3.30	-0.50	1.29	-1.26	62.70	-78.77
C	3.30	-0.70	1.29	-1.48	46.20	-68.16
D	3.30	-0.70	1.29	-1.48	46.20	-68.16

Walls load case 2 - Wind 0, GC_{pi} -0.55, + c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A	3.30	0.80	1.29	1.58	62.72	99.41
B	3.30	-0.50	1.29	0.16	62.70	10.10

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Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
C	3.30	-0.70	1.29	-0.06	46.20	-2.68
D	3.30	-0.70	1.29	-0.06	46.20	-2.68

Walls load case 3 - Wind 90, GC_{pi} 0.55, - c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A	3.30	0.80	1.29	0.17	46.21	7.74
B	3.30	-0.43	1.29	-1.18	46.20	-54.43
C	3.30	-0.70	1.29	-1.48	62.70	-92.51
D	3.30	-0.70	1.29	-1.48	62.70	-92.51

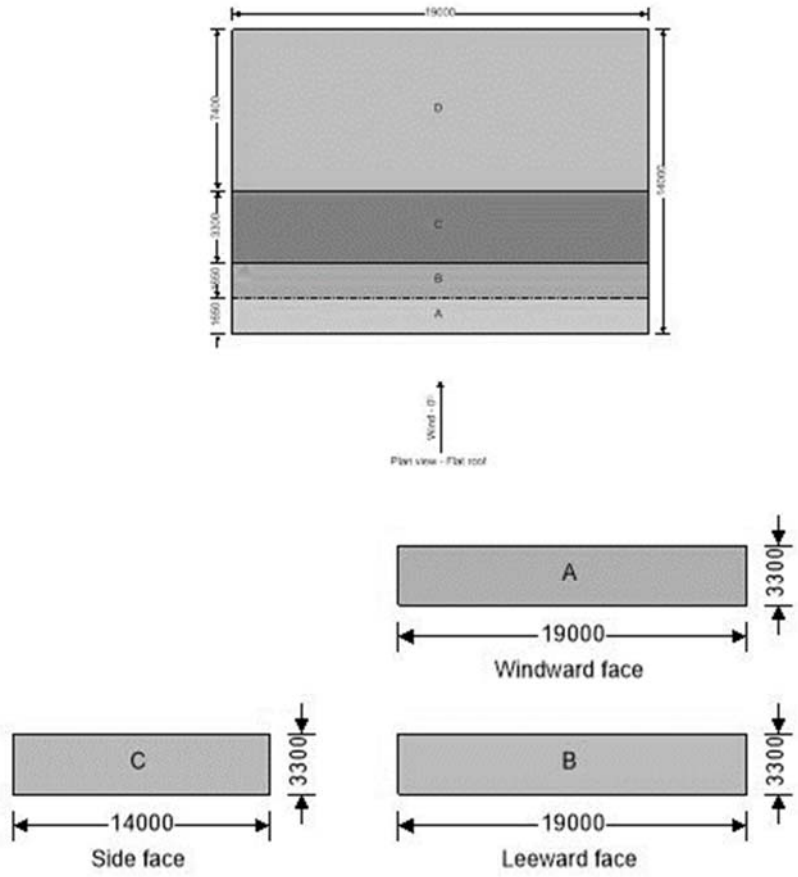
Walls load case 4 - Wind 90, GC_{pi} -0.55, + c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A	3.30	0.80	1.29	1.58	46.21	73.25
B	3.30	-0.43	1.29	0.24	46.20	11.06
C	3.30	-0.70	1.29	-0.06	62.70	-3.64
D	3.30	-0.70	1.29	-0.06	62.70	-3.64

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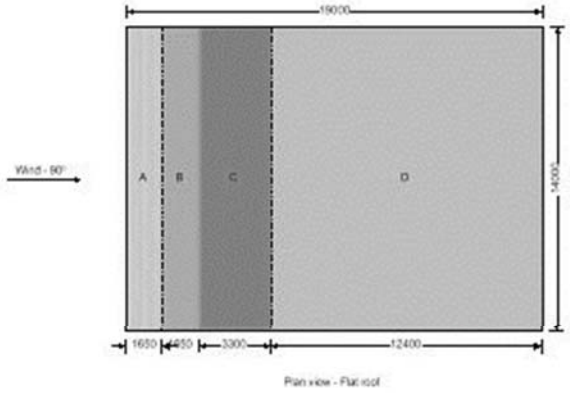
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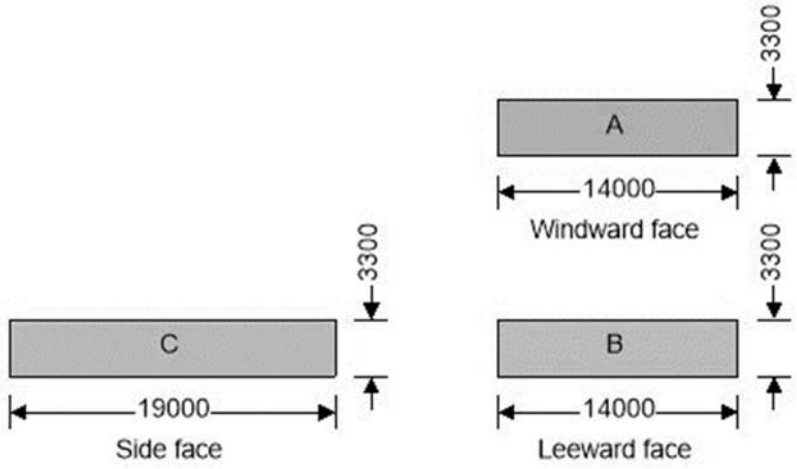


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So from the above analysis, the Wind Loads generated in STAAD are in accordance with SBC-301-2007-Structural Loading & Forces.

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5.5. Seismic Load

Earthquake loads were automatically generated in STAAD as per IBC 2003. The basic parameters affecting the generation of seismic loads is shown below:

Importance Factor, I = 1.0(As per SBC-301, Section-9.5)

Design Spectral Response Acceleration (Refer Clause-4.2.7b, MD-216-1000-EG-ST- CRT-1001)

Ss = 0.1g (Spectral Response Acceleration 0.2 second)

S1 = 0.02g (Spectral Response Acceleration 1 second)

S_{D_1} = 2/3*Fv*S1 = 2/3 * 2.4 * 0.02 = 0.032 (As per SBC-301, Section-9.4.4)

S_{DS} = 2/3*Fa*S1 = 2/3 * 1.60 * 0.10 = 0.10 (As per SBC-301, Section-9.4.4)

Where S_{D_1} Damped Design Spectral Response Acceleration@1 second

S_{DS} Damped Design Spectral Response Acceleration@short period

Site Class = D

Site Coefficient, Fa = 1.60 (As per SBC-301, Table 9.4.3a)

Site Coefficient, Fv = 2.40 (As per SBC-301, Table 9.4.3b)

Building period Coefficient, Ct=0.04 (As per SBC-301, Table 10.9.3.2)

For Ordinary reinforced concrete moment frame,

Response Modification Factor, Rx = 2.5 (As per SBC-301, Table 10.9.3.2)

Response Modification Factor, Rz = 2.5 (As per SBC-301, Table 10.2)

System Over Strength Factor, Omega = 3 (As per SBC-301, Table 10.2)

Deflection Amplification factor, Cd = 2.5 (As per SBC-301, Table 10.2)

5.6. Thermal/Temperature Loads (Self-restraining and Thermal Expansion and Contraction Loads)

An ambient temperature ranges from a maximum of 50° C to a low of 0° C. In direct exposure to sunlight, the temperature shall be assumed to rise a minimum of 15° C above ambient. As a minimum, the temperature gradient for thermal loads shall be +25° C and -15° C with the most severe condition governing the design.

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6. DESIGN COMBINATIONS

Service Loads (Allowable Stress Design) (SBC-301, Section 2.4.1)

D

D + T

D +Lr

D +0.75Lr

D + (W or 0.7E)

D + 0.75W + 0.75Lr

0.6D + W

0.6D + 0.7E

Factored loads (Strength Design) (SBC-301, Section 2.3.2)

1.4D

1.4D+1.4T

1.2(D + T) + 0.5Lr

1.2D + 1.6Lr + 0.8W

1.2D + 1.6W + 0.5Lr

1.2D + 1.0E

0.9D + 1.6W

0.9D + 1.0E

Where:

D = Dead Load

Lr = Live Load

W = Wind Load

E = Seismic Load

T = Temperature Load

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7. CRACKED SECTIONS

In lieu of a more accurate analysis of the effects of cracking of concrete, provisions of SBC-304-Section 10.11 was used.

For Beams:	I = 0.35Ig
For Columns:	I = 0.70Ig
For Walls:	
Un-cracked	I = 0.70Ig
Cracked	I = 0.35Ig
For Slabs	I = 0.25Ig

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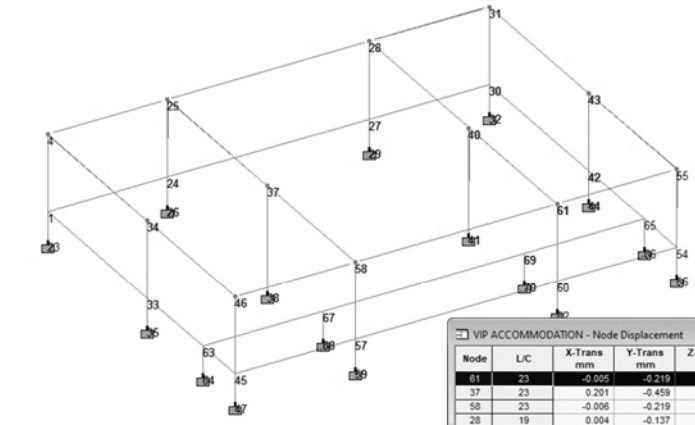
8. DEFLECTIONS/DRIFT CHECK

8.1. Maximum Allowable Deflection

L/240: for member load with live load plus dead load

L/600: for members supporting masonry under total loads

Where, L = span of member between support



VIP ACCOMMODATION - Node Displacement

Node	L/C	X-Trans mm	Y-Trans mm	Z-Trans mm	Absolute mm	X-Rotan rad	Y-Rotan rad	Z-Rotan rad
61	23	-0.005	-0.219	9.691	9.693	-0.001	0.000	0.000
37	23	0.201	-0.459	9.687	9.700	-0.001	0.000	-0.000
58	23	-0.006	-0.219	9.678	9.681	-0.001	-0.000	-0.000
28	19	0.004	-0.137	9.375	9.376	0.002	0.000	0.000
25	19	-0.009	-0.137	9.362	9.363	0.002	-0.000	-0.000
40	19	-0.161	-0.275	9.344	9.349	-0.000	-0.000	0.000
61	19	0.001	-0.135	9.341	9.342	-0.001	0.000	0.000
37	19	0.162	-0.275	9.330	9.336	-0.000	0.000	-0.000
58	19	-0.014	-0.135	9.328	9.329	-0.001	-0.000	-0.000
61	42	0.001	-0.118	-8.324	8.325	-0.002	-0.000	0.000
58	42	-0.009	-0.118	-8.312	8.313	-0.002	0.000	-0.000
40	42	-0.160	-0.278	-8.301	8.307	-0.000	0.000	0.000
28	42	0.003	-0.151	-8.294	8.296	0.002	-0.000	0.000
37	42	0.161	-0.278	-8.288	8.295	-0.000	-0.000	-0.000
25	42	-0.014	-0.151	-8.282	8.283	0.002	0.000	-0.000
61	46	-0.005	-0.202	-7.974	7.977	-0.002	-0.000	0.000
58	46	-0.001	-0.202	-7.961	7.964	-0.002	0.000	-0.000
40	46	-0.197	-0.462	-7.945	7.961	-0.001	0.000	0.000

Maximum attained Drift as per Analysis from above figure = 9.5 mm

Allowable Drift as per Table-10.12, SBC-301-2007 is H/100 = 3300/100 = 33mm

Hence the Structure is safe considering drift check…

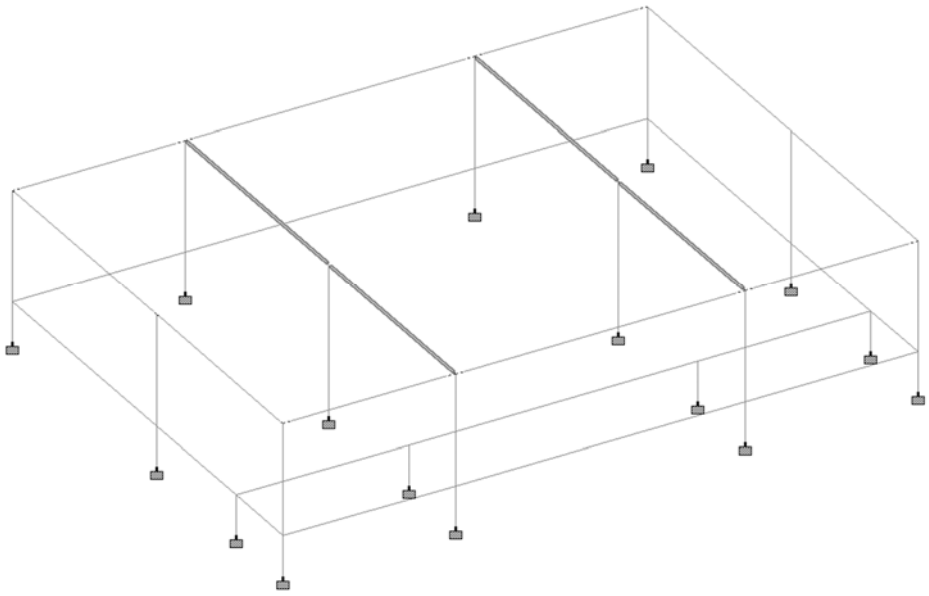
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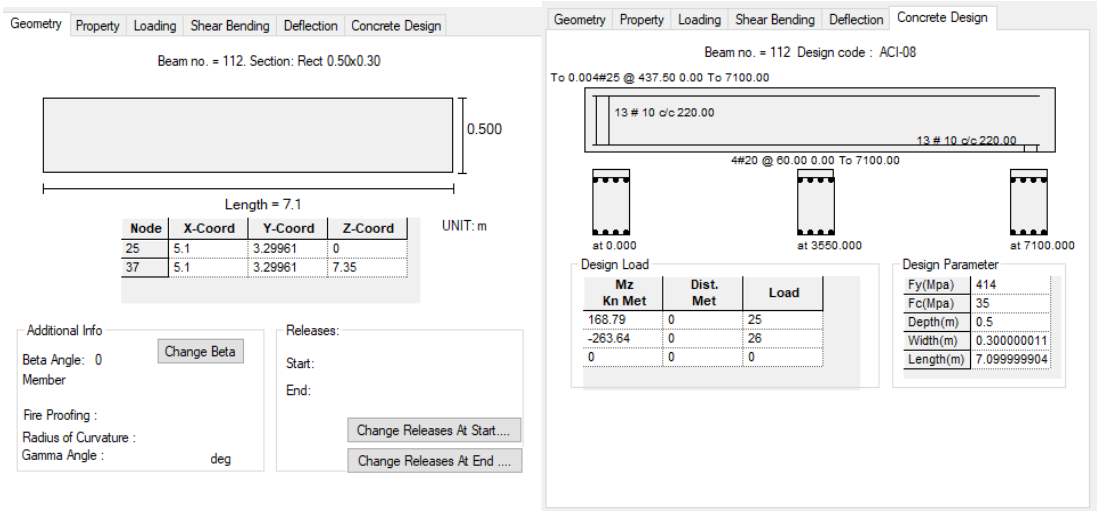
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9. CONCRETE ROOF BEAM DESIGN

9.1. Design of Ledger Beam(LB1)





Design of Ledger Beam (LB1)

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Maximum Moment required from Analysis = **263 Knm (Hogging)**

Therefore Ast.req = $$A_s = 0.85 f_c' \frac{b}{Fy}\left(d - \sqrt{d^2 - \frac{2M_u}{0.85\Phi f_c' b}}\right)$$

Where b=300mm, Fy=414 Mpa, d=500-40-10-20/2=440mm, Fc'=35 Mpa & ϕ =0.9

Ast1.req = **1769 mm²**

Ast.min= (As per clause-10.5-ACI-318)

$$A_{s,\,min} = \frac{0.25\sqrt{f_c'}}{f_y} b_w d \qquad (10\text{-}3)$$

and not less than $1.4 b_w d / f_y$.

Ast.min= **471mm²**

Maximum Moment required from Analysis = **168 Knm (Sagging)**

Ast2.req = **1087 mm²**

However Provide 3-25 dia + 2-25 dia -Full Straight Bars Top & 3-25 dia-Full Straight Bottom & Additional 3-25 dia Reinf@Bottom considering Long term deflection check as shown below in section 8.1.1.

Ast1.pro**>>>>>>(Ast.min, Ast1.req)**

Ast2.pro**>>>>>>(Ast.min, Ast2.req)**

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9.1.1. Long Term Deflection Check for LB1 Beam:

(LB1-BEAM-LONGTERM DEFLECTION CHECK)

INPUT DATA :

Conc. Compressive stress, fc`	=	35.00	MPa
Reinforcement bar tensile stress, fy	=	414.00	MPa
Modulus Elasticity of concrete,	=	27,806	Mpa
Modulus Elasticity of steel, *Es*	=	200,000	Mpa
Unit wt. of concrete, Wconc	=	24.00	kN/m³
Unit wt. of steel	=	77.00	kN/m³
Depth of beam, h	=	500	mm
width of beam, b	=	300	mm
Length of beam, L	=	7.30	m
Bottom bar diam., db	=	25.00	mm
area of 1 bar Ab	=	490.87	mm^2
no. of bottom bars	=	6.00	pcs.
Top bar diam., db	=	25.00	mm
area of 1 bar Ab	=	490.87	mm^2
no. of top bars	=	5.00	pcs.
Stirrups diam. Dv	=	10.00	mm
area of 1 bar Av	=	78.54	mm^2
concrete cover, cc	=	40.00	mm
$d_{eff.bot}$ (@ bottom), d	=	437.50	mm
(FROM STAAD RESULTS) Mdead	=	109.00	kN-m
Mlive	=	11.00	kN-m

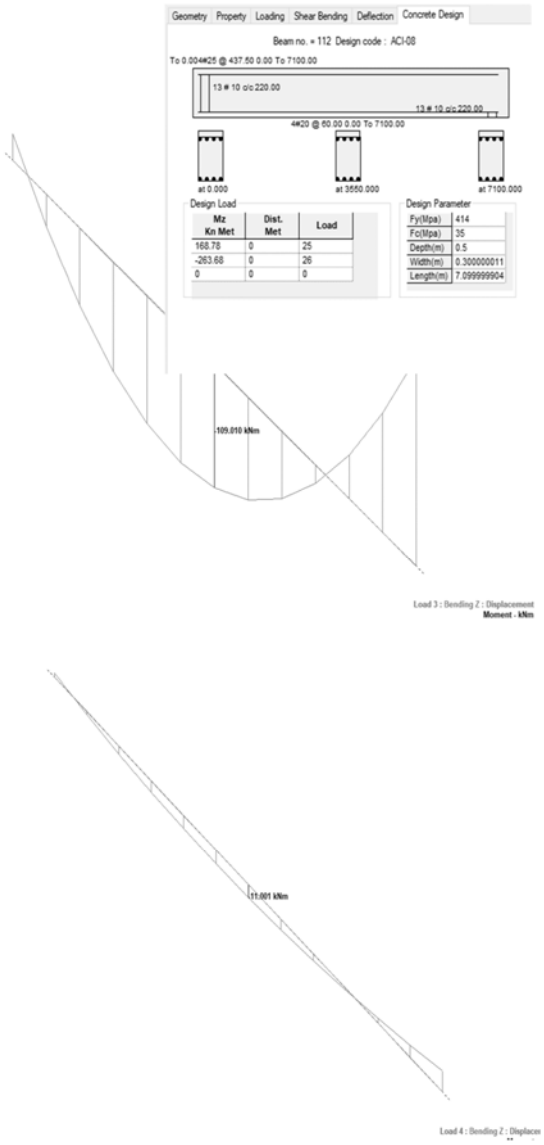
CHECK DEFLECTION: $M_{d+\ell}$ = 120.00 kN-m

$$M_{sus} = M_d + 0.50M_\ell \quad = \quad 114.50 \text{ kN-m}$$

modulus of rupture, modulus of elasticity, modular ratio

$$fr = 0.62\lambda\sqrt{fc'} \quad = \quad 3.67 \text{ Mpa}$$

$$n_s = \frac{Es}{Ec} \quad = \quad 7.19$$



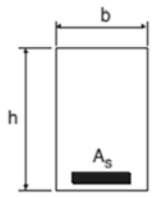
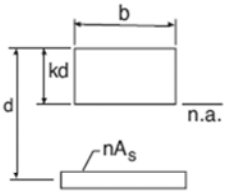
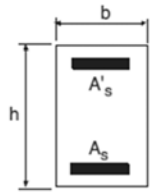
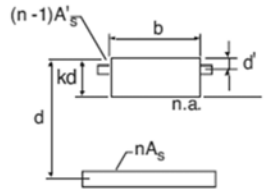
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gross and crack section of moment of inertia

Table 10-2 Gross and Cracked Moment of Inertia of Rectangular and Flanged Section

Gross Section	Cracked Transformed Section	Gross and Cracked Moment of Inertia
	 Without compression steel	$n = \frac{E_s}{E_c}$ $B = \frac{b}{(nA_s)}$ $I_g = \frac{bh^3}{12}$ Without compression steel $kd = \left(\sqrt{2dB+1}-1\right)/B$ $I_{cr} = b(kd)^3/3 + nA_s(d-kd)^2$
	 With compression steel	With compression steel $r = (n-1)A'_s/(nA_s)$ $kd = \left[\sqrt{2dB(1+rd'/d)+(1+r)^2}-(1+r)\right]/B$ $I_{cr} = b(kd)^3/3 + nA_s(d-kd)^2 + (n-1)A'_s(kd-d')^2$

$$I_g = \frac{bh^3}{12} = 3.13E+09 \text{ mm}^4$$

$$B = \frac{b}{nAs} = 0.01 \text{ mm/mm}^2$$

$$kd = \left(\sqrt{2dB+1}-1\right)/B = 187.79 \text{ mm}$$

$$I_{cr} = \frac{b(kd)^3}{3} + nAs(d-kd)^2 = 1.98E+09 \text{ mm}^4$$

$$\frac{I_g}{I_{cr}} = 1.58$$

effective moment of inertia

$$M_{cr} = \frac{frI_g}{y_t} = 45.85 \text{ kN-m}$$ ←*ACI(318M-08) Sec. 9.5.2.3*

under dead load

$$\frac{M_{cr}}{M_d} = 0.42$$

$$(I_e)_d = \left(\frac{M_{cr}}{M_d}\right)^3 I_g + \left[1-\left(\frac{M_{cr}}{M_d}\right)^3\right] I_{cr} = 2.07E+09 \text{ mm}^4$$ ←*ACI(318M-08) Sec. 9.5.2.3*

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under sustained load

$$\frac{M_{cr}}{M_{sus}} = 0.40$$

$$(I_e)_{sus} = \left(\frac{M_{cr}}{M_{sus}}\right)^3 I_g + \left[1-\left(\frac{M_{cr}}{M_{sus}}\right)^3\right] I_{cr} = 2.06E+09 \text{ mm}^4$$ ←ACI(318M-08) Sec. 9.5.2.3

under dead + live load

$$\frac{M_{cr}}{M_{d+\ell}} = 0.38$$

$$(I_e)_{d+\ell} = \left(\frac{M_{cr}}{M_{d+\ell}}\right)^3 I_g + \left[1-\left(\frac{M_{cr}}{M_{d+\ell}}\right)^3\right] I_{cr} = 2.05E+09 \text{ mm}^4$$ ←ACI(318M-08) Sec. 9.5.2.3

initial or short-term deflection

$K = 1.00$

$$(\Delta_i)_d = \frac{K(5/48)M_d\ell^2}{Ec(I_e)_d} = 10.522 \text{ mm}$$

$$(\Delta_i)_{sus} = \frac{K(5/48)M_{sus}\ell^2}{Ec(I_e)_{sus}} = 11.115 \text{ mm}$$

$$(\Delta_i)_{d+\ell} = \frac{K(5/48)M_{d+\ell}\ell^2}{Ec(I_e)_{d+\ell}} = 11.704 \text{ mm}$$

$$(\Delta_i)_\ell = (\Delta_i)_{d+\ell} - (\Delta_i)_d = 1.182 \text{ mm}$$

Table 10-3 Deflection Coeffcient K

		K
1.	Cantilevers (deflection due to rotation at supports not included)	2.40
2.	Simple beams	1.0
3.	Continuous beams	1.2-0.2 M_o/M_a
4.	Fixed-hinged beams (midspan deflection)	0.80
5.	Fixed-hinged beams (maximum deflection using maximum moment)	0.74
6.	Fixed-fixed beams	0.60

For other types of loading, K values are given in Ref. 10.2.

M_o = Simple span moment at midspan $\left(\frac{w\ell^2}{8}\right)$

M_a = Net midspan moment.

TABLE 9.5(b) — MAXIMUM PERMISSIBLE COMPUTED DEFLECTIONS

Type of member	Deflection to be considered	Deflection limitation
Flat roofs not supporting or attached to nonstructural elements likely to be damaged by large deflections	Immediate deflection due to live load *L*	$\ell/180^*$
Floors not supporting or attached to nonstructural elements likely to be damaged by large deflections	Immediate deflection due to live load *L*	$\ell/360$
Roof or floor construction supporting or attached to nonstructural elements likely to be damaged by large deflections	That part of the total deflection occurring after attachment of nonstructural elements (sum of the long-term deflection due to all sustained loads and the immediate deflection due to any additional live load)†	$\ell/480^‡$
Roof or floor construction supporting or attached to nonstructural elements not likely to be damaged by large deflections		$\ell/240^§$

*Limit not intended to safeguard against ponding. Ponding should be checked by suitable calculations of deflection, including added deflections due to ponded water, and considering long-term effects of all sustained loads, camber, construction tolerances, and reliability of provisions for drainage.

†Long-term deflection shall be determined in accordance with 9.5.2.5 or 9.5.4.3, but may be reduced by amount of deflection calculated to occur before attachment of nonstructural elements. This amount shall be determined on basis of accepted engineering data relating to time-deflection characteristics of members similar to those being considered.

‡Limit may be exceeded if adequate measures are taken to prevent damage to supported or attached elements.

§Limit shall not be greater than tolerance provided for nonstructural elements. Limit may be exceeded if camber is provided so that total deflection minus camber does not exceed limit.

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$$(\Delta_i)_\ell \le \frac{\ell}{180} = 40.56 \text{ mm} > 1.18 \text{ mm} \quad ok!$$

$$(\Delta_i)_\ell \le \frac{\ell}{360} = 20.28 \text{ mm} > 1.18 \text{ mm} \quad ok!$$

additional long-term deflection

$\xi = 2.00$

5 years or more 2.0
12 months.....1.4
6 months.....1.2
3 months.....1.0

$$\rho = \frac{As}{bd} = 0.0224$$

$$\lambda_\Delta = \frac{\xi}{1+50\rho'} = 0.94 \quad \leftarrow ACI(318M\text{-}08)\ Sec.\ 9.5.2.5$$

$$\Delta_{cp} + \Delta_{sh} = \lambda(\Delta_i)_{sus} = 10.48 \text{ mm}$$

$$\Delta_{cp} + \Delta_{sh} + (\Delta_i)_\ell = 11.66 \text{ mm}$$

$$\Delta_{cp} + \Delta_{sh} + (\Delta_i)_\ell \le \frac{\ell}{480} = 15.21 \text{ mm} > 11.66 \text{ mm} \quad ok!$$

$$\Delta_{cp} + \Delta_{sh} + (\Delta_i)_\ell \le \frac{\ell}{240} = 30.42 \text{ mm} > 11.66 \text{ mm} \quad ok!$$

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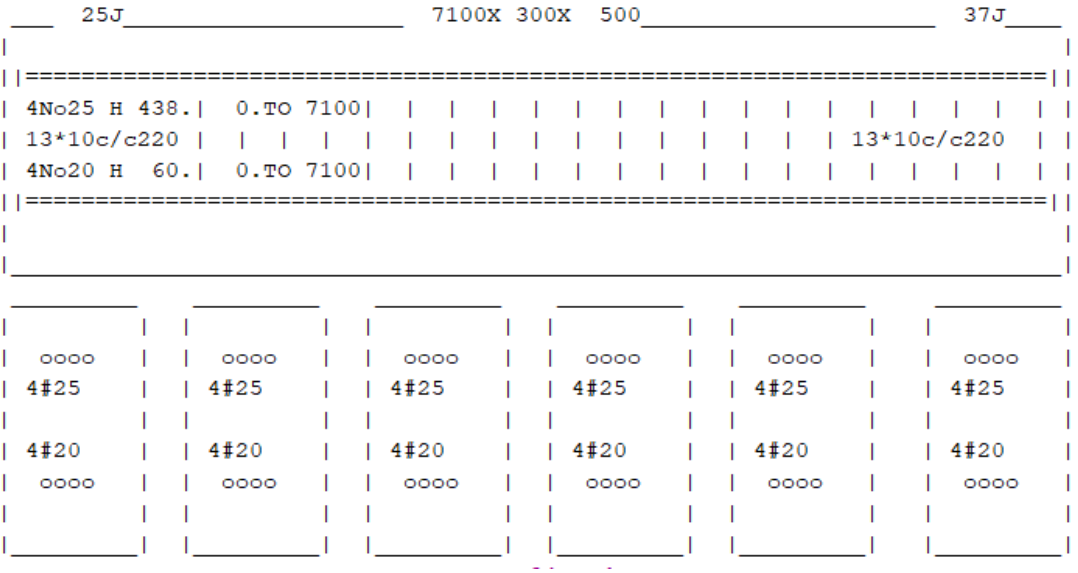
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9.1.2. Staad Output For LB1

```
        B E A M  N O.   112 D E S I G N  R E S U L T S - SHEAR


AT START SUPPORT - Vu=   65.16 KNS  Vc=  134.05 KNS  Vs=    0.00 KNS
Tu=    0.15 KN-MET  Tc=    5.3 KN-MET  Ts=    0.0 KN-MET  LOAD     3
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 220. MM  C/C FOR 2610. MM

AT END   SUPPORT - Vu=  159.45 KNS  Vc=  137.65 KNS  Vs=   74.94 KNS
Tu=    0.21 KN-MET  Tc=    5.3 KN-MET  Ts=    0.0 KN-MET  LOAD    25
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT IS REQUIRED FOR SHEAR.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 220. MM  C/C FOR 2610. MM
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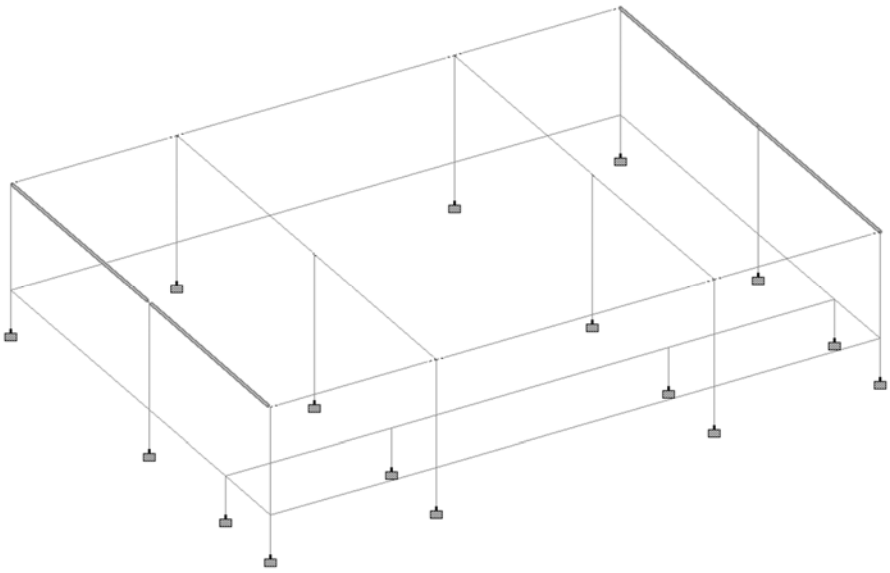


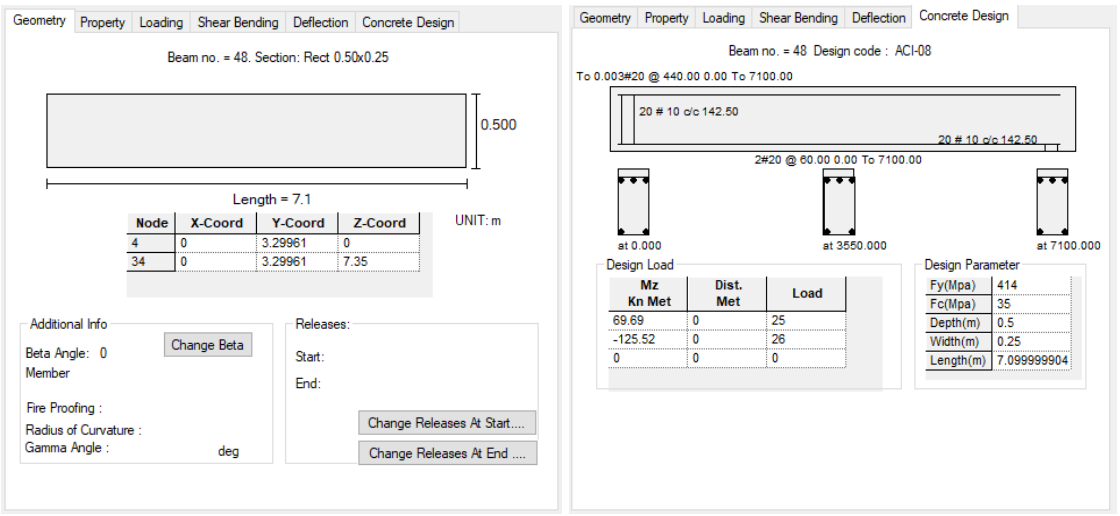
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9.2. Design of Ledger Beam(LB2)





Design of Ledger Beam (LB2)

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Maximum Moment required from Analysis = **125 Knm (Hogging)**

Therefore Ast.req = $$A_s = 0.85 f_c' \frac{b}{Fy}\left(d - \sqrt{d^2 - \frac{2M_u}{0.85\Phi f_c' b}}\right)$$

Where b=250mm, Fy=414 Mpa, d=500-40-10-20/2=440mm, Fc'=35 Mpa & ϕ =0.9

Ast1.req = **803 mm²**

Ast.min= (As per clause-10.5-ACI-318)

$$A_{s,min} = \frac{0.25\sqrt{f_c'}}{f_y} b_w d \qquad (10\text{-}3)$$

and not less than $1.4 b_w d / f_y$.

Ast.min= **471mm²**

Maximum Moment required from Analysis = **69 Knm (Sagging)**

Ast2.req = **432 mm²**

Hence Provide 3-20 dia-Full Straight Bars Top & Bottom considering Long term deflection check as shown below in section 8.2.1.

Ast1.pro= **600mm² >>>>>>(Ast.min, Ast1.req)**

Ast2.pro= **600mm² >>>>>>(Ast.min, Ast2.req)**

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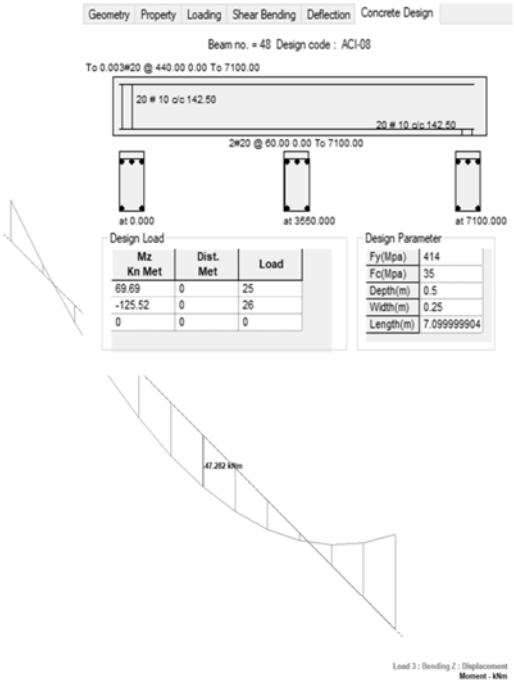
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9.2.1. Long Term Deflection Check for LB2 Beam:

(LB2-BEAM-LONGTERM DEFLECTION CHECK)

INPUT DATA : (LB2-BEAM)

Conc. Compressive stress, fc`	=	35.00	MPa
Reinforcement bar tensile stress, fy	=	414.00	MPa
Modulus Elasticity of concrete,	=	27,806	Mpa
Modulus Elasticity of steel, *Es*	=	200,000	Mpa
Unit wt. of concrete, Wconc	=	24.00	kN/m^3
Unit wt. of steel	=	77.00	kN/m^3
Depth of beam, h	=	500	mm
width of beam, b	=	250	mm
Length of beam, L	=	7.05	m
Bottom bar diam., db	=	20.00	mm
area of 1 bar Ab	=	314.16	mm^2
no. of bottom bars	=	2.00	pcs.
Top bar diam., db	=	20.00	mm
area of 1 bar Ab	=	314.16	mm^2
no. of top bars	=	3.00	pcs.
Stirrups diam. Dv	=	10.00	mm
area of 1 bar Av	=	78.54	mm^2
concrete cover, cc	=	40.00	mm
$d_{eff.bot}$ (@ bottom), d	=	440.00	mm
(FROM STAAD RESULTS) Mdead	=	47.00	kN-m
Mlive	=	4.00	kN-m



CHECK DEFLECTION: $M_{d+\ell}$ = 51.00 kN-m

$$M_{sus} = M_d + 0.50M_\ell \quad = \quad 49.00 \text{ kN-m}$$

modulus of rupture, modulus of elasticity, modular ratio

$$fr = 0.62\lambda\sqrt{fc'} \quad = \quad 3.67 \text{ Mpa}$$

$$n_s = \frac{Es}{Ec} \quad = \quad 7.19 \qquad \leftarrow ACI(318M\text{-}08)\ Sec.\ 9.5.2.3$$

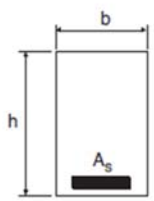
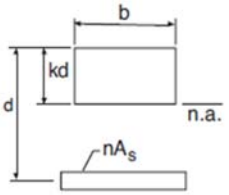
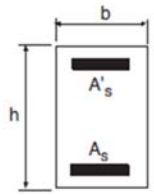
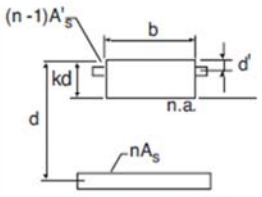
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gross and crack section of moment of inertia

Table 10-2 Gross and Cracked Moment of Inertia of Rectangular and Flanged Section

Gross Section	Cracked Transformed Section	Gross and Cracked Moment of Inertia
	 Without compression steel	$n = \frac{E_s}{E_c}$ $B = \frac{b}{(nA_s)}$ $I_g = \frac{bh^3}{12}$ Without compression steel $kd = \left(\sqrt{2dB+1}-1\right)/B$ $I_{cr} = b(kd)^3/3 + nA_s(d-kd)^2$
	 With compression steel	With compression steel $r = (n-1)A'_s/(nA_s)$ $kd = \left[\sqrt{2dB(1+rd'/d)+(1+r)^2}-(1+r)\right]/B$ $I_{cr} = b(kd)^3/3 + nA_s(d-kd)^2 + (n-1)A'_s(kd-d')^2$

$$I_g = \frac{bh^3}{12} = 2.60E+09 \text{ mm}^4$$

$$B = \frac{b}{nAs} = 0.06 \text{ mm/mm}^2$$

$$kd = \left(\sqrt{2dB+1}-1\right)/B = 109.34 \text{ mm}$$

$$I_{cr} = \frac{b(kd)^3}{3} + nAs(d-kd)^2 = 6.03E+08 \text{ mm}^4$$

$$\frac{I_g}{I_{cr}} = 4.32$$

effective moment of inertia

$$M_{cr} = \frac{frI_g}{y_t} = 38.21 \text{ kN-m} \quad \leftarrow ACI(318M\text{-}08)\ Sec.\ 9.5.2.3$$

under dead load

$$\frac{M_{cr}}{M_d} = 0.81$$

$$(I_e)_d = \left(\frac{M_{cr}}{M_d}\right)^3 I_g + \left[1-\left(\frac{M_{cr}}{M_d}\right)^3\right] I_{cr} = 1.68E+09 \text{ mm}^4 \quad \leftarrow ACI(318M\text{-}08)\ Sec.\ 9.5.2.3$$

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under sustained load

$$\frac{M_{cr}}{M_{sus}} = 0.78$$

$$(I_e)_{sus} = \left(\frac{M_{cr}}{M_{sus}}\right)^3 I_g + \left[1-\left(\frac{M_{cr}}{M_{sus}}\right)^3\right] I_{cr} = 1.55E+09 \text{ mm}^4$$ ←ACI(318M-08) Sec. 9.5.2.3

under dead + live load

$$\frac{M_{cr}}{M_{d+\ell}} = 0.75$$

$$(I_e)_{d+\ell} = \left(\frac{M_{cr}}{M_{d+\ell}}\right)^3 I_g + \left[1-\left(\frac{M_{cr}}{M_{d+\ell}}\right)^3\right] I_{cr} = 1.44E+09 \text{ mm}^4$$ ←ACI(318M-08) Sec. 9.5.2.3

initial or short-term deflection

$K = 1.00$

$$(\Delta_i)_d = \frac{K(5/48)M_d\ell^2}{Ec(I_e)_d} = 5.215 \text{ mm}$$

$$(\Delta_i)_{sus} = \frac{K(5/48)M_{sus}\ell^2}{Ec(I_e)_{sus}} = 5.879 \text{ mm}$$

$$(\Delta_i)_{d+\ell} = \frac{K(5/48)M_{d+\ell}\ell^2}{Ec(I_e)_{d+\ell}} = 6.574 \text{ mm}$$

$$(\Delta_i)_\ell = (\Delta_i)_{d+\ell} - (\Delta_i)_d = 1.359 \text{ mm}$$

Table 10-3 Deflection Coeffcient K

		K
1.	Cantilevers (deflection due to rotation at supports not included)	2.40
2.	Simple beams	1.0
3.	Continuous beams	1.2-0.2 M_o/M_a
4.	Fixed-hinged beams (midspan deflection)	0.80
5.	Fixed-hinged beams (maximum deflection using maximum moment)	0.74
6.	Fixed-fixed beams	0.60

For other types of loading, K values are given in Ref. 10.2.

M_o = Simple span moment at midspan $\left(\frac{w\ell^2}{8}\right)$.

M_a = Net midspan moment.

TABLE 9.5(b) — MAXIMUM PERMISSIBLE COMPUTED DEFLECTIONS

Type of member	Deflection to be considered	Deflection limitation
Flat roofs not supporting or attached to nonstructural elements likely to be damaged by large deflections	Immediate deflection due to live load *L*	$\ell/180^*$
Floors not supporting or attached to nonstructural elements likely to be damaged by large deflections	Immediate deflection due to live load *L*	$\ell/360$
Roof or floor construction supporting or attached to nonstructural elements likely to be damaged by large deflections	That part of the total deflection occurring after attachment of nonstructural elements (sum of the long-term deflection due to all sustained loads and the immediate deflection due to any additional live load)†	$\ell/480^‡$
Roof or floor construction supporting or attached to nonstructural elements not likely to be damaged by large deflections		$\ell/240^§$

*Limit not intended to safeguard against ponding. Ponding should be checked by suitable calculations of deflection, including added deflections due to ponded water, and considering long-term effects of all sustained loads, camber, construction tolerances, and reliability of provisions for drainage.

†Long-term deflection shall be determined in accordance with 9.5.2.5 or 9.5.4.3, but may be reduced by amount of deflection calculated to occur before attachment of nonstructural elements. This amount shall be determined on basis of accepted engineering data relating to time-deflection characteristics of members similar to those being considered.

‡Limit may be exceeded if adequate measures are taken to prevent damage to supported or attached elements.

§Limit shall not be greater than tolerance provided for nonstructural elements. Limit may be exceeded if camber is provided so that total deflection minus camber does not exceed limit.

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$$(\Delta_i)_\ell \le \frac{\ell}{180} = 39.17 \text{ mm} > 1.36 \text{ mm} \quad ok!$$

$$(\Delta_i)_\ell \le \frac{\ell}{360} = 19.58 \text{ mm} > 1.36 \text{ mm} \quad ok!$$

additional long-term deflection

$\xi = 2.00$

5 years or more 2.0
12 months.....1.4
6 months.....1.2
3 months.....1.0

$$\rho = \frac{As}{bd} = 0.0057$$

$$\lambda_\Delta = \frac{\xi}{1+50\rho'} = 1.56$$ ←*ACI(318M-08) Sec. 9.5.2.5*

$$\Delta_{cp} + \Delta_{sh} = \lambda(\Delta_i)_{sus} = 9.15 \text{ mm}$$

$$\Delta_{cp} + \Delta_{sh} + (\Delta_i)_\ell = 10.51 \text{ mm}$$

$$\Delta_{cp} + \Delta_{sh} + (\Delta_i)_\ell \le \frac{\ell}{480} = 14.69 \text{ mm} > 10.51 \text{ mm} \quad ok!$$

$$\Delta_{cp} + \Delta_{sh} + (\Delta_i)_\ell \le \frac{\ell}{240} = 29.38 \text{ mm} > 10.51 \text{ mm} \quad ok!$$

9.2.2. Staad Output For LB2

```
        B E A M   N O.     48 D E S I G N   R E S U L T S - SHEAR

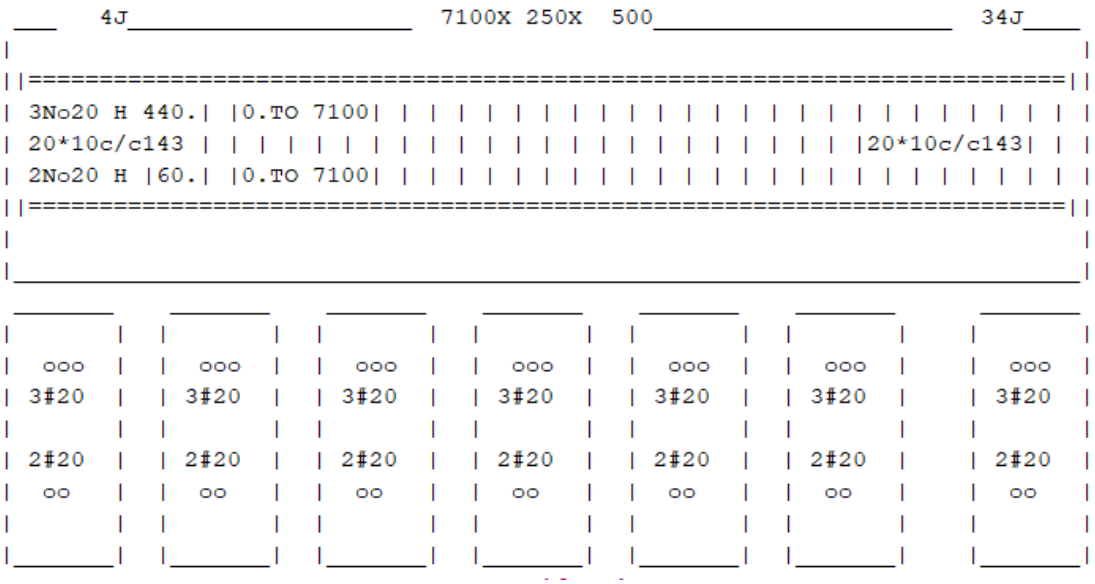

AT START SUPPORT - Vu=    49.82 KNS   Vc=  193.21 KNS   Vs=     0.00 KNS
Tu=    13.10 KN-MET   Tc=     4.0 KN-MET   Ts=    17.5 KN-MET   LOAD     48
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR 2610. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =    4.32 SQ.CM.

AT END    SUPPORT - Vu=    64.88 KNS   Vc=  108.73 KNS   Vs=     0.00 KNS
Tu=    11.05 KN-MET   Tc=     4.0 KN-MET   Ts=    14.7 KN-MET   LOAD     29
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR 2610. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =    3.64 SQ.CM.
```


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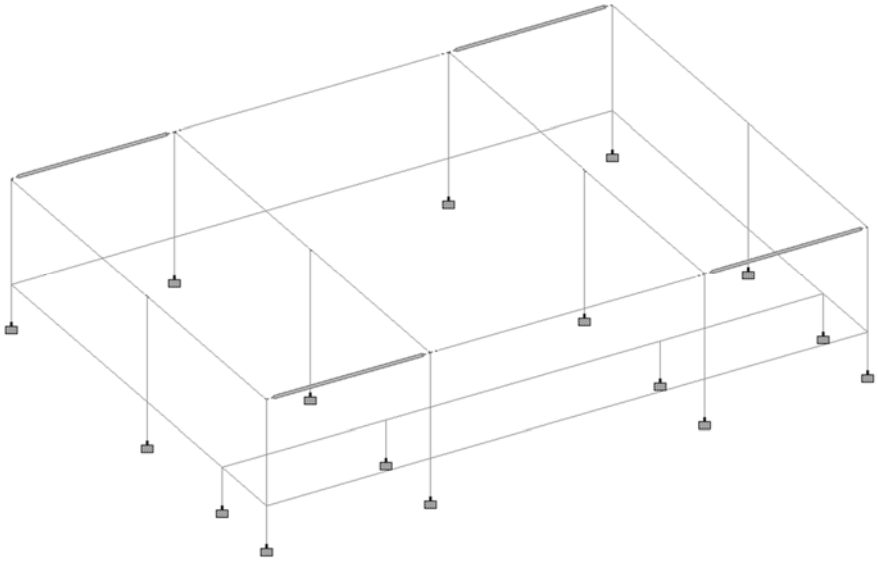


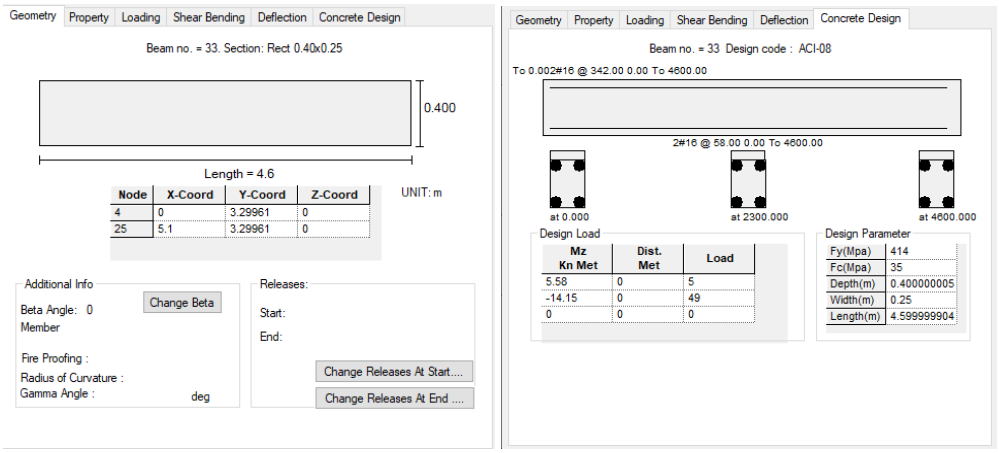
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9.3. Design of Beam-(B1)





Design of Beam (B1)

Maximum Moment required from Analysis = **14 Knm (Hogging)**

Therefore Ast.req =

$$A_s = 0.85 f_c' \frac{b}{Fy}\left(d - \sqrt{d^2 - \frac{2M_u}{0.85\Phi f_c' b}}\right)$$

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Where b=250mm, Fy=414 Mpa, d=400-40-10-12/2=340mm, Fc'=35 Mpa & ϕ=0.9

Ast.min=
(As per clause-10.5-ACI-318)

$$A_{s,min} = \frac{0.25\sqrt{f'_c}}{f_y} b_w d \qquad (10\text{-}3)$$

and not less than $1.4 b_w d / f_y$.

Ast.min= **303mm²**

Maximum Moment required from Analysis = **5 Knm (Sagging)**

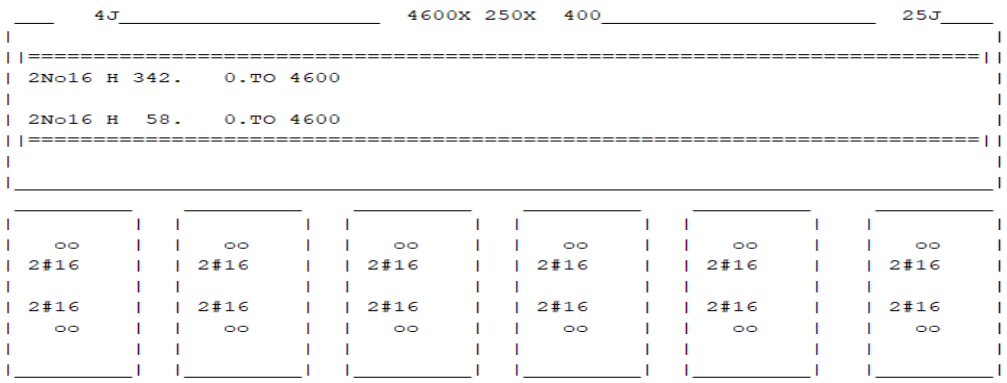
Hence Provide 3-16 dia-Full Straight Bars Top & Bottom

Ast.pro= **600mm² >>>>>>(Ast.min, Ast.req)**

9.3.1. Staad Output For B1

```
            B E A M   N O.     33 D E S I G N   R E S U L T S - SHEAR

   AT START SUPPORT - Vu=     3.48 KNS   Vc=    82.47 KNS   Vs=     0.00 KNS
   Tu=     1.52 KN-MET   Tc=     2.9 KN-MET   Ts=     0.0 KN-MET   LOAD      3
                     STIRRUPS ARE NOT REQUIRED.
   AT END   SUPPORT - Vu=     3.42 KNS   Vc=    83.06 KNS   Vs=     0.00 KNS
   Tu=     1.40 KN-MET   Tc=     2.9 KN-MET   Ts=     0.0 KN-MET   LOAD      3
                     STIRRUPS ARE NOT REQUIRED.
```

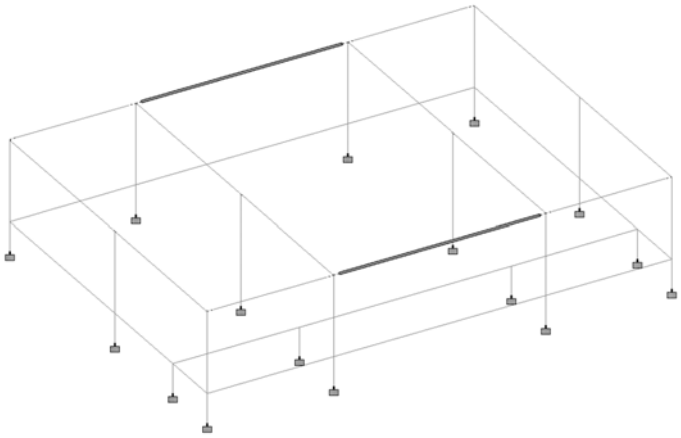


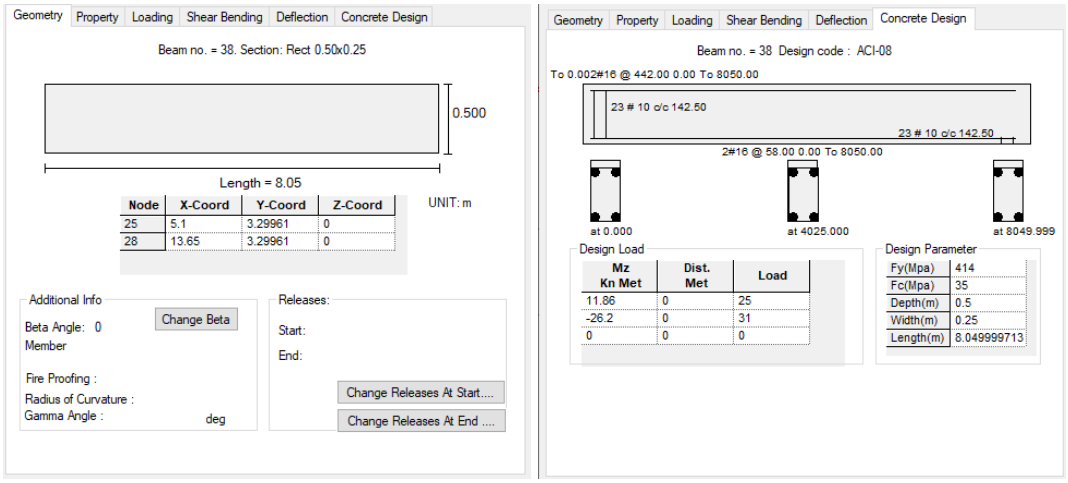
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9.4. Design of Beam-(B2)





Design of Beam (B2)

Maximum Moment required from Analysis = **26 Knm (Hogging)**

Therefore Ast.req = $$A_s = 0.85 f_c' \frac{b}{Fy}\left(d - \sqrt{d^2 - \frac{2M_u}{0.85\Phi f_c' b}}\right)$$

Where b=250mm, Fy=414 Mpa, d=500-40-10-10/2=440mm, Fc'=35 Mpa & ϕ=0.9

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Ast.min= (As per clause-10.5-ACI-318)

$$A_{s,\,min} = \frac{0.25\sqrt{f'_c}}{f_y} b_w d \qquad (10\text{-}3)$$

and not less than $1.4 b_w d / f_y$.

Ast.min= **392mm²**

Maximum Moment required from Analysis = **11 Knm (Sagging)**

However Provide 3-16 dia-Full Straight Bars Top & Bottom

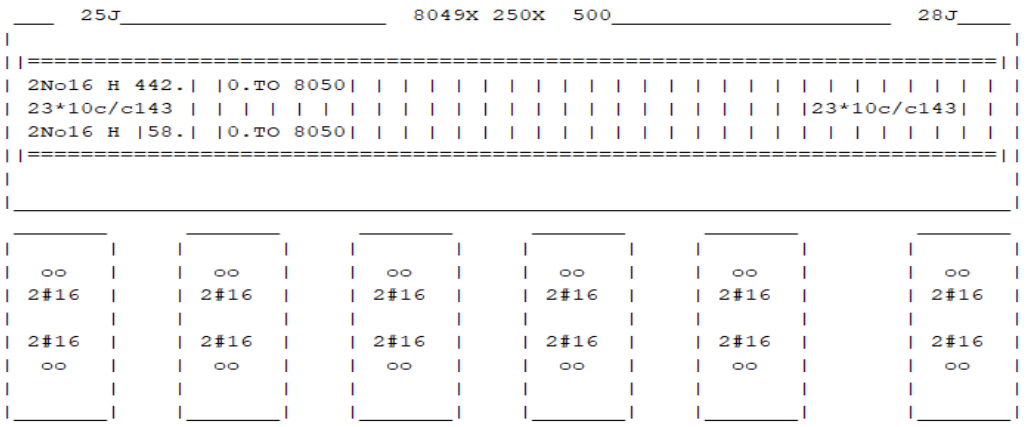
Ast.pro= **600mm² >>>>>>(Ast.min, Ast.req)**

9.4.1. Staad Output For B2

```
                B E A M   N O.     38 D E S I G N   R E S U L T S - SHEAR

 AT START SUPPORT - Vu=    10.09 KNS   Vc=  108.79 KNS   Vs=     0.00 KNS
 Tu=     4.60 KN-MET   Tc=     4.0 KN-MET   Ts=     6.1 KN-MET   LOAD     27
 STIRRUPS ARE REQUIRED FOR TORSION.
 REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
 PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR 3085. MM
 ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =    1.51 SQ.CM.

 AT END    SUPPORT - Vu=    10.09 KNS   Vc=  108.66 KNS   Vs=     0.00 KNS
 Tu=     4.60 KN-MET   Tc=     4.0 KN-MET   Ts=     6.1 KN-MET   LOAD     47
 STIRRUPS ARE REQUIRED FOR TORSION.
 REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
 PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR 3085. MM
 ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =    1.52 SQ.CM.
```



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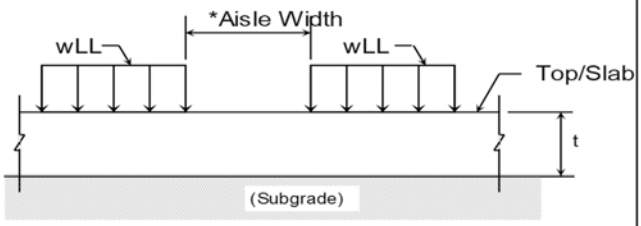
10. CONCRETE GRADE SLAB DESIGN

CONCRETE SLAB ON GRADE ANALYSIS

For Slab Subjected to Stationary Uniformly Distributed Live Loads

Input Data:

Slab Thickness, t =	3.937	in.	100.000	mm		
Concrete Strength, f 'c =	5000	psi	35.000	kpa		
Subgrade Modulus, k =	39.7866	pci	10800	Kn/m3	(AS PER SECTION-7.7-SOIL REPORT)	
Factor of Safety, FS =	1.500		1.500			
Uniform Live Load, wLL =	104.45	psf	5.00	kpa		



Concrete Slab on Grade with Uniform Loads

Results:

Design Parameters:

Modulus of Rupture, MR =	636.40	psi	4387.79	kpa	MR = 9*SQRT(f 'c)
Low. Bending Stress, Fb =	424.26	psi	2925.19	kpa	Fb = MR/FS
Modulus of Elasticity, Ec =	4030509	psi	27789329	kpa	Ec = 57000*SQRT(f 'c)
Poisson's Ratio, μ =	0.15		0.15		μ = 0.15 (assumed for concrete)
Radius of Stiffness, Lr =	26.94	in.	684.37	mm	Lr = (Ec*t^3/(12*(1-μ^2)*k))^0.25
Critical Aisle Width, Wcr =	4.96	ft.	1511.77	mm	Wcr = (2.209*Lr)/12

Stationary Uniformly Distributed Live Loads:

wLL(allow) =	682.05	psf	wLL(allow) = 257.876*Fb*SQRT(k*t/Ec)
	32.65	kpa	**wLL(allow) >= wLL, O.K.**

Provide Minimum Reinforcement of 0.14%@both faces = 0.14%(h) = 140 mm2/M

Table 24.4.3.2—Minimum ratios of deformed shrinkage and temperature reinforcement area to gross concrete area

Reinforcement type	f_y, psi	Minimum reinforcement ratio	
Deformed bars	< 60,000	0.0020	
Deformed bars or welded wire reinforcement	≥ 60,000	Greater of:	$\frac{0.0018 \times 60,000}{f_y}$
			0.0014

Provide WWF-8mm Dia@200 c-c Both Ways-TOP LAYER

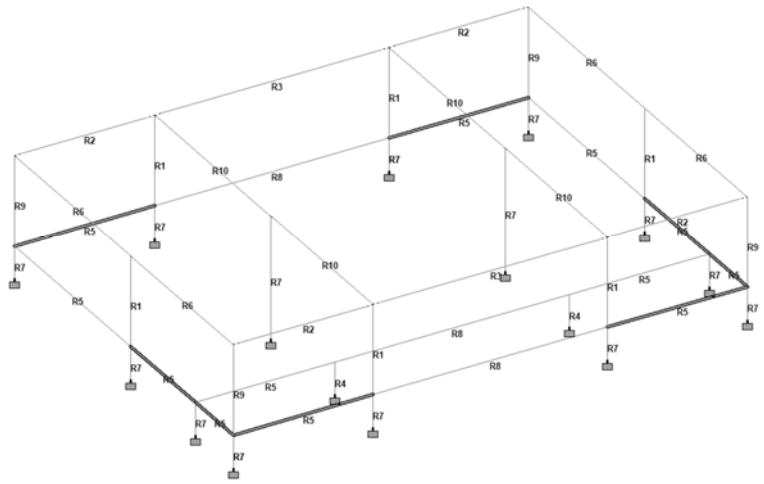
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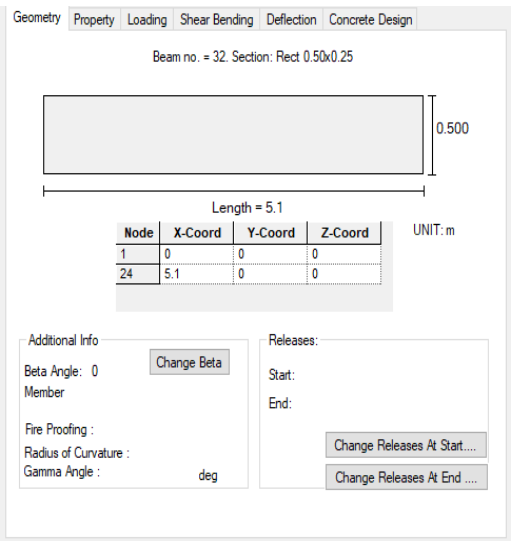
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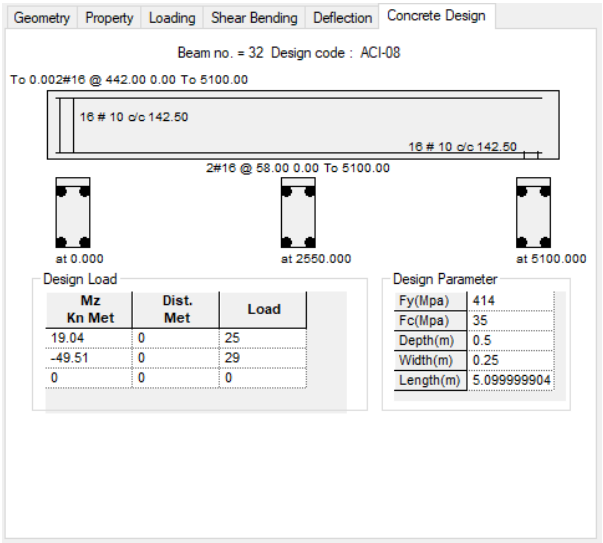
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11. DESIGN OF BEAM-(GB1)

11.1. DESIGN OF BEAM-(GB1)







Maximum Moment required from Analysis = **49 Knm (Hogging)**

Therefore Ast.req =

$$A_s = 0.85 f_c' \frac{b}{Fy}\left(d - \sqrt{d^2 - \frac{2M_u}{0.85\Phi f_c' b}}\right)$$

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Where b=250mm, Fy=414 Mpa, d=500-50-10-16/2=432mm, Fc'=35 Mpa & ϕ=0.9

Ast.req = **310 mm²**

Ast.min= $$A_{s,min} = \frac{0.25\sqrt{f'_c}}{f_y} b_w d \qquad (10\text{-}3)$$

and not less than $1.4 b_w d / f_y$.

Ast.min= **392mm²**

Maximum Moment required from Analysis = **19 Knm (Sagging)**

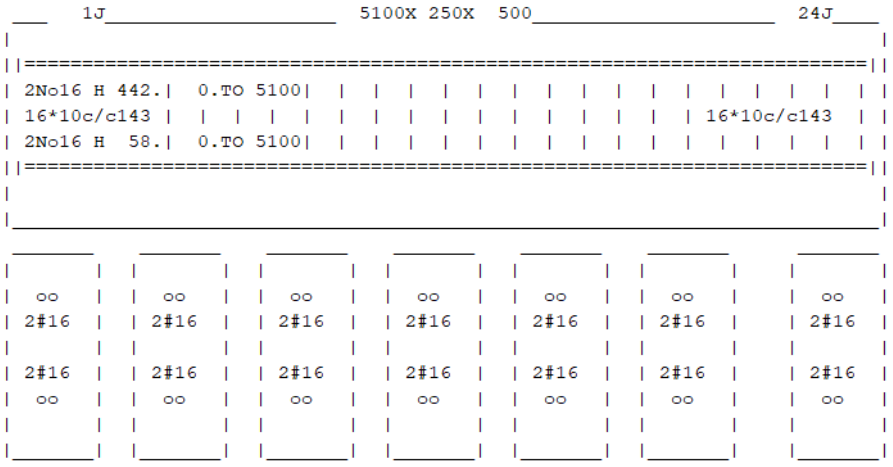
Provide 3-16 dia-Full Straight Bars Top & Bottom

Ast.pro= **600mm² >>>>>>(Ast.min, Ast.req)**

11.1.1. STAAD OUTPUT FOR GB1

```
AT START SUPPORT - Vu=   30.30 KNS  Vc=  105.94 KNS  Vs=    0.00 KNS
Tu=   14.42 KN-MET  Tc=    3.9 KN-MET  Ts=   19.2 KN-MET  LOAD    48
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR 2110. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   4.75 SQ.CM.

AT END   SUPPORT - Vu=   35.28 KNS  Vc=  104.92 KNS  Vs=    0.00 KNS
Tu=   14.42 KN-MET  Tc=    3.9 KN-MET  Ts=   19.2 KN-MET  LOAD    48
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR 2110. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   4.75 SQ.CM.
```

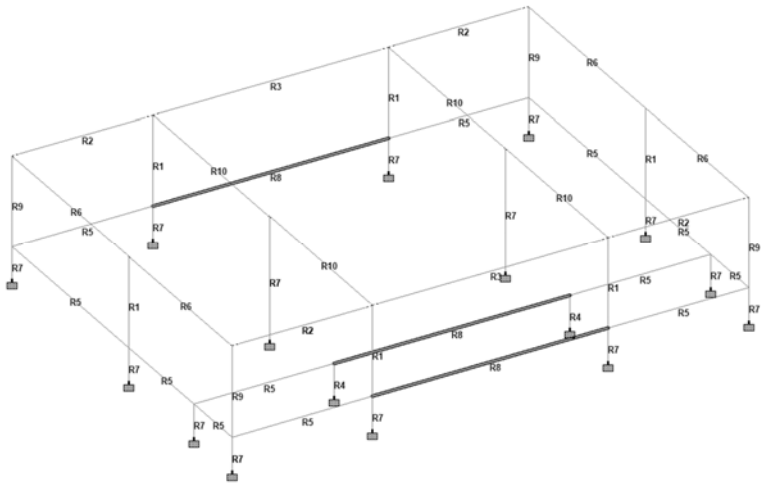


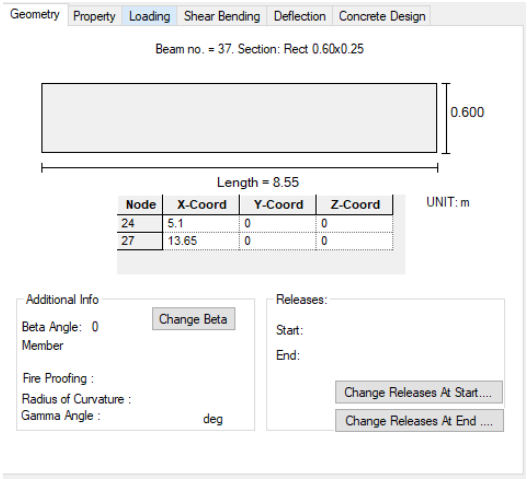
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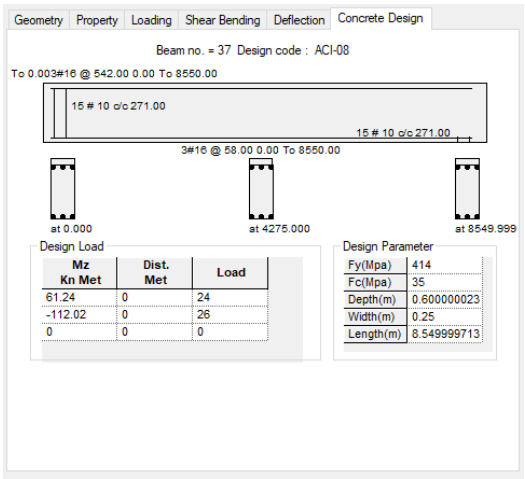
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11.2. DESIGN OF BEAM-(GB2)







Maximum Moment required from Analysis = **112 Knm (Hogging)**

Therefore Ast.req = $A_s = 0.85 f_c' \frac{b}{Fy}\left(d - \sqrt{d^2 - \frac{2M_u}{0.85\Phi f_c' b}}\right)$

Where b=250mm, Fy=414 Mpa, d=600-50-10-16/2=532mm, Fc'=35 Mpa & ϕ=0.9

Ast.req = **582 mm²**

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Ast.min= $A_{s,min} = \frac{0.25\sqrt{f'_c}}{f_y} b_w d$ (10-3)

and not less than $1.4 b_w d / f_y$.

Ast.min= **535mm²**

Maximum Moment required from Analysis = **61 Knm (Sagging)**

Provide 3-20 dia-Full Straight Bars Top & Bottom

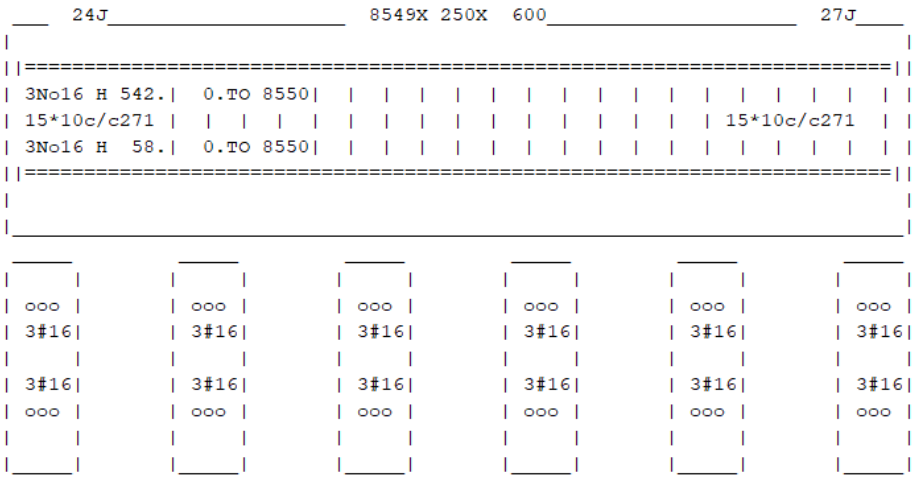
Ast.pro= **942mm² >>>>>>(Ast.min, Ast.req)**

11.2.1. STAAD OUTPUT FOR GB2

```
        B E A M   N O.    37 D E S I G N   R E S U L T S - SHEAR


AT START SUPPORT - Vu=   50.58 KNS  Vc=  130.85 KNS  Vs=    0.00 KNS
Tu=    0.00 KN-MET  Tc=    5.1 KN-MET  Ts=    0.0 KN-MET  LOAD     3
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 271. MM  C/C FOR 3735. MM

AT END   SUPPORT - Vu=   50.58 KNS  Vc=  130.85 KNS  Vs=    0.00 KNS
Tu=    0.00 KN-MET  Tc=    5.1 KN-MET  Ts=    0.0 KN-MET  LOAD     3
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 271. MM  C/C FOR 3735. MM
```

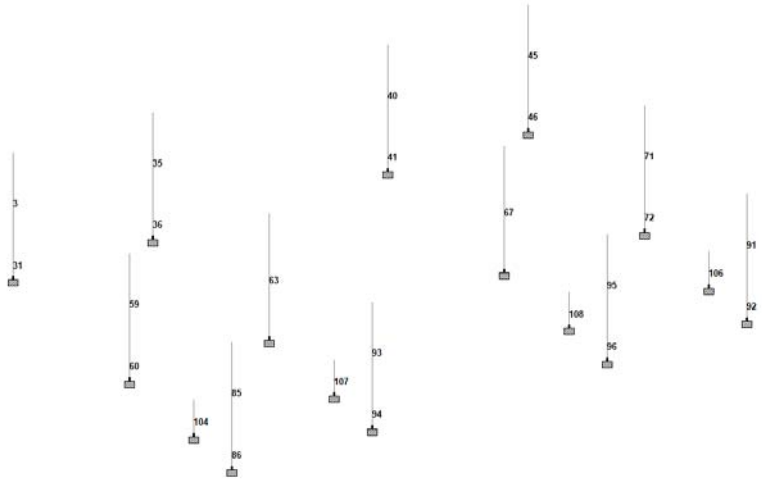


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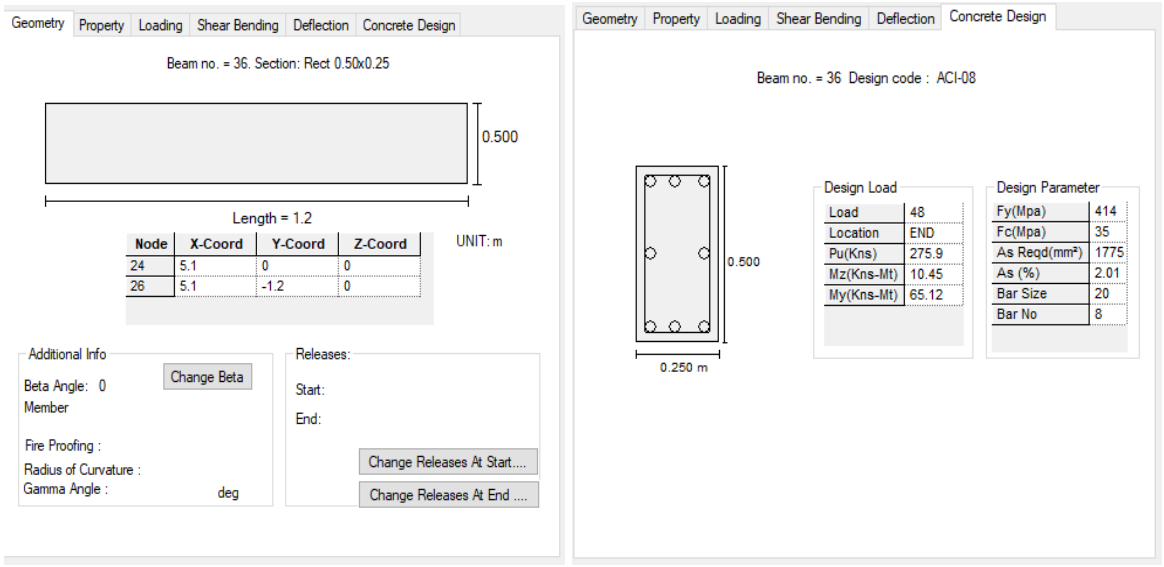
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12. CONCRETE COLUMN DESIGN



Column numbers

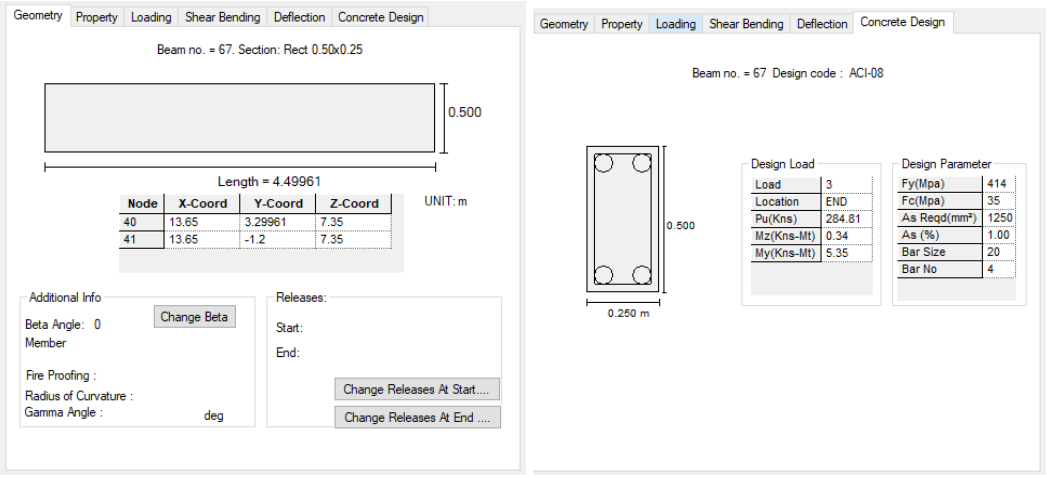


Column Design CR

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Column Design C1

12.1. Staad Output For Column CR & C1

```
      COLUMN  NO.    36  DESIGN PER ACI 318-08 - AXIAL + BENDING


FY - 413.7  FC -  35.0 MPA,  RECT SIZE - 250.0 X 500.0 MMS, TIED
        AREA OF STEEL REQUIRED =  1775.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
-------------------------------------------------------------

  8 - 20 MM               2.011       48      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER   10 SPACING 250.00 MM


     COLUMN  NO.    63  DESIGN PER ACI 318-08 - AXIAL + BENDING


FY - 413.7  FC -  35.0 MPA,  RECT SIZE - 250.0 X 500.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =  1250.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
-------------------------------------------------------------

  4 - 20 MM               1.005        3      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER   10 SPACING 250.00 MM
```


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13. CANTILEVER SLAB DESIGN

DESIGN OF CANTILEVER SLAB

Geometrical Properties			Material Properties			
Span to center of support "sp_1"	0.8	met	Weight of reinforced concrete "γ_r	24		kN/m^3
Clear Span	0.6	met	Strength of concrete "f'_c"	35		N/mm^2
Clear span + depth of slab "sp_2"	0.75	met	Strength of steel "f_y"	420		N/mm^2
Max. eff span le = min (sp_1;sp_2)	0.8	met	Clear cover to concrete("c")	20		mm
Width of Slab "b"	1000	mm	Flexure strength reduction factor	0.9		
Thickness of slab "t"	150	mm	Shear strength reduction factor	0.75		
Eff. Depth "d"	125		Modulus of Elasticity of steel "Es"	2E+05		N/mm^2
Loading			Modulus of Elasticity of concrete	27806		N/mm^2
Self Weight of slab	3.6	kN/m^2	Modular Ratio "n = Es/Ec"	7.19		
Floor Finish	3.5	kN/m^2	**Provided Reinforcement**			
Wall load	0	kN/m^2	Top layer Nos.	6	Φ 10	mm
Total dead Load "U_{dl}"	7.1	kN/m^2	Bottom layer Nos.	0	Φ 12	mm
Live load	1	kN/m^2	Area of Top steel "$A_{s\ t}$"		468	mm^2
Total Ultimate Load "$Uu_{,tot}$"$_{(1.2D+1.6L)}$	10.12	kN/m^2	Area of Bottom steel "$A_{s\ b}$"		0	mm^2
Moment due to Overhang Point Load(0.125*0.15*24*0.6)	0.27	kNm	"d_b' = 10 mm			
Calculations						
Shear force Vu = U*le	**8.096**	**kN**				
Neg. Moment Mu = U*le^2/2	**3.51**	**kNm**				

Check for Section capacity							
Depth of comp. block "a" = $A_{s\ t}$*f_y/(0.85*f'_c*b)	6.61	mm					
Section capacity ΦMn = Φ*$A_{s\ t}$*f_y*(d-a/2)	**21.94**	**kNm**	**O.K**	**>**	**3.51**	**kNm**	(Mu)
Shear requirements							
Shear capacity ΦVc = Φ*0.17*√f'c*b*d	**94.29**	**kN**	**O.K**	**>**	**8.096**	**kN**	(Vu)

Serviceability requirements		
d' = c+d_b/2	26	mm
B = b/(n*$A_{s\ t}$)	0.297	
r = (n-1)*$A_{s\ b}$/(n*$A_{s\ t}$)	0	
k_d = (√(2*d*B(1+r*d'/d)+(1+r)^2)-(1+r))/B	29.01	mm
I_{cr} = bkd^3/3+n*$A_{s\ t}$*(d-k_d)^2+(n-1)*$A_{s\ b}$*(k_d-d'^{2}	3.915E+07	mm^4
Calculate Gross moment of inertia		
Area of slab A = t*b	2E+05	mm^2
Depth of neutral axis "y"	75	mm
Distance of top fiber from N.A "y_t"	75	mm
Gross moment of inertia "I_g"	2.813E+08	mm^4
Allowable stress fr = 0.62√f'c	3.67	mm^2
Cracking moment M_{cr} = fr*I_g/y_t	13.75	kNm

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Service moments

Moment due to dead load "M_{dl}"	2.54	kNm
Moment due to live load "Mll"	0.32	kNm
Total Moment Ms = $M_{dl}+M_{ll}$	2.86	kNm

Deflection due to dead + live loads

R(dl+ll) = M_{cr}/Ms	1.000		
I_e(dl+ll) = $R^3*I_g+(1-R^3)*I_{cr}$	2.813E+08	mm^4	
δ = K*5/48*Ms*le^2/(Ec*I_e)	0.08	mm	**K = 3.2 (conservative)**

Deflection due to dead load

R(dl) = M_{cr}/Md	5.411	
I_e(dl) = $R^3*I_g+(1-R^3)*I_{cr}$	3.839E+10	mm^4
δ = K*5/48*Md*le^2/(Ec*I_e)	0.00	mm

Deflection due to live load

Live load deflection = dl+ll deflection - dl deflection 0.08 mm

Allowable immediate deflection due to Live load = per ACI 318 Tab 9.5b I_e/360 **2.22 mm > 0.08 mm O.K**

Long term deflection

% of liveload sustained "SL"	50	%
M_{sus} = $M_{dl}+SL*M_{ll}$	2.70	kNm
R(sus) = M_{cr}/Msus	5.091	
I_e(sus) = $R^3*I_g+(1-R^3)*I_{cr}$	3.198E+10	
δ = K*5/48*Msus*le^2/(Ec*I_e)	0.00	mm

Magnification factor for creep and shrinkage

$\lambda_\Delta = \xi / (1+50*\rho')$ 2.000

Where:-

ξ = 2 for 5 years or more cl.9.5.2.5 ACI 318

ρ' = A_s b/(b*d) 0

Total deflection = λ_Δ*sus deflection + ll deflection 0.079 mm

Allowable deflection = I_e/240 3.333 mm > 0.079 mm O.K

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14. FOUNDATION DESIGN

Foundations have been designed based on the Soil Report Recommendations. The minimum Embedment Depth of Foundations has been considered as 1.2m & applicable SBC of 350 Kpa. (Refer Doc-MD-216-3000-EG-CV-RPT-1008-A01 for Details)

TABLE 4: RECOMMENDED NET ALLOWABLE BEARING CAPACITIES

PRPJECT: MANSOURAH - MANSSARAH GOLD PROJECT
CLIENT: L & T CONSTRUCTION

Area	Borehole No.	Proposed Structures	Foundation Type	Depth of Foundation below FGL	Footing Size (B x L)	Net Allowable Bearing Capacity Based on		Recommended Net Allowable Bearing Capacity	Estimated Total Settlement	Vertical Modulus of Subgrade Reaction (Ks)	Horizontal Modulus of Subgrade Reaction (Kh)
						Settlement	Shear Failure				
				(m)	(m)	(kN/m^2)	(kN/m^2)	(kN/m^2)	(mm)	($kN/m^2/m$)	(kN/m^3)
Village Area	BH-21	Village	Isolated Square	1.20	1.0 x 1.0	1000	350	350	25	42,000	32,200
				1.50		1100	500	500	25	60,000	38,900
				1.20	2.0 x 2.0	700	580	580	25	69,600	37,500
				1.50		850	750	750	25	90,000	44,200
				1.20	3.0 x 3.0	600	750	600	25	72,000	42,800
				1.50		700	930	700	25	84,000	49,500
				1.20	4.0 x 4.0	550	910	550	25	66,000	48,100
				1.50		625	1090	625	25	75,000	54,800
				1.20	5.0 x 5.0	475	1060	475	25	57,000	53,400
				1.50		550	1240	550	25	66,000	60,200

Refer Attachment- II for Foundation Design calculations

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REFERENCES

1.)	BUILDING PLANS	MD-216-6000-EG-AR-GAD-3092
2.)	OVERALL LAYOUT PLAN	MD-216-6000-EG-GAD-1535
3.)	LAYOUT & DETAILS OF FOUNDATION, COLUMN & GRADE BEAM	MD-216-6000-EG-CV-DED-1490
4.)	LAYOUT & DETAILS OF ROOF & GRADE SLAB	MD-216-6000-EG-CV-DED-1491
5.)	GEOTECHNICAL REPORT	MD-216-3000-EG-CV-RPT-1008
6.)	STRUCTURAL DESIGN CRITERIA	MD-216-0000-EG-ST-CRT-0001

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ATTACHMENT 1 STAAD INPUT DATA

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STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 18-Oct-19
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0 0 0; 4 0 3.29961 0; 23 0 -1.2 0; 24 5.1 0 0; 25 5.1 3.29961 0;
26 5.1 -1.2 0; 27 13.65 0 0; 28 13.65 3.29961 0; 29 13.65 -1.2 0; 30 18.75 0 0;
31 18.75 3.29961 0; 32 18.75 -1.2 0; 33 0 0 7.35; 34 0 3.29961 7.35;
35 0 -1.2 7.35; 37 5.1 3.29961 7.35; 38 5.1 -1.2 7.35;
40 13.65 3.29961 7.35;
41 13.65 -1.2 7.35; 42 18.75 0 7.35; 43 18.75 3.29961
7.35; 44 18.75 -1.2 7.35;
45 0 0 13.82; 46 0 3.29961 13.82; 47 0 -1.2 13.82;
54 18.75 0 13.82;
55 18.75 3.29961 13.82; 56 18.75 -1.2 13.82;
57 5.1 0 13.82;
58 5.1 3.29961 13.82; 59 5.1 -1.2 13.82; 60 13.65 0 13.82;
61 13.65 3.29961 13.82; 62 13.65 -1.2 13.82; 63 0 0 11.42;
64 0 -1.2 11.42;
65 18.75 0 11.42; 66 18.75 -1.2 11.42; 67 5.1 0 11.42;
68 5.1 -1.2 11.42;
69 13.65 0 11.42; 70 13.65 -1.2 11.42;
MEMBER INCIDENCES
3 1 4; 31 1 23; 32 1 24; 33 4 25; 35 24 25; 36 24 26; 37 24 27; 38 25 28;
40 27 28; 41 27 29; 42 27 30; 43 28 31; 45 30 31; 46 30 32; 47 1 33; 48 4 34;
56 30 42; 57 31 43; 59 33 34; 60 33 35; 63 37 38; 67 40 41; 71 42 43; 72 42 44;
73 33 63; 74 34 46; 82 42 65; 83 43 55; 85 45 46; 86 45 47; 91 54 55; 92 54 56;
93 57 58; 94 57 59; 95 60 61; 96 60 62; 97 46 58; 98 58 61; 99 61 55;
100 45 57; 101 57 60; 102 60 54; 103 63 45; 104 63 64; 105 65 54; 106 65 66;
107 67 68; 108 69 70; 109 63 67; 110 67 69; 111 69 65; 112 25 37; 113 37 58;
114 28 40; 115 40 61;
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17184e+007
POISSON 0.17
DENSITY 23.6158
ALPHA 5e-006
DAMP 0.05

TYPE CONCRETE
STRENGTH FCU 27578.9
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
35 40 59 71 93 95 PRIS YD 0.5 ZD 0.25
MEMBER PROPERTY AMERICAN
33 43 97 99 PRIS YD 0.4 ZD 0.25
38 98 PRIS YD 0.5 ZD 0.25
107 108 PRIS YD 0.25 ZD 0.25
32 42 47 56 73 82 100 102 103 105 109 111 PRIS YD 0.5 ZD 0.25
48 57 74 83 PRIS YD 0.5 ZD 0.25
MEMBER PROPERTY AMERICAN
31 36 41 46 60 63 67 72 86 92 94 96 104 106 PRIS YD 0.5 ZD 0.25
MEMBER PROPERTY AMERICAN
37 101 110 PRIS YD 0.6 ZD 0.25
MEMBER PROPERTY AMERICAN
3 45 85 91 PRIS YD 0.5 ZD 0.25
MEMBER PROPERTY AMERICAN
112 TO 115 PRIS YD 0.5 ZD 0.3
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
23 26 29 32 35 38 41 44 47 56 59 62 64 66 68 70 FIXED
****
MEMBER CRACKED
3 31 35 36 40 41 45 46 59 60 63 67 71 72 85 86 91 TO 96 104 106 TO 107 -
108 REDUCTION RIY 0.700000 RIZ 0.700000
32 33 37 38 42 43 47 48 56 57 73 74 82 83 97 99 TO 103 105 109 TO 110 -
111 REDUCTION RIY 0.350000 RIZ 0.350000
MEMBER OFFSET
33 38 43 97 TO 99 START 0.25 0 0
33 38 43 97 TO 99 END -0.25 0 0
48 74 112 TO 115 START 0 0 0.125
48 74 112 TO 115 END 0 0 -0.125
DEFINE IBC 2003
SDS 0.106 SD1 0.032 S1 0.02 IE 1 RX 2.5 RZ 2.5 SCLASS 4
SELFWEIGHT 1
MEMBER WEIGHT
48 57 74 83 UNI 14.4
112 TO 115 UNI 30

32 37 42 47 56 73 82 100 TO 103 105 109 TO 111 UNI 10
DEFINE WIND LOAD
TYPE 1 WIND X WINDWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 4.000 M 20.000 M 15.000 M 2.000 0.010 0 -
0 0 0 0 1.030 1.000 1.000 0.850 0 -
1 0 0 0.850 0.800 -0.550
!> END GENERATED DATA BLOCK
INT 1.59699 1.59699 HEIG 0 4.572
TYPE 2 WIND X LEEWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 4.000 M 20.000 M 15.000 M 2.000 0.010 1 -
0 0 1 0 1.030 1.000 1.000 0.850 0 -
1 0 0 0.850 -0.433 0.550
!> END GENERATED DATA BLOCK
INT -1.19234 -1.19234 HEIG 0 4
TYPE 3 SIDEWARD X
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 4.000 M 20.000 M 15.000 M 2.000 0.010 2 -
0 0 1 0 1.030 1.000 1.000 0.850 0 -
1 0 0 0.850 -0.700 0.550
!> END GENERATED DATA BLOCK
INT -1.48663 -1.48663 HEIG 0 4
TYPE 4 WIND Z WINDWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 4.000 M 15.000 M 20.000 M 2.000 0.010 0 -
0 0 1 0 1.030 1.000 1.000 0.850 0 -
1 0 0 0.850 0.800 -0.550
!> END GENERATED DATA BLOCK
INT 1.59699 1.59699 HEIG 0 4.572
TYPE 5 WIND Z LEEWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 4.000 M 15.000 M 20.000 M 2.000 0.010 1 -
0 0 1 0 1.030 1.000 1.000 0.850 0 -
1 0 0 0.850 -0.500 0.550

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!> END GENERATED DATA BLOCK
INT -1.26591 -1.26591 HEIG  0  4
TYPE 6 SIDEWARD Z
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 4.000 M 15.000 M 20.000 M 2.000 0.010 2 -
0 0 1 0 1.030 1.000 1.000 0.850 0 -
1 0 0 0.850 -0.700 0.550
!> END GENERATED DATA BLOCK
INT -1.48663 -1.48663 HEIG  0  4
LOAD 1 LOADTYPE Seismic  TITLE EQK LOAD Z
IBC LOAD Z 1
PERFORM ANALYSIS
CHANGE
LOAD 2 LOADTYPE Seismic  TITLE EQK LOAD X
IBC LOAD X 1
PERFORM ANALYSIS
CHANGE
LOAD 3 LOADTYPE Dead  TITLE DL
SELFWEIGHT Y -1
MEMBER LOAD
48 57 74 83 UNI GY -14.4
112 TO 115 UNI GY -30
48 74 UMOM GZ 1
57 83 UMOM GZ -1
33 38 43 UMOM GX -1
97 TO 99 UMOM GX 1
32 37 42 47 56 73 82 100 TO 103 105 109 TO 111 UNI GY -10
LOAD 4 LOADTYPE Live  TITLE LL
MEMBER LOAD
48 74 UMOM GZ 0.18
57 83 UMOM GZ -0.18
33 38 43 UMOM GX -0.18
97 TO 99 UMOM GX 0.18
48 57 74 83 UNI GY -1.275
112 TO 115 UNI GY -3.42
LOAD 5 LOADTYPE Wind  TITLE WL+X
WIND LOAD X 1 TYPE 1 YR 0 4.001
WIND LOAD -X 1 TYPE 2 YR 0 4
WIND LOAD -Z 1 TYPE 3 YR 0 4
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WIND LOAD -Z -1 TYPE 3 YR 0 4

LOAD 6 LOADTYPE Wind TITLE WL+Z

WIND LOAD Z 1 TYPE 4 YR 0 4

WIND LOAD -Z 1 TYPE 5 YR 0 4

WIND LOAD -X 1 TYPE 6 YR 0 4

WIND LOAD -X -1 TYPE 6 YR 0 4

LOAD 7 LOADTYPE Wind TITLE WL-X

WIND LOAD X -1 TYPE 1 YR 0 4

WIND LOAD -X -1 TYPE 2 YR 0 4

WIND LOAD -Z 1 TYPE 3 YR 0 4

WIND LOAD -Z -1 TYPE 3 YR 0 4

LOAD 8 LOADTYPE Wind TITLE WL-Z

WIND LOAD Z -1 TYPE 4 YR 0 4

WIND LOAD -Z -1 TYPE 5 YR 0 4

WIND LOAD -X 1 TYPE 6 YR 0 4

WIND LOAD -X -1 TYPE 6 YR 0 4

LOAD 9 LOADTYPE Temperature TITLE TEMP

TEMPERATURE LOAD

3 33 35 38 40 43 45 48 57 59 63 67 71 74 83 85 91 93 95 97 TO 99 112 TO 114
-

115 TEMP 25

LOAD COMB 10 SLS: (DL)

3 1.0

LOAD COMB 11 SLS: (DL+T)

3 1.0

LOAD COMB 12 SLS: (D+LR)

3 1.0 4 1.0

LOAD COMB 13 SLS: (D+0.75LR)

3 1.0 4 0.75

LOAD COMB 14 SLS: (D+0.7EZ)

3 1.0 1 0.7

LOAD COMB 15 SLS: (D+0.7EX)

3 1.0 2 0.7

LOAD COMB 16 SLS: (D+0.75WX1+0.75LR)

3 1.0 5 0.75 4 0.75

LOAD COMB 17 SLS: (D+0.75WZ1+0.75LR)

3 1.0 6 0.75 4 0.75

LOAD COMB 18 SLS: (0.6D+WX1)

3 0.6 5 1.0

LOAD COMB 19 SLS: (0.6D+WZ1)

3 0.6 6 1.0
LOAD COMB 20 SLS: (0.6D+0.7EX)
3 0.6 2 0.7
LOAD COMB 21 SLS: (0.6D+0.7EZ)
3 0.6 1 0.7
LOAD COMB 22 SLS: (D+WX1)
3 1.0 5 1.0
LOAD COMB 23 SLS: (D+WZ1)
3 1.0 6 1.0
LOAD COMB 24 ULS: (1.4 DL)
3 1.4
LOAD COMB 25 ULS: (1.4D+1.4T+0.5LR)
3 1.4 4 0.5
LOAD COMB 26 ULS: (1.4D+1.4T+0.5LR)
3 1.4 4 0.5
LOAD COMB 27 ULS: (1.2D+0.80WX1+1.6LR)
3 1.2 5 0.8 4 1.6
LOAD COMB 28 ULS: (1.2D+1.6WZ1+0.5LR)
3 1.2 6 1.6 4 0.5
LOAD COMB 29 ULS: (1.2D+1.6WX1+0.5LR)
3 1.2 4 0.5 5 1.6
LOAD COMB 30 ULS: (1.2D+1.6WZ1+0.5LR)
3 1.2 4 0.5 6 1.6
LOAD COMB 31 ULS: (1.2D+EX)
3 1.2 2 1.0
LOAD COMB 32 ULS: (1.2D+EZ)
3 1.2 1 1.0
LOAD COMB 33 ULS: (0.90D+1.6WX1)
3 0.9 5 1.6
LOAD COMB 34 ULS: (0.90D+1.6WZ1)
3 0.9 6 1.6
LOAD COMB 35 ULS: (0.90D+EZ)
3 0.9 1 1.0
LOAD COMB 36 ULS: (0.90D+EX)
3 0.9 2 1.0
LOAD COMB 39 SLS: (D+0.75WX2+0.75LR)
3 1.0 7 0.75 4 0.75
LOAD COMB 40 SLS: (D+0.75WZ2+0.75LR)
3 1.0 8 0.75 4 0.75
LOAD COMB 41 SLS: (0.6D+WX2)

3 0.6 7 1.0

LOAD COMB 42 SLS: (0.6D+WZ2)

3 0.6 8 1.0

LOAD COMB 45 SLS: (D+WX2)

3 1.0 7 1.0

LOAD COMB 46 SLS: (D+WZ2)

3 1.0 8 1.0

LOAD COMB 47 ULS: (1.2D+0.80WX2+1.6LR)

3 1.2 7 0.8 4 1.6

LOAD COMB 48 ULS: (1.2D+1.6WZ2+0.5LR)

3 1.2 8 1.6 4 0.5

LOAD COMB 49 ULS: (1.2D+1.6WX2+0.5LR)

3 1.2 4 0.5 7 1.6

LOAD COMB 50 ULS: (1.2D+1.6WZ2+0.5LR)

3 1.2 4 0.5 8 1.6

LOAD COMB 53 ULS: (0.90D+1.6WX2)

3 0.9 7 1.6

LOAD COMB 54 ULS: (0.90D+1.6WZ2)

3 0.9 8 1.6

PERFORM ANALYSIS

START CONCRETE DESIGN

CODE ACI

FC 35000 ALL

CLB 0.04 ALL

CLS 0.04 ALL

CLT 0.04 ALL

MINSEC 10 ALL

SFACE 0.5 MEMB 33 38 43 48 57 74 83 97 TO 99 112 TO 115

EFACE 0.5 MEMB 33 38 43 48 57 74 83 97 TO 99 112 TO 115

MAXMAIN 16 MEMB 32 37 42 47 56 73 82 100 TO 103 105 109 TO 111

MINMAIN 16 MEMB 32 33 37 38 42 43 47 56 73 82 97 TO 103 105 109 TO 111

MAXMAIN 20 MEMB 3 31 33 35 36 38 40 41 43 45 46 48 57 59 60 63 67 71 72 74 -

83 85 86 91 TO 99 104 106 TO 108

MINMAIN 20 MEMB 3 31 35 36 40 41 45 46 48 57 59 60 63 67 71 72 74 83 85 86 -

91 TO 96 104 106 TO 108 112 TO 115

DESIGN BEAM 32 33 37 38 42 43 47 48 56 57 73 74 82 83 97 TO 103 105 109 TO 115

DESIGN COLUMN 3 31 35 36 40 41 45 46 59 60 63 67 71 72 85 86 91 TO 96 104 -

106 TO 108
END CONCRETE DESIGN
FINISH

ATTACHMENT 2 FOUNDATION DESIGN CALCULATIONS

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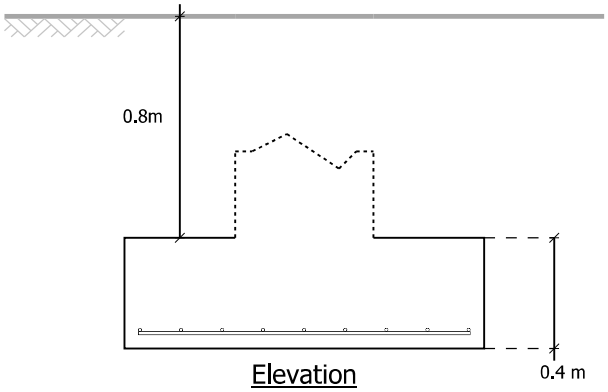
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FOUNDATION F1 ALONG GRID-1 & A

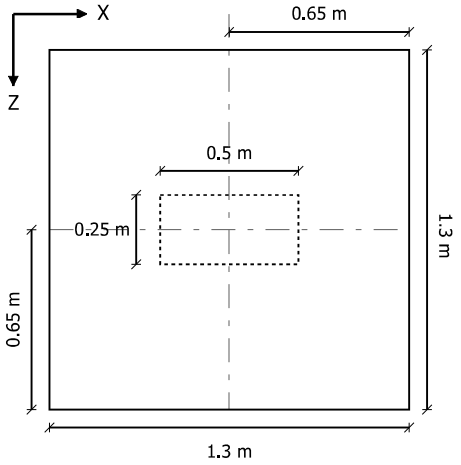
Isolated Footing Design(ACI 318-08)

Footing No.	Group ID	Foundation Geometry		
-	-	Length	Width	Thickness
23	1	1.300m	1.300m	0.400m

Footing No.	Footing Reinforcement				Pedestal Reinforcement	
-	Bottom Reinforcement(M_z)	Bottom Reinforcement(M_x)	Top Reinforcement(M_z)	Top Reinforcement(M_x)	Main Steel	Trans Steel
23	9 - ϕ12	9 -ϕ12	N/A	N/A	N/A	N/A



Elevation



Plan

Input Values

Footing Geomtery

Design Type : Set Dimension
Footing Thickness (Ft) : 400.000mm
Footing Length - X (Fl) : 1300.000mm
Footing Width - Z (Fw) : 1300.000mm
Eccentricity along X (Oxd) : 0.000mm
Eccentricity along Z (Ozd) : 0.000mm

Column Dimensions

Column Shape : Rectangular
Column Length - X (D_{col}) : 0.500m

Column Width - Z (B_{col}) : 0.250m

Pedestal

Include Pedestal? No
Pedestal Shape : N/A
Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete : 24.000kN/m3
Strength of Concrete : 35.000N/mm2
Yield Strength of Steel : 420.000N/mm2
Minimum Bar Size : #12
Maximum Bar Size : #12
Top Footing Minimum Bar Size : #12
Top Footing Maximum Bar Size : #12
Pedestal Minimum Bar Size : #16
Pedestal Maximum Bar Size : #16
Minimum Bar Spacing : 125.000mm
Maximum Bar Spacing : 150.000mm
Pedestal Clear Cover (P, CL) : 50.000mm
Bottom Footing Clear Cover (F, CL) : 50.000mm

Soil Properties

Soil Type : Cohesionless Soil
Unit Weight : 18.000kN/m3
Soil Bearing Capacity : 350.000kN/m2
Soil Bearing Capacity Type: Net Bearing Capacity
Soil Surcharge : 0.000kN/m2
Depth of Soil above Footing : 800.000mm
Type of Depth : Fixed Top
Undrained Shear Strength : 0.000kN/m2
Bearing Capacity Input Method: Fixed Bearing Capacity

Sliding and Overturning

Coefficient of Friction : 0.500
Factor of Safety Against Sliding : 1.500
Factor of Safety Against Overturning : 1.500

Global Settings

Top Reinforcement Option : Calculate only when foundation is subjected to uplift forces
Concrete Design Option : Gross Pressure
Top Reinforcement Factor : 1.000

Design Calculations

Footing Size

Initial Length (L_o) = 1.300m

Initial Width (W_o) = 1.300m

Load Combinations

Load Combination/s- Service Stress Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
10	SLS: (DL)	1.00	1.00	1.00
11	SLS: (DL+T)	1.00	1.00	1.00
12	SLS: (D+LR)	1.00	1.00	1.00
13	SLS: (D+0.75LR)	1.00	1.00	1.00
14	SLS: (D+0.7EZ)	1.00	1.00	1.00
15	SLS: (D+0.7EX)	1.00	1.00	1.00
16	SLS: (D+0.75WX1+0.75LR)	1.00	1.00	1.00
17	SLS: (D+0.75WZ1+0.75LR)	1.00	1.00	1.00
18	SLS: (0.6D+WX1)	1.00	1.00	1.00
19	SLS: (0.6D+WZ1)	1.00	1.00	1.00
20	SLS: (0.6D+0.7EX)	1.00	1.00	1.00
21	SLS: (0.6D+0.7EZ)	1.00	1.00	1.00
22	SLS: (D+WX1)	1.00	1.00	1.00
23	SLS: (D+WZ1)	1.00	1.00	1.00
39	SLS: (D+0.75WX2+0.75LR)	1.00	1.00	1.00
40	SLS: (D+0.75WZ2+0.75LR)	1.00	1.00	1.00
41	SLS: (0.6D+WX2)	1.00	1.00	1.00
42	SLS: (0.6D+WZ2)	1.00	1.00	1.00
45	SLS: (D+WX2)	1.00	1.00	1.00
46	SLS: (D+WZ2)	1.00	1.00	1.00

Load Combination/s- Strength Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
24	ULS: (1.4 DL)	1.00	1.00	1.00
25	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
26	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
27	ULS: (1.2D+0.80WX1+1.6LR)	1.00	1.00	1.00
28	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
29	ULS: (1.2D+1.6WX1+0.5LR)	1.00	1.00	1.00
30	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
31	ULS: (1.2D+EX)	1.00	1.00	1.00
32	ULS: (1.2D+EZ)	1.00	1.00	1.00
33	ULS: (0.90D+1.6WX1)	1.00	1.00	1.00
34	ULS: (0.90D+1.6WZ1)	1.00	1.00	1.00
35	ULS: (0.90D+EZ)	1.00	1.00	1.00
36	ULS: (0.90D+EX)	1.00	1.00	1.00
47	ULS: (1.2D+0.80WX2+1.6LR)	1.00	1.00	1.00
48	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
49	ULS: (1.2D+1.6WX2+0.5LR)	1.00	1.00	1.00
50	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
53	ULS: (0.90D+1.6WX2)	1.00	1.00	1.00
54	ULS: (0.90D+1.6WZ2)	1.00	1.00	1.00

Applied Loads

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
10	150.913	-12.463	-26.090	-9.069	0.946
11	150.913	-12.463	-26.090	-9.069	0.946
12	155.006	-12.556	-25.237	-8.838	0.925
13	153.983	-12.533	-25.451	-8.896	0.930
14	148.280	-12.454	-26.504	-6.308	0.931
15	146.541	-7.411	-26.068	-9.069	-9.801
16	150.714	-6.680	-25.501	-9.136	-8.732
17	151.671	-13.360	-25.306	-5.457	1.961
18	86.189	0.326	-15.722	-5.762	-12.315
19	87.465	-8.580	-15.461	-0.856	1.942

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
20	86.176	-2.426	-15.632	-5.441	-10.179
21	87.915	-7.469	-16.068	-2.680	0.552
22	146.554	-4.659	-26.158	-9.389	-11.937
23	147.830	-13.565	-25.898	-4.483	2.321
39	157.230	-18.163	-25.495	-9.123	10.310
40	156.897	-13.382	-25.596	-12.282	1.997
41	94.878	-14.984	-15.713	-5.745	13.075
42	94.434	-8.610	-15.848	-9.956	1.991
45	155.243	-19.969	-26.149	-9.372	13.453
46	154.799	-13.595	-26.285	-13.583	2.369

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
24	211.278	-17.448	-36.527	-12.697	1.325
25	213.325	-17.495	-36.100	-12.581	1.314
26	213.325	-17.495	-36.100	-12.581	1.314
27	184.157	-8.862	-29.998	-10.770	-9.205
28	178.210	-16.766	-30.573	-3.430	3.324
29	176.168	-2.516	-30.990	-11.280	-19.488
30	178.210	-16.766	-30.573	-3.430	3.324
31	174.850	-7.738	-31.277	-10.883	-14.218
32	177.334	-14.943	-31.899	-6.938	1.113
33	128.848	1.270	-23.589	-8.674	-19.761
34	130.889	-12.980	-23.173	-0.825	3.051
35	132.060	-11.204	-24.072	-4.217	0.829
36	129.576	-3.999	-23.449	-8.162	-14.501
47	191.108	-21.110	-29.991	-10.756	11.107
48	189.360	-16.814	-31.193	-17.990	3.402
49	190.070	-27.012	-30.976	-11.253	21.136
50	189.360	-16.814	-31.193	-17.990	3.402
53	142.750	-23.226	-23.575	-8.648	20.863
54	142.040	-13.028	-23.792	-15.385	3.128

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o = 1.690m^2$

Min. area required from bearing pressure, $A_{min} = P / q_{max} = 1.035m^2$

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) = 1.300 m Governing Load Case : # 10

Width (W_2) = 1.300 m Governing Load Case : # 10

Depth (D_2) = 0.400 m Governing Load Case : # 49

Depth is governed by Ultimate Load Case

(Service check is performed with footing thickness requirements from concrete check)

Area (A_2) = 1.690 m^2

Final Soil Height = 0.800 m

Footing Self Weight = 16.223 kN

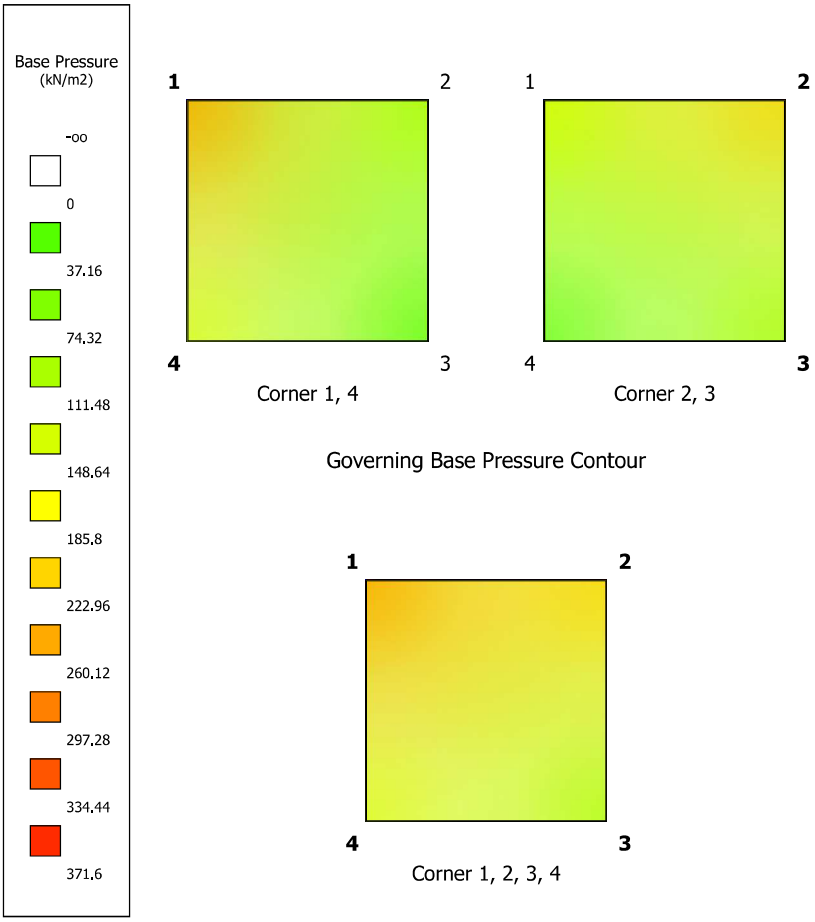
Gross Soil Bearing Capacity = 371.60 kN/m2

Soil Weight On Top Of Footing = 22.535 kN

Pressures at 4 Corners

Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

Governing Base Pressure Contour for Each Corner



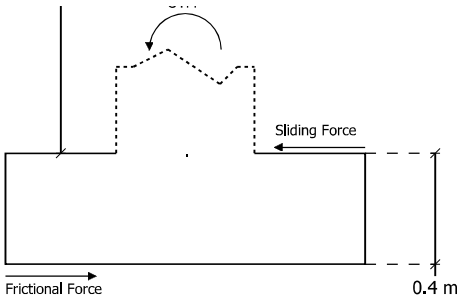
Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
45	**227.5106**	110.4005	2.0771	119.1873	0.000
22	136.3590	**191.3802**	82.9462	27.9250	0.000
22	136.3590	191.3802	**82.9462**	27.9250	0.000
45	227.5106	110.4005	2.0771	**119.1873**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
45	**227.5106**	110.4005	2.0771	119.1873
22	136.3590	**191.3802**	82.9462	27.9250
22	136.3590	191.3802	**82.9462**	27.9250
45	227.5106	110.4005	2.0771	**119.1873**

Stability Check



-	Factor of safety against sliding			Factor of safety against overturning			
Load Case No.	Along X-Direction	Along Z-Direction	Resultant	Required FOS	About X-Direction	About Z-Direction	Required FOS
10	7.609	3.635	4.638	1.500	6.366	20.785	1.500
11	7.609	3.635	4.638	1.500	6.366	20.785	1.500
12	7.716	3.839	4.861	1.500	6.698	21.177	1.500
13	7.689	3.787	4.804	1.500	6.613	21.079	1.500
14	7.509	3.529	4.516	1.500	7.242	20.562	1.500
15	12.502	3.554	4.835	1.500	6.223	17.617	1.500
16	14.182	3.715	5.082	1.500	6.414	20.322	1.500
17	7.127	3.763	4.706	1.500	8.001	16.945	1.500
18	191.567	3.974	5.619	1.500	6.813	6.525	1.500
19	7.356	4.082	5.048	1.500	11.778	15.267	1.500
20	25.754	3.996	5.585	1.500	7.019	8.818	1.500
21	8.480	3.942	5.055	1.500	9.137	23.260	1.500
22	19.887	3.542	4.932	1.500	6.112	11.957	1.500
23	6.877	3.602	4.513	1.500	8.230	15.656	1.500
39	5.395	3.844	4.427	1.500	6.639	7.248	1.500
40	7.310	3.822	4.790	1.500	5.686	17.302	1.500
41	4.459	4.252	4.352	1.500	7.293	4.555	1.500
42	7.735	4.202	5.222	1.500	5.367	15.930	1.500
45	4.858	3.710	4.169	1.500	6.403	5.881	1.500
46	7.119	3.682	4.625	1.500	5.257	16.115	1.500

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : 41
Governing Disturbing Force : -14.984kN
Governing Restoring Force : 66.818kN
Minimum Sliding Ratio for the Critical Load Case : 4.459
Critical Load Case for Overturning about X-Direction : 46
Governing Overturning Moment : -24.097kNm
Governing Resisting Moment : 126.688kNm
Minimum Overturning Ratio for the Critical Load Case : 5.257

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction : 14
Governing Disturbing Force : -26.504kN
Governing Restoring Force : 93.519kN

Minimum Sliding Ratio for the Critical Load Case : 3.529

Critical Load Case for Overturning about Z-Direction : 41

Governing Overturning Moment : 19.068kNm

Governing Resisting Moment : 86.862kNm

Minimum Overturning Ratio for the Critical Load Case : 4.555

Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

Critical Load Case for Sliding along Resultant Direction : 45

Governing Disturbing Force : 32.902kN

Governing Restoring Force : 137.180kN

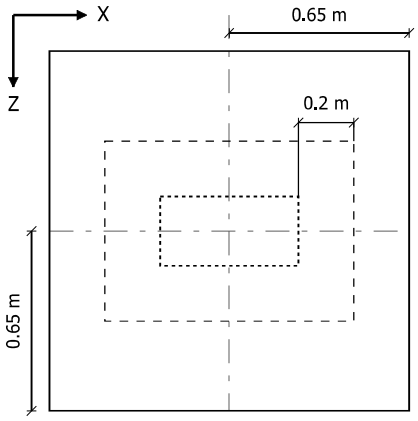
Minimum Sliding Ratio for the Critical Load Case : 4.169

Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimnesion Task Pane)

Shear Calculation

Punching Shear Check



Plan

Total Footing Depth, D = 0.400m

Calculated Effective Depth, d_{eff} = D - C_{cover} - 0.5 * d_b = 0.344m

For rectangular column, β_c = B_{col} / D_{col} = 2.000

Effective depth, d_{eff}, increased until $0.75XV_c \geq$ Punching Shear Force

Punching Shear Force, Vu = 177.303kN, Load Case # 25

From ACI Cl.11.11.2, b_o for column= $2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ = 2.876m

Equation 11-31, V_{c1} = $0.17 \times \left(1 + \frac{2}{\beta}\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 1944.021kN

Equation 11-32, V_{c2} = $0.083 \times \left(\frac{\alpha_s \times d}{b_o} + 2\right) \times \lambda \times \left(\sqrt{f_c'} \times b_o \times d_{eff}\right)$ 3297.265kN

Equation 11-33, V_{c3} = $0.33 \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 1944.021kN

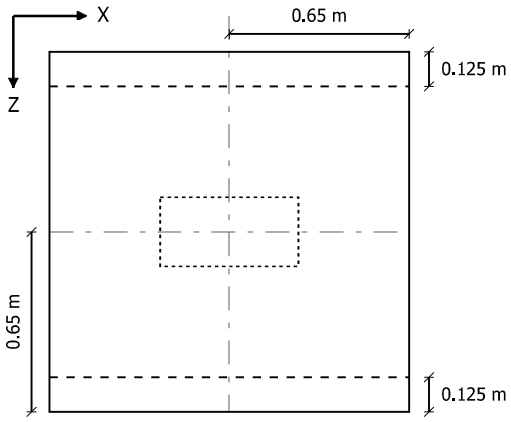
Punching shear strength, V_c = 0.75 X minimum of (V_{c1}, V_{c2}, V_{c3}) = 1458.016kN

0.75 X V_c > V_u hence, OK

One-Way Shear Check

One-Way Shear Along X

(Shear Plane Parallel to Global X Axis)



Plan

From ACI Cl.11.2.1.1, V_c = $2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 439.365kN

Distance along X to design for shear, D_x = $0.5 \times (W \pm D_{col}) - d_{eff} + O_{xd}$ = 0.181m

Check that 0.75 X V_c > V_{ux} where V_{ux} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the X axis.

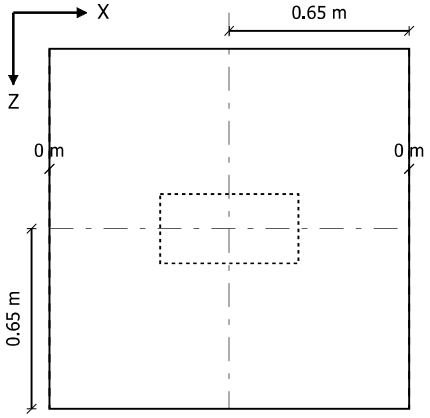
From above calculations, 0.75 X V_c = 329.524 kN

Critical load case for V_{ux} is # 25 $V_{ux} = V_{ux}|_{x=D_x}$ = 50.044 kN

0.75 X V_c > V_{ux} hence, OK

One-Way Shear Along Z

(Shear Plane Parallel to Global Z Axis)



Plan

From ACI Cl.11.2.1.1, V_c =

$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 439.365 kN

Distance along X to design for shear, D_z = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{xd}$ = 0.056 m

Check that **0.75 X V_c > V_{uz}** where V_{uz} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the Z axis.

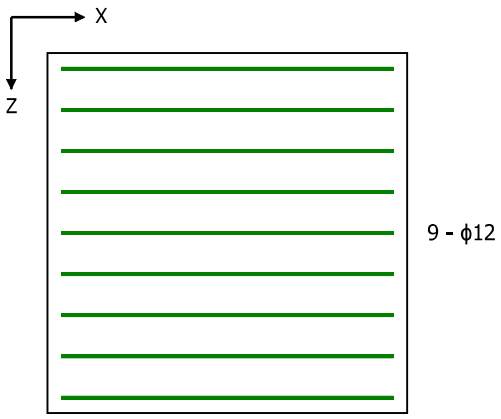
From above calculations, 0.75 X V_c = 329.524 kN

Critical load case for V_{uz} is # 49 $V_{uz} = V_{uz}|_{z=D_z}$ = 15.934 kN

0.75 X V_c > V_{uz} hence, OK

Flexure About Z-Axis

Design For Bottom Reinforcement Parallel to X Axis



Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 49

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.344	m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796	
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317	
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488	
From ACI Cl. 7.12.2, ρ_{min} =		0.00172	
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	14.118	

Calculate reinforcement ratio ρ for critical load case

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	0.400	m
Ultimate moment,	$M_u\big\vert_{z=D_z}$ =	21.291	kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	23.657	kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00037	
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00032	
Since	ρ= ρ_{min}	ρ_{min} Governs	
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	896.155	mm2

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = = 125.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars = =0.410 m

$\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}} =$

Available development length for bars, D_L = $0.5 \times (L - D_{col}) \cdot - C_{cover} =$ 0.350 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}} =$ 9

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1017.869 mm2

d_{eff} = D - C_{cover} - 0.5 X (dia. of one bar) = 0.344 m

Reinforcement ratio, ρ = $\frac{A_{s_total}}{(d_{eff} \times W)} =$ 0.00228

From ACI Cl.7.6.1, minimum req'd clear distance between bars

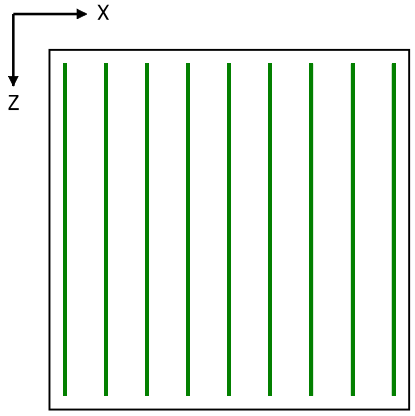
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 125.000mm

Provided Steel Area / Required Steel Area = 1.136

Check to see if width is sufficient to accomodate bars

Flexure About X-Axis

Design For Bottom Reinforcement Parallel to Z Axis



9 - ϕ12

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 25

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} = 0.332 m

Factor β_1 from ACI Cl.10.2.7.3 = 0.796

From ACI Cl. 10.3.2, ρ_{bal} = $0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$ 0.03317

From ACI Cl. 10.3.3, ρ_{max} = $0.75 \times \rho_{bal} =$ 0.02488

From ACI Cl.7.12.2, ρ_{min} = 0.00176

From Ref. 1, Eq. 3.8.4a, constant m = $\frac{F_y}{(0.85 \times F_c')} =$ 14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_z = $0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ = 0.525 m

Ultimate moment, $M_u|_{x=D_x}$ = 36.381 kNm

Nominal moment capacity, M_n = $\frac{M_u}{\Phi}$ = 40.424 kNm

(Based on effective depth) Required ρ = $\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ = 0.00067

(Based on gross depth) ρ x d_{eff} / Depth = 0.00056

Since $\rho = \rho_{min}$ ρ_{min} Governs

Area of Steel Required, A_s = $\rho \times W \times d_{eff}$ = 914.436 mm2

Selected Bar Size = #12

Minimum spacing allowed (S_{min}) = 125.000mm

Selected spacing (S) = 148.500mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 145mm o.c.

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ = =0.410 m

Available development length for bars, D_L = $0.5 \times (L - D_{col}) - C_{cover}$ = 0.475 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 9

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1017.869 mm2

d_{eff} = D - C_{cover} - 1.5 X (dia. of one bar) = 0.332 m

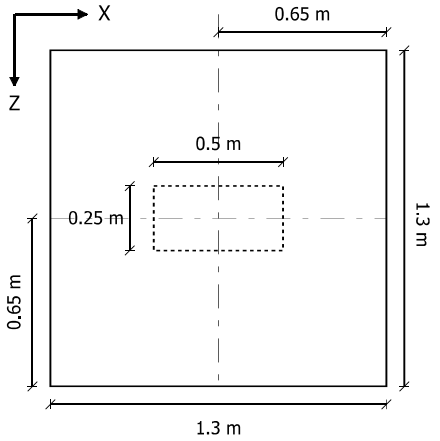
Reinforcement ratio, ρ = $\frac{A_{s_total}}{(d_{eff} \times W)}$ = 0.00236

From ACI Cl.7.6.1, minimum req'd clear distance between bars

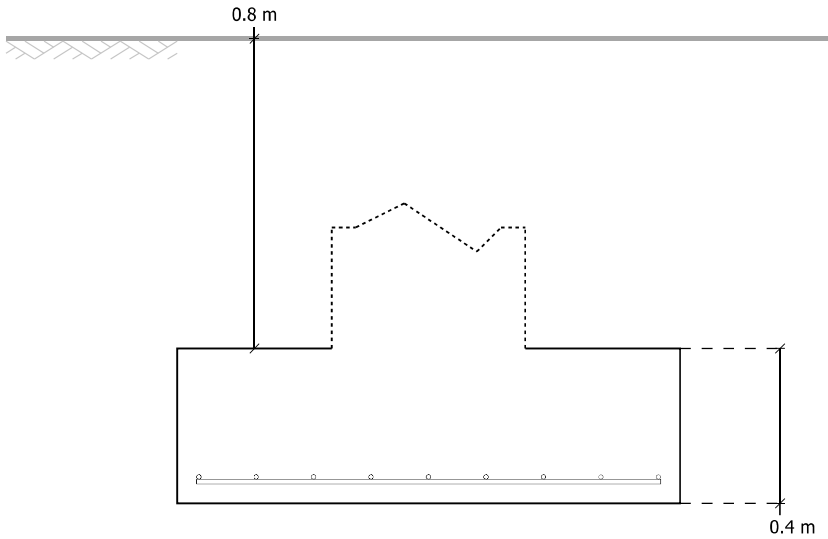
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 125.000mm

Provided Steel Area / Required Steel Area = 1.113

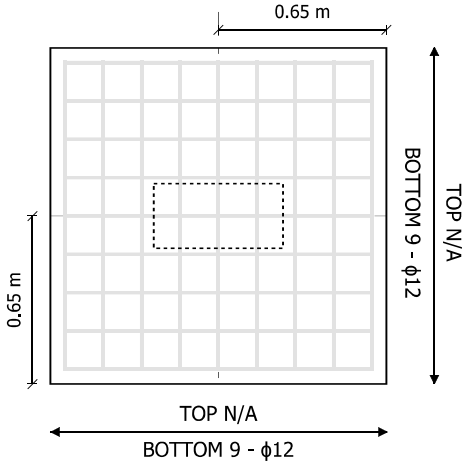
Check to see if width is sufficient to accomodate bars



Plan



Elevation



Plan

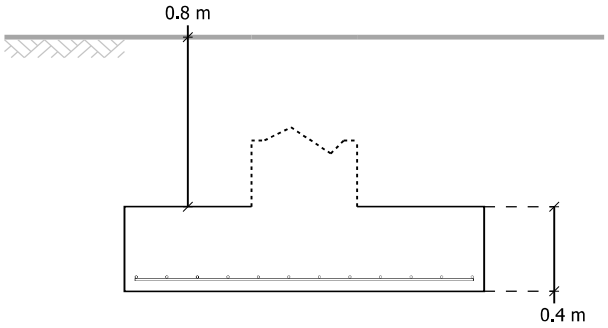
	Dimensions (m)			Reinforcement			
	L	B	D	AstX(T)	AstZ(T)	AstX(B)	AstZ(B)
Foundation	1.3 m	1.3 m	0.4 m	N/A	N/A	9 - ϕ12	9 - ϕ12
	l	b	h	Main Reinforcement		Trans Reinforcement	
Pedestal	N/A	N/A	N/A	N/A		N/A	

Print Calculation Sheet

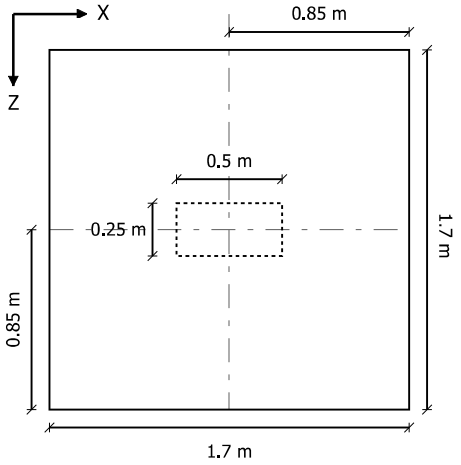
FOUNDATION F2 ALONG GRID-1 & B

Footing No.	Group ID	Foundation Geometry		
-	-	Length	Width	Thickness
26	1	1.700m	1.700m	0.400m

Footing No.	Footing Reinforcement				Pedestal Reinforcement	
-	Bottom Reinforcement(M_z)	Bottom Reinforcement(M_x)	Top Reinforcement(M_z)	Top Reinforcement(M_x)	Main Steel	Trans Steel
26	12 - ϕ12	12 - ϕ12	N/A	N/A	N/A	N/A



Elevation



Plan

Input Values

Footing Geomtery

Design Type : Set Dimension
Footing Thickness (Ft) : 400.000mm
Footing Length - X (Fl) : 1700.000mm
Footing Width - Z (Fw) : 1700.000mm
Eccentricity along X (Oxd) : 0.000mm
Eccentricity along Z (Ozd) : 0.000mm

Column Dimensions

Column Shape : Rectangular
Column Length - X (D_{col}) : 0.500m

Column Width - Z (B_{col}) : 0.250m

Pedestal

Include Pedestal? No
Pedestal Shape : N/A
Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete : 24.000kN/m3
Strength of Concrete : 35.000N/mm2
Yield Strength of Steel : 420.000N/mm2
Minimum Bar Size : #12
Maximum Bar Size : #12
Top Footing Minimum Bar Size : #12
Top Footing Maximum Bar Size : #12
Pedestal Minimum Bar Size : #16
Pedestal Maximum Bar Size : #16
Minimum Bar Spacing : 125.000mm
Maximum Bar Spacing : 150.000mm
Pedestal Clear Cover (P, CL) : 50.000mm
Bottom Footing Clear Cover (F, CL) : 50.000mm

Soil Properties

Soil Type : Cohesionless Soil
Unit Weight : 18.000kN/m3
Soil Bearing Capacity : 350.000kN/m2
Soil Bearing Capacity Type: Net Bearing Capacity
Soil Surcharge : 0.000kN/m2
Depth of Soil above Footing : 800.000mm
Type of Depth : Fixed Top
Undrained Shear Strength : 0.000kN/m2
Bearing Capacity Input Method: Fixed Bearing Capacity

Sliding and Overturning

Coefficient of Friction : 0.500
Factor of Safety Against Sliding : 1.500
Factor of Safety Against Overturning : 1.500

Global Settings

Top Reinforcement Option : Calculate only when foundation is subjected to uplift forces
Concrete Design Option : Gross Pressure
Top Reinforcement Factor : 1.000

Design Calculations

Footing Size

Initial Length (L_o) = 1.700m

Initial Width (W_o) = 1.700m

Load Combinations

Load Combination/s- Service Stress Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
10	SLS: (DL)	1.00	1.00	1.00
11	SLS: (DL+T)	1.00	1.00	1.00
12	SLS: (D+LR)	1.00	1.00	1.00
13	SLS: (D+0.75LR)	1.00	1.00	1.00
14	SLS: (D+0.7EZ)	1.00	1.00	1.00
15	SLS: (D+0.7EX)	1.00	1.00	1.00
16	SLS: (D+0.75WX1+0.75LR)	1.00	1.00	1.00
17	SLS: (D+0.75WZ1+0.75LR)	1.00	1.00	1.00
18	SLS: (0.6D+WX1)	1.00	1.00	1.00
19	SLS: (0.6D+WZ1)	1.00	1.00	1.00
20	SLS: (0.6D+0.7EX)	1.00	1.00	1.00
21	SLS: (0.6D+0.7EZ)	1.00	1.00	1.00
22	SLS: (D+WX1)	1.00	1.00	1.00
23	SLS: (D+WZ1)	1.00	1.00	1.00
39	SLS: (D+0.75WX2+0.75LR)	1.00	1.00	1.00
40	SLS: (D+0.75WZ2+0.75LR)	1.00	1.00	1.00
41	SLS: (0.6D+WX2)	1.00	1.00	1.00
42	SLS: (0.6D+WZ2)	1.00	1.00	1.00
45	SLS: (D+WX2)	1.00	1.00	1.00
46	SLS: (D+WZ2)	1.00	1.00	1.00

Load Combination/s- Strength Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
24	ULS: (1.4 DL)	1.00	1.00	1.00
25	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
26	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
27	ULS: (1.2D+0.80WX1+1.6LR)	1.00	1.00	1.00
28	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
29	ULS: (1.2D+1.6WX1+0.5LR)	1.00	1.00	1.00
30	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
31	ULS: (1.2D+EX)	1.00	1.00	1.00
32	ULS: (1.2D+EZ)	1.00	1.00	1.00
33	ULS: (0.90D+1.6WX1)	1.00	1.00	1.00
34	ULS: (0.90D+1.6WZ1)	1.00	1.00	1.00
35	ULS: (0.90D+EZ)	1.00	1.00	1.00
36	ULS: (0.90D+EX)	1.00	1.00	1.00
47	ULS: (1.2D+0.80WX2+1.6LR)	1.00	1.00	1.00
48	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
49	ULS: (1.2D+1.6WX2+0.5LR)	1.00	1.00	1.00
50	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
53	ULS: (0.90D+1.6WX2)	1.00	1.00	1.00
54	ULS: (0.90D+1.6WZ2)	1.00	1.00	1.00

Applied Loads

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
10	220.441	-30.243	-9.789	-11.326	7.847
11	220.441	-30.243	-9.789	-11.326	7.847
12	230.370	-30.168	-10.704	-12.558	7.804
13	227.888	-30.187	-10.475	-12.250	7.815
14	217.593	-30.238	-2.297	3.046	7.836
15	222.356	-22.315	-9.907	-11.511	-3.817
16	229.045	-22.656	-21.142	-20.482	-2.086
17	224.387	-30.691	9.186	13.517	8.287
18	133.808	-8.105	-20.096	-17.772	-8.493
19	127.598	-18.818	20.342	27.561	5.337

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
20	134.180	-10.218	-5.991	-6.980	-6.955
21	129.416	-18.141	1.619	7.577	4.697
22	221.985	-20.203	-24.011	-22.303	-5.355
23	215.775	-30.915	16.426	23.030	8.476
39	225.953	-37.578	-21.075	-20.506	17.586
40	230.889	-30.700	-27.648	-36.114	8.309
41	129.686	-28.000	-20.007	-17.803	17.735
42	136.266	-18.830	-28.770	-38.615	5.366
45	217.862	-40.098	-23.923	-22.334	20.874
46	224.443	-30.927	-32.686	-43.145	8.505

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
24	308.618	-42.340	-13.705	-15.857	10.986
25	313.582	-42.303	-14.162	-16.473	10.965
26	313.582	-42.303	-14.162	-16.473	10.965
27	281.649	-28.139	-24.588	-24.343	-1.213
28	262.027	-37.329	29.741	40.763	10.402
29	271.963	-20.189	-34.960	-31.770	-11.727
30	262.027	-37.329	29.741	40.763	10.402
31	267.265	-24.966	-11.915	-13.855	-7.246
32	260.460	-36.284	-1.044	6.941	9.401
33	200.867	-11.154	-31.566	-27.756	-14.060
34	190.930	-28.294	33.135	44.777	8.069
35	194.328	-27.211	1.893	10.339	7.047
36	201.133	-15.893	-8.978	-10.457	-9.600
47	278.352	-44.055	-24.517	-24.368	19.770
48	275.896	-37.349	-48.839	-65.118	10.448
49	265.367	-52.021	-34.818	-31.819	30.239
50	275.896	-37.349	-48.839	-65.118	10.448
53	194.271	-42.986	-31.424	-27.805	27.906
54	204.800	-28.314	-45.445	-61.104	8.115

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o = 2.890 m^2$

Min. area required from bearing pressure, $A_{min} = P / q_{max} = 1.518 m^2$

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) = 1.700 m Governing Load Case : # 10

Width (W_2) = 1.700 m Governing Load Case : # 10

Depth (D_2) = 0.400 m Governing Load Case : # 49

Depth is governed by Ultimate Load Case

(Service check is performed with footing thickness requirements from concrete check)

Area (A_2) = 2.890 m^2

Final Soil Height = 0.800 m

Footing Self Weight = 27.743 kN

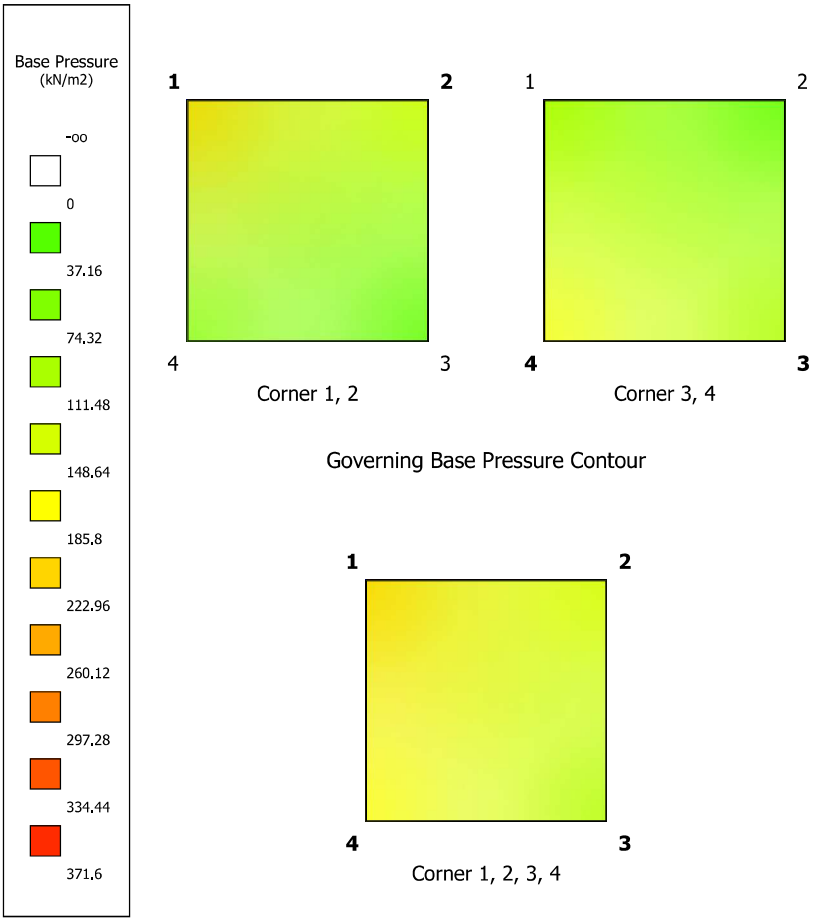
Gross Soil Bearing Capacity = 371.60 kN/m2

Soil Weight On Top Of Footing = 39.815 kN

Pressures at 4 Corners

Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

Governing Base Pressure Contour for Each Corner



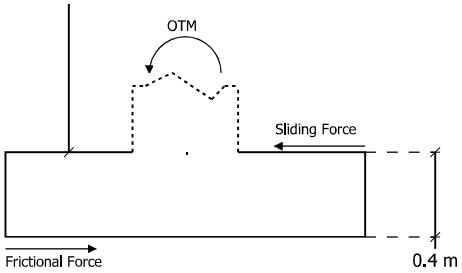
Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
46	**195.1929**	144.2024	6.8836	57.8740	0.000
46	195.1929	**144.2024**	6.8836	57.8740	0.000
23	87.3419	36.4347	**108.7357**	159.6429	0.000
23	87.3419	36.4347	108.7357	**159.6429**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
46	**195.1929**	144.2024	6.8836	57.8740
46	195.1929	**144.2024**	6.8836	57.8740
23	87.3419	36.4347	**108.7357**	159.6429
23	87.3419	36.4347	108.7357	**159.6429**

Stability Check



-	Factor of safety against sliding			Factor of safety against overturning			
Load Case No.	Along X-Direction	Along Z-Direction	Resultant	Required FOS	About X-Direction	About Z-Direction	Required FOS
10	4.761	14.710	6.406	1.500	16.136	12.274	1.500
11	4.761	14.710	6.406	1.500	16.136	12.274	1.500
12	4.938	13.917	6.581	1.500	15.106	12.744	1.500
13	4.894	14.102	6.538	1.500	15.345	12.626	1.500
14	4.715	62.072	6.649	1.500	114.462	12.161	1.500
15	6.496	14.632	8.396	1.500	16.000	48.230	1.500
16	6.546	7.015	6.768	1.500	8.751	36.138	1.500
17	4.756	15.890	6.444	1.500	14.501	12.068	1.500
18	12.422	5.010	6.571	1.500	6.676	32.593	1.500
19	5.185	4.797	4.980	1.500	4.679	12.895	1.500
20	9.872	16.837	12.043	1.500	18.410	59.782	1.500
21	5.429	60.840	7.647	1.500	20.497	14.007	1.500
22	7.166	6.029	6.525	1.500	7.749	90.272	1.500
23	4.582	8.624	5.723	1.500	8.175	11.555	1.500
39	3.905	6.963	4.817	1.500	8.662	7.649	1.500
40	4.861	5.397	5.108	1.500	5.402	12.321	1.500
41	3.522	4.929	4.053	1.500	6.541	5.794	1.500
42	5.412	3.542	4.192	1.500	3.479	13.432	1.500
45	3.559	5.966	4.322	1.500	7.640	6.572	1.500
46	4.721	4.467	4.588	1.500	4.435	11.889	1.500

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : 41
Governing Disturbing Force : -28.000kN
Governing Restoring Force : 98.622kN
Minimum Sliding Ratio for the Critical Load Case : 3.522
Critical Load Case for Overturning about X-Direction : 42
Governing Overturning Moment : -50.123kNm
Governing Resisting Moment : 174.395kNm
Minimum Overturning Ratio for the Critical Load Case : 3.479

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction : 42
Governing Disturbing Force : -28.770kN
Governing Restoring Force : 101.912kN

Minimum Sliding Ratio for the Critical Load Case : 3.542

Critical Load Case for Overturning about Z-Direction : 41

Governing Overturning Moment : 28.935kNm

Governing Resisting Moment : 167.654kNm

Minimum Overturning Ratio for the Critical Load Case : 5.794

Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

Critical Load Case for Sliding along Resultant Direction : 41

Governing Disturbing Force : 34.414kN

Governing Restoring Force : 139.472kN

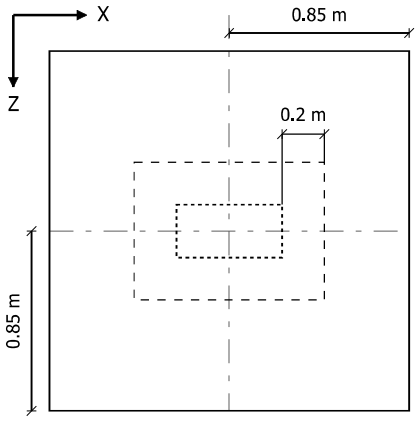
Minimum Sliding Ratio for the Critical Load Case : 4.053

Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimnesion Task Pane)

Shear Calculation

Punching Shear Check



Plan

Total Footing Depth, D = 0.400m

Calculated Effective Depth, d_{eff} = D - C_{cover} - 0.5 * d_b = 0.344m

For rectangular column, β_c = B_{col} / D_{col} = 2.000

Effective depth, d_{eff}, increased until $0.75XV_c \geq$ Punching Shear Force

Punching Shear Force, Vu = 315.022kN, Load Case # 25

From ACI Cl.11.11.2, b_o for column= $2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ = 2.876m

Equation 11-31, V_{c1} = $0.17 \times \left(1 + \frac{2}{\beta}\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 1944.021kN

Equation 11-32, V_{c2} = $0.083 \times \left(\frac{\alpha_s \times d}{b_o} + 2\right) \times \lambda \times \left(\sqrt{f_c'} \times b_o \times d_{eff}\right)$ 3297.265kN

Equation 11-33, V_{c3} = $0.33 \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 1944.021kN

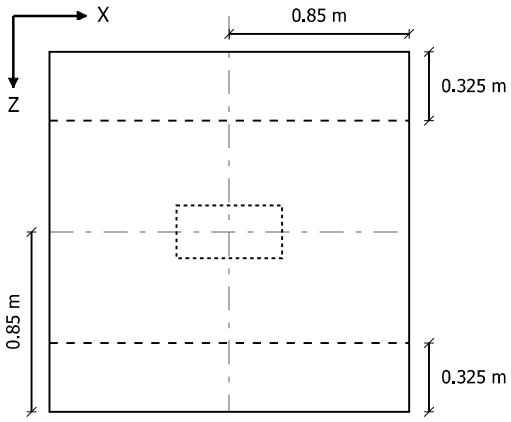
Punching shear strength, V_c = 0.75 X minimum of (V_{c1}, V_{c2}, V_{c3}) = 1458.016kN

0.75 X V_c > V_u hence, OK

One-Way Shear Check

One-Way Shear Along X

(Shear Plane Parallel to Global X Axis)



Plan

From ACI Cl.11.2.1.1, V_c = $2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 574.554kN

Distance along X to design for shear, D_x = $0.5 \times (W \pm D_{col}) - d_{eff} + O_{xd}$ = 0.381m

Check that 0.75 X V_c > V_{ux} where V_{ux} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the X axis.

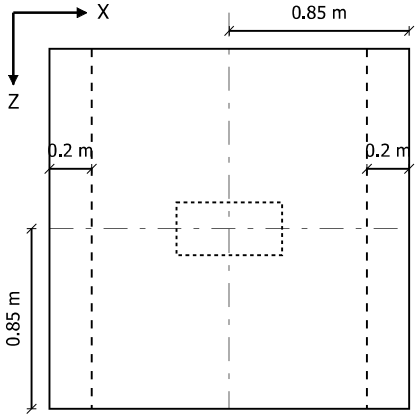
From above calculations, 0.75 X V_c = 430.916 kN

Critical load case for V_{ux} is # 48 $V_{ux} = V_{ux}|_{x=D_x}$ = 128.929 kN

0.75 X V_c > V_{ux} hence, OK

One-Way Shear Along Z

(Shear Plane Parallel to Global Z Axis)



Plan

From ACI Cl.11.2.1.1, V_c =

$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 574.554 kN

Distance along X to design for shear, D_z = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{xd}$ = 0.256 m

Check that **0.75 X V_c > V_{uz}** where V_{uz} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the Z axis.

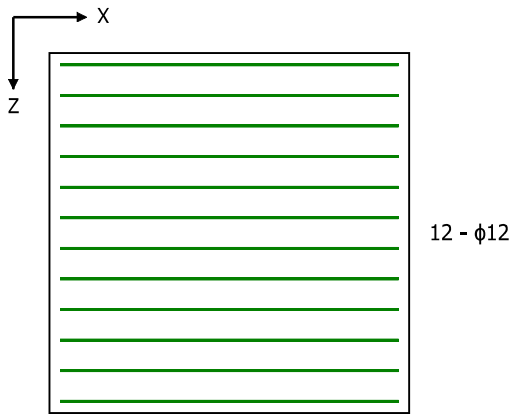
From above calculations, 0.75 X V_c = 430.916 kN

Critical load case for V_{uz} is # 49 $V_{uz} = V_{uz}|_{z=D_z}$ = 73.180 kN

0.75 X V_c > V_{uz} hence, OK

Flexure About Z-Axis

Design For Bottom Reinforcement Parallel to X Axis



Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 49

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.344 m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488
From ACI Cl. 7.12.2, ρ_{min} =		0.00165
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	0.600 m
Ultimate moment,	$M_u\big\vert_{z=D_z}$ =	49.834 kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	55.372 kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00066
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00057
Since	ρ= ρ_{min}	ρ_{min} Governs
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	1124.583 mm2

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = = 125.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars = =0.410 m

$\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}} =$

Available development length for bars, D_L = $0.5 \times (L - D_{col}) - C_{cover} =$ 0.550 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, $N_{bar} = \frac{A_s}{A_{bar}} =$ 12

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1357.159 mm2

d_{eff} = D - C_{cover} - 0.5 X (dia. of one bar) = 0.344 m

Reinforcement ratio, $\rho = \frac{A_{s_total}}{(d_{eff} \times W)} =$ 0.00232

From ACI Cl.7.6.1, minimum req'd clear distance between bars

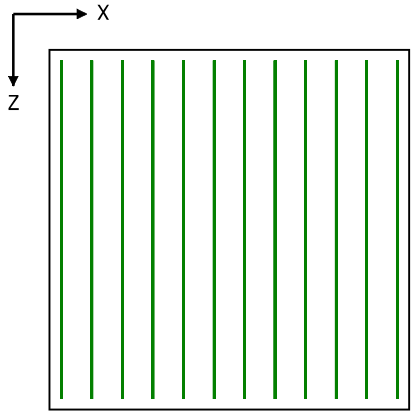
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 125.000mm

Provided Steel Area / Required Steel Area = 1.207

Check to see if width is sufficient to accomodate bars

Flexure About X-Axis

Design For Bottom Reinforcement Parallel to Z Axis



12 - ϕ12

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 48

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} = 0.332 m

Factor β_1 from ACI Cl.10.2.7.3 = 0.796

From ACI Cl. 10.3.2, $\rho_{bal} = 0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$ 0.03317

From ACI Cl. 10.3.3, $\rho_{max} = 0.75 \times \rho_{bal} =$ 0.02488

From ACI Cl.7.12.2, ρ_{min} = 0.00171

From Ref. 1, Eq. 3.8.4a, constant m = $\frac{F_y}{(0.85 \times F_c')} =$ 14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_z = $0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ = 0.725 m

Ultimate moment, $M_u|_{x=D_x}$ = 86.142 kNm

Nominal moment capacity, M_n = $\frac{M_u}{\Phi}$ = 95.714 kNm

(Based on effective depth) Required ρ = $\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ = 0.00123

(Based on gross depth) ρ x d_{eff} / Depth = 0.00102

Since $\rho = \rho_{min}$ ρ_{min} Governs

Area of Steel Required, A_s = $\rho \times W \times d_{eff}$ = 1160.523 mm2

Selected Bar Size = #12

Minimum spacing allowed (S_{min}) = 125.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ = =0.410 m

Available development length for bars, D_L = $0.5 \times (L - D_{col}) - C_{cover}$ = 0.675 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 12

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1357.159 mm2

d_{eff} = D - C_{cover} - 1.5 X (dia. of one bar) = 0.332 m

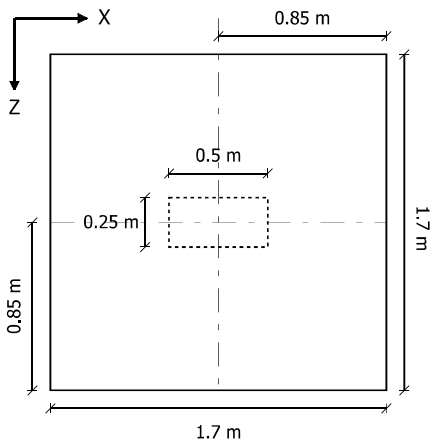
Reinforcement ratio, ρ = $\frac{A_{s_total}}{(d_{eff} \times W)}$ = 0.00240

From ACI Cl.7.6.1, minimum req'd clear distance between bars

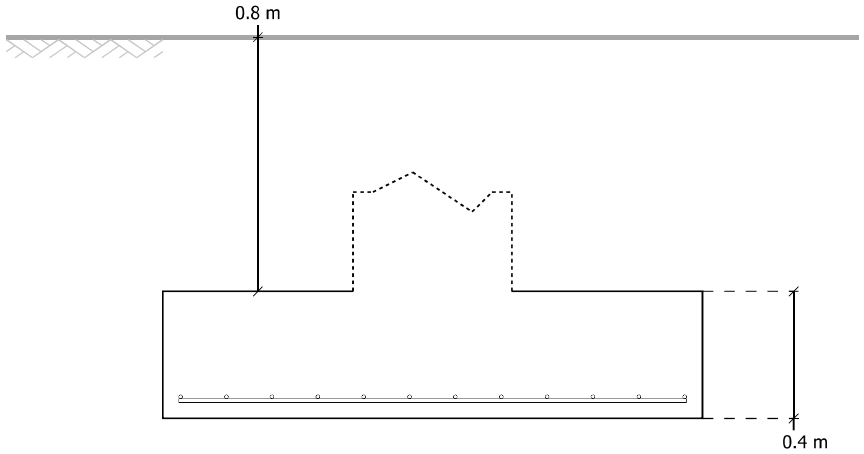
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 125.000mm

Provided Steel Area / Required Steel Area = 1.169

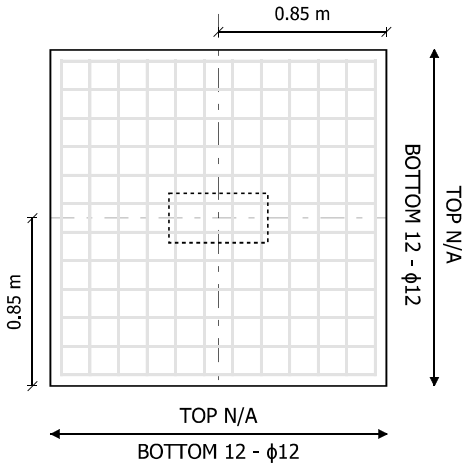
Check to see if width is sufficient to accomodate bars



Plan



Elevation



Plan

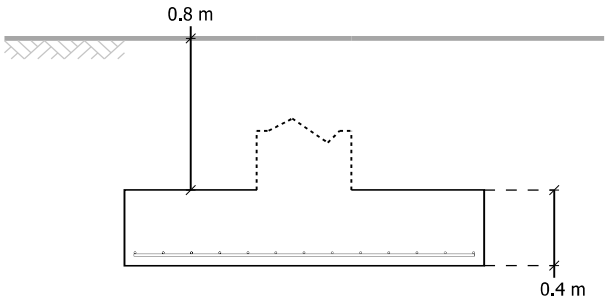
	Dimensions (m)			Reinforcement			
	L	B	D	AstX(T)	AstZ(T)	AstX(B)	AstZ(B)
Foundation	1.7 m	1.7 m	0.4 m	N/A	N/A	12 - ϕ12	12 - ϕ12
	l	b	h	Main Reinforcement		Trans Reinforcement	
Pedestal	N/A	N/A	N/A	N/A		N/A	

Print Calculation Sheet

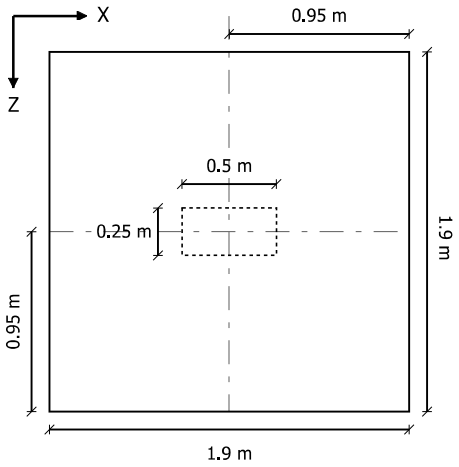
FOUNDATION F3 ALONG GRID-2 & A

Footing No.	Group ID	Foundation Geometry		
-	-	Length	Width	Thickness
35	1	1.900m	1.900m	0.400m

Footing No.	Footing Reinforcement				Pedestal Reinforcement	
-	Bottom Reinforcement(M_z)	Bottom Reinforcement(M_x)	Top Reinforcement(M_z)	Top Reinforcement(M_x)	Main Steel	Trans Steel
35	13 - ϕ12	13 - ϕ12	N/A	N/A	N/A	N/A



Elevation



Plan

Input Values

Footing Geomtery

Design Type : Set Dimension
Footing Thickness (Ft) : 400.000mm
Footing Length - X (Fl) : 1900.000mm
Footing Width - Z (Fw) : 1900.000mm
Eccentricity along X (Oxd) : 0.000mm
Eccentricity along Z (Ozd) : 0.000mm

Column Dimensions

Column Shape : Rectangular
Column Length - X (D_{col}) : 0.500m

Column Width - Z (B_{col}) : 0.250m

Pedestal

Include Pedestal? No
Pedestal Shape : N/A
Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete : 24.000kN/m3
Strength of Concrete : 35.000N/mm2
Yield Strength of Steel : 420.000N/mm2
Minimum Bar Size : #12
Maximum Bar Size : #12
Top Footing Minimum Bar Size : #12
Top Footing Maximum Bar Size : #12
Pedestal Minimum Bar Size : #16
Pedestal Maximum Bar Size : #16
Minimum Bar Spacing : 125.000mm
Maximum Bar Spacing : 150.000mm
Pedestal Clear Cover (P, CL) : 50.000mm
Bottom Footing Clear Cover (F, CL) : 50.000mm

Soil Properties

Soil Type : Cohesionless Soil
Unit Weight : 18.000kN/m3
Soil Bearing Capacity : 350.000kN/m2
Soil Bearing Capacity Type: Net Bearing Capacity
Soil Surcharge : 0.000kN/m2
Depth of Soil above Footing : 800.000mm
Type of Depth : Fixed Top
Undrained Shear Strength : 0.000kN/m2
Bearing Capacity Input Method: Fixed Bearing Capacity

Sliding and Overturning

Coefficient of Friction : 0.500
Factor of Safety Against Sliding : 1.500
Factor of Safety Against Overturning : 1.500

Global Settings

Top Reinforcement Option : Calculate only when foundation is subjected to uplift forces
Concrete Design Option : Gross Pressure
Top Reinforcement Factor : 1.000

Design Calculations

Footing Size

Initial Length (L_o) = 1.900m

Initial Width (W_o) = 1.900m

Load Combinations

Load Combination/s- Service Stress Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
10	SLS: (DL)	1.00	1.00	1.00
11	SLS: (DL+T)	1.00	1.00	1.00
12	SLS: (D+LR)	1.00	1.00	1.00
13	SLS: (D+0.75LR)	1.00	1.00	1.00
14	SLS: (D+0.7EZ)	1.00	1.00	1.00
15	SLS: (D+0.7EX)	1.00	1.00	1.00
16	SLS: (D+0.75WX1+0.75LR)	1.00	1.00	1.00
17	SLS: (D+0.75WZ1+0.75LR)	1.00	1.00	1.00
18	SLS: (0.6D+WX1)	1.00	1.00	1.00
19	SLS: (0.6D+WZ1)	1.00	1.00	1.00
20	SLS: (0.6D+0.7EX)	1.00	1.00	1.00
21	SLS: (0.6D+0.7EZ)	1.00	1.00	1.00
22	SLS: (D+WX1)	1.00	1.00	1.00
23	SLS: (D+WZ1)	1.00	1.00	1.00
39	SLS: (D+0.75WX2+0.75LR)	1.00	1.00	1.00
40	SLS: (D+0.75WZ2+0.75LR)	1.00	1.00	1.00
41	SLS: (0.6D+WX2)	1.00	1.00	1.00
42	SLS: (0.6D+WZ2)	1.00	1.00	1.00
45	SLS: (D+WX2)	1.00	1.00	1.00
46	SLS: (D+WZ2)	1.00	1.00	1.00

Load Combination/s- Strength Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
24	ULS: (1.4 DL)	1.00	1.00	1.00
25	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
26	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
27	ULS: (1.2D+0.80WX1+1.6LR)	1.00	1.00	1.00
28	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
29	ULS: (1.2D+1.6WX1+0.5LR)	1.00	1.00	1.00
30	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
31	ULS: (1.2D+EX)	1.00	1.00	1.00
32	ULS: (1.2D+EZ)	1.00	1.00	1.00
33	ULS: (0.90D+1.6WX1)	1.00	1.00	1.00
34	ULS: (0.90D+1.6WZ1)	1.00	1.00	1.00
35	ULS: (0.90D+EZ)	1.00	1.00	1.00
36	ULS: (0.90D+EX)	1.00	1.00	1.00
47	ULS: (1.2D+0.80WX2+1.6LR)	1.00	1.00	1.00
48	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
49	ULS: (1.2D+1.6WX2+0.5LR)	1.00	1.00	1.00
50	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
53	ULS: (0.90D+1.6WX2)	1.00	1.00	1.00
54	ULS: (0.90D+1.6WZ2)	1.00	1.00	1.00

Applied Loads

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
10	222.584	0.373	21.784	9.068	1.818
11	222.584	0.373	21.784	9.068	1.818
12	232.140	0.434	21.366	8.887	2.324
13	229.751	0.419	21.471	8.933	2.198
14	221.780	0.306	25.595	13.264	1.826
15	222.582	5.821	21.766	9.052	-15.851
16	229.753	24.092	21.273	8.808	-50.000
17	229.015	-21.500	26.879	14.092	49.074
18	133.553	31.788	12.807	5.275	-68.506
19	132.569	-29.002	20.282	12.321	63.593

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
20	133.548	5.672	13.053	5.424	-16.578
21	132.746	0.157	16.881	9.637	1.098
22	222.587	31.937	21.520	8.902	-67.779
23	221.602	-28.852	28.995	15.948	64.320
39	229.752	-17.292	21.282	8.825	41.583
40	230.482	-21.286	16.109	3.811	49.037
41	133.551	-23.391	12.819	5.298	53.605
42	134.524	-28.717	5.921	-1.388	63.543
45	222.584	-23.242	21.532	8.925	54.332
46	223.558	-28.568	14.634	2.240	64.271

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
24	311.618	0.522	30.497	12.696	2.546
25	316.396	0.553	30.289	12.605	2.799
26	316.396	0.553	30.289	12.605	2.799
27	282.393	25.797	25.262	10.459	-52.687
28	270.309	-46.282	37.470	21.799	102.438
29	271.883	50.980	25.510	10.525	-108.921
30	270.309	-46.282	37.470	21.799	102.438
31	267.098	8.230	26.115	10.858	-23.059
32	265.952	0.352	31.585	16.876	2.193
33	200.330	50.838	19.184	7.896	-109.719
34	198.755	-46.425	31.144	19.169	101.639
35	199.177	0.240	25.050	14.155	1.647
36	200.322	8.118	19.580	8.138	-23.605
47	282.391	-18.346	25.272	10.478	45.002
48	273.437	-45.827	14.493	-0.134	102.358
49	271.880	-37.306	25.530	10.562	86.457
50	273.437	-45.827	14.493	-0.134	102.358
53	200.326	-37.448	19.203	7.932	85.658
54	201.884	-45.969	8.166	-2.764	101.560

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o = 3.610m^2$

Min. area required from bearing pressure, $A_{min} = P / q_{max} = 1.703m^2$

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) = 1.900 m Governing Load Case : # 10

Width (W_2) = 1.900 m Governing Load Case : # 10

Depth (D_2) = 0.400 m Governing Load Case : # 29

Depth is governed by Ultimate Load Case

(Service check is performed with footing thickness requirements from concrete check)

Area (A_2) = 3.610 m^2

Final Soil Height = 0.800 m

Footing Self Weight = 34.655 kN

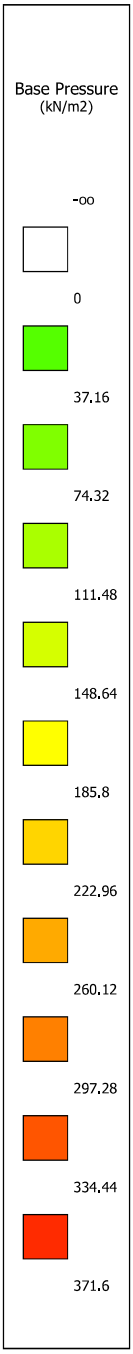
Gross Soil Bearing Capacity = 371.60 kN/m2

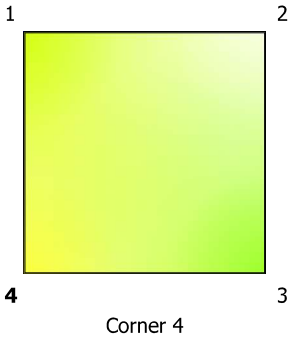
Soil Weight On Top Of Footing = 50.182 kN

Pressures at 4 Corners

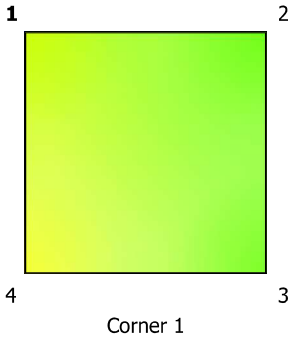
Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

Governing Base Pressure Contour for Each Corner

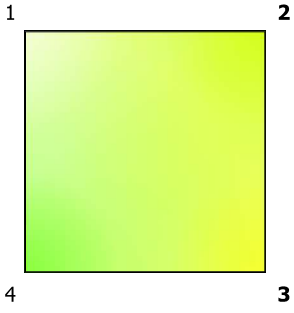




Corner 4

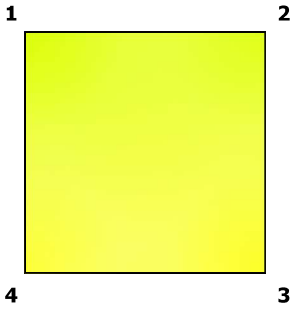


Corner 1



Corner 2, 3

Governing Base Pressure Contour



Corner 1, 2, 3, 4

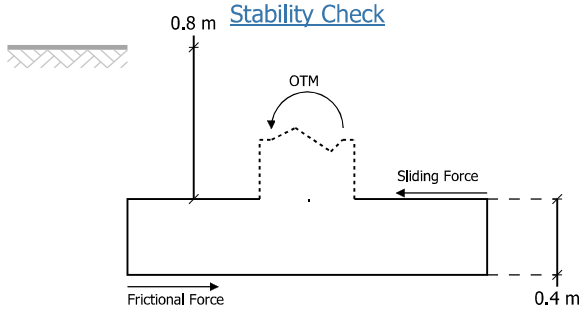
Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
46	**144.5664**	12.1295	26.2894	158.7264	0.000
22	-0.6251	**140.3079**	170.9428	30.0099	0.000
22	-0.6251	140.3079	**170.9428**	30.0099	0.000
23	127.1510	-5.5719	42.6216	**175.3445**	0.012

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
46	**144.5664**	12.1295	26.2894	158.7264
22	0.0000	**140.3080**	170.9430	30.0097
22	0.0000	140.3080	**170.9430**	30.0097
23	127.1574	0.0000	42.6223	**175.3719**

Stability Check



-	Factor of safety against sliding			Factor of safety against overturning			
Load Case No.	Along X-Direction	Along Z-Direction	Resultant	Required FOS	About X-Direction	About Z-Direction	Required FOS
10	411.979	7.056	9.977	1.500	16.496	174.963	1.500
11	411.979	7.056	9.977	1.500	16.496	174.963	1.500
12	364.851	7.418	10.488	1.500	17.346	140.029	1.500
13	375.340	7.326	10.359	1.500	17.130	147.210	1.500
14	500.561	5.990	8.470	1.500	12.449	171.020	1.500
15	26.407	7.062	9.648	1.500	16.518	16.065	1.500
16	6.529	7.394	6.921	1.500	17.332	5.011	1.500
17	7.299	5.838	6.448	1.500	12.053	5.170	1.500
18	3.435	8.526	4.506	1.500	20.077	2.554	1.500
19	3.748	5.360	4.344	1.500	10.170	2.747	1.500
20	19.253	8.366	10.851	1.500	19.609	11.008	1.500
21	692.804	6.445	9.114	1.500	12.690	199.610	1.500
22	4.813	7.143	5.645	1.500	16.752	3.626	1.500
23	5.310	5.284	5.297	1.500	10.615	3.837	1.500
39	9.096	7.391	8.112	1.500	17.311	6.162	1.500
40	7.407	9.787	8.352	1.500	29.337	5.205	1.500
41	4.668	8.518	5.789	1.500	20.023	3.295	1.500
42	3.819	18.524	5.290	1.500	213.799	2.777	1.500
45	6.614	7.139	6.861	1.500	16.725	4.590	1.500
46	5.398	10.537	6.794	1.500	36.357	3.870	1.500

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : 18

Governing Disturbing Force : 31.788kN

Governing Restoring Force : 109.195kN

Minimum Sliding Ratio for the Critical Load Case : 3.435

Critical Load Case for Overturning about X-Direction : 19

Governing Overturning Moment : 20.433kNm

Governing Resisting Moment : 207.814kNm

Minimum Overturning Ratio for the Critical Load Case : 10.170

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction : 23

Governing Disturbing Force : 28.995kN

Governing Restoring Force : 153.220kN

Minimum Sliding Ratio for the Critical Load Case : 5.284

Critical Load Case for Overturning about Z-Direction : 18

Governing Overturning Moment : -81.221kNm

Governing Resisting Moment : 207.467kNm

Minimum Overturning Ratio for the Critical Load Case : 2.554

Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

Critical Load Case for Sliding along Resultant Direction : 19

Governing Disturbing Force : 35.390kN

Governing Restoring Force : 153.729kN

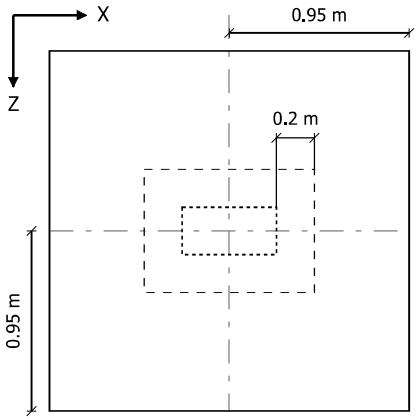
Minimum Sliding Ratio for the Critical Load Case : 4.344

Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimnesion Task Pane)

Shear Calculation

Punching Shear Check



Plan

Total Footing Depth, D = 0.400m

Calculated Effective Depth, d_{eff} = D - C_{cover} - 0.5 * d_b = 0.344m

For rectangular column, β_c = B_{col} / D_{col} = 2.000

Effective depth, d_{eff}, increased until 0.75XV_c $\geq$ Punching Shear Force

Punching Shear Force, Vu = 345.512kN, Load Case # 25

From ACI Cl.11.11.2, b_o for column= $2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ = 2.876m

Equation 11-31, V_{c1} = $0.17 \times \left(1 + \frac{2}{\beta}\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 1944.021kN

Equation 11-32, V_{c2} = $0.083 \times \left(\frac{\alpha_s \times d}{b_o} + 2\right) \times \lambda \times \left(\sqrt{f_c'} \times b_o \times d_{eff}\right)$ 3297.265kN

Equation 11-33, V_{c3} = $0.33 \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 1944.021kN

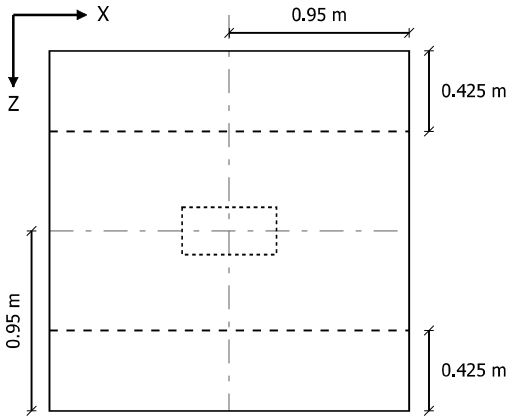
Punching shear strength, V_c = 0.75 X minimum of (V_{c1}, V_{c2}, V_{c3}) = 1458.016kN

0.75 X V_c > V_u hence, OK

One-Way Shear Check

One-Way Shear Along X

(Shear Plane Parallel to Global X Axis)



Plan

From ACI Cl.11.2.1.1, V_c = $2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 642.149kN

Distance along X to design for shear, D_x = $0.5 \times (W \pm D_{col}) - d_{eff} + O_{zd}$ = 0.481m

Check that 0.75 X V_c > V_{ux} where V_{ux} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the X axis.

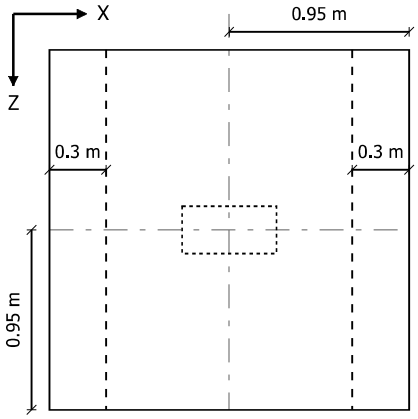
From above calculations, 0.75 X V_c = 481.612 kN

Critical load case for V_{ux} is # 25 $V_{ux} = V_{ux}|_{x=D_x}$ = 116.335 kN

0.75 X V_c > V_{ux} hence, OK

One-Way Shear Along Z

(Shear Plane Parallel to Global Z Axis)



Plan

From ACI Cl.11.2.1.1, V_c =

$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 642.149 kN

Distance along X to design for shear, D_z = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{xd}$ = 0.356 m

Check that **0.75 X V_c > V_{uz}** where V_{uz} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the Z axis.

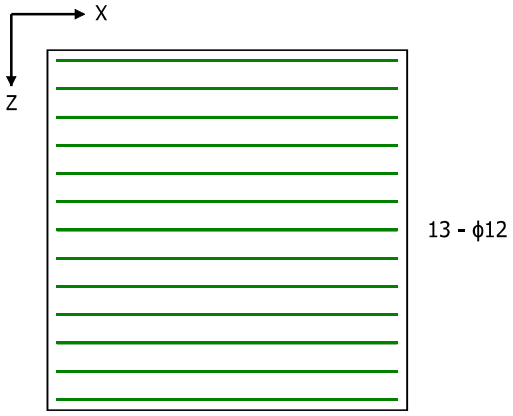
From above calculations, 0.75 X V_c = 481.612 kN

Critical load case for V_{uz} is # 29 $V_{uz} = V_{uz}|_{z=D_z} =$ 126.545 kN

0.75 X V_c > V_{uz} hence, OK

Flexure About Z-Axis

Design For Bottom Reinforcement Parallel to X Axis



Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 29

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.344 m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488
From ACI Cl. 7.12.2, ρ_{min} =		0.00161
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	1.200 m
Ultimate moment,	$M_u\|_{z=D_z}$ =	88.576 kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	98.418 kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00105
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00090
Since	ρ= ρ_{min}	ρ_{min} Governs
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	1224.076 mm2

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = = 125.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars =	$\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f'_c}}$ =	=0.410	m
Available development length for bars, D_L =	$0.5 \times (L - D_{col}) \cdot - C_{cover}$ =	0.650	m
Try bar size # 12	Area of one bar =	113.097	mm2
Number of bars required, N_{bar} =	$\frac{A_s}{A_{bar}}$ =	13	

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} =	N_{bar} X (Area of one bar) =	1470.255	mm2
d_{eff} =	D - C_{cover} - 0.5 X (dia. of one bar) =	0.344	m
Reinforcement ratio, ρ =	$\frac{A_{s_total}}{(d_{eff} \times W)}$ =	0.00225	

From ACI Cl.7.6.1, minimum req'd clear distance between bars

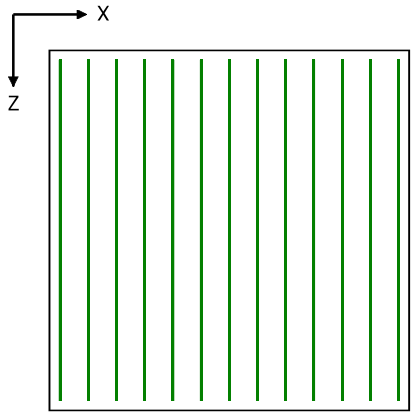
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 125.000mm

Provided Steel Area / Required Steel Area = 1.201

Check to see if width is sufficient to accomodate bars

Flexure About X-Axis

Design For Bottom Reinforcement Parallel to Z Axis



13 - ϕ12

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 25

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.332 m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f'_c \times \left(\frac{600}{600 + f_y}\right)$	0.03317
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488

From ACI Cl.7.12.2, ρ_{min} = 0.00167

From Ref. 1, Eq. 3.8.4a, constant m = $\frac{F_y}{\left(0.85 \times F_c'\right)}$ = 14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_z = $0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ = 1.075 m

Ultimate moment, $M_u\big|_{x=D_x}$ = 81.806 kNm

Nominal moment capacity, M_n = $\frac{M_u}{\Phi}$ = 90.896 kNm

(Based on effective depth) Required ρ = $\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ = 0.00104

(Based on gross depth) ρ x d_{eff} / Depth = 0.00086

Since ρ= ρ_{min} ρ_{min} Governs

Area of Steel Required, A_s = $\rho \times W \times d_{eff}$ = 1271.412 mm2

Selected Bar Size = #12

Minimum spacing allowed (S_{min}) = 125.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ = =0.410 m

Available development length for bars, D_L = $0.5 \times \left(L - D_{col}\right) - C_{cover}$ = 0.775 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 13

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1470.255 mm2

d_{eff} = D - C_{cover} - 1.5 X (dia. of one bar) = 0.332 m

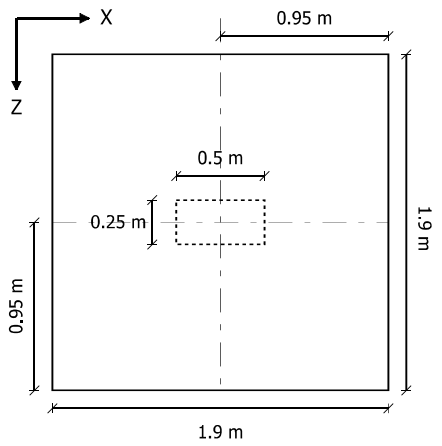
Reinforcement ratio, ρ = $\frac{A_{s_total}}{\left(d_{eff} \times W\right)}$ = 0.00233

From ACI Cl.7.6.1, minimum req'd clear distance between bars

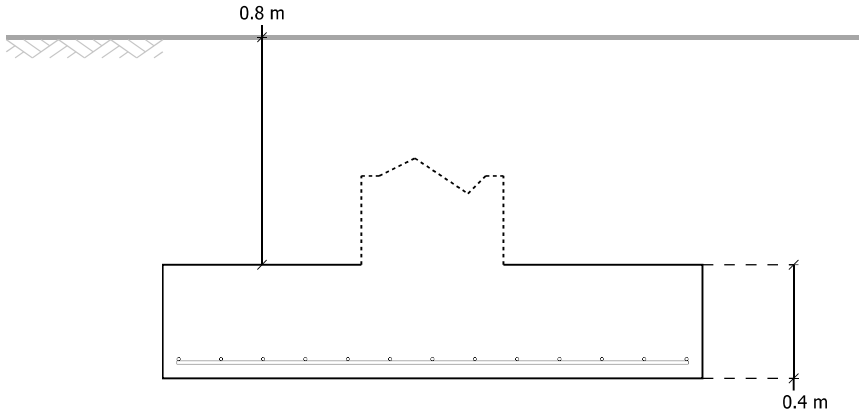
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 125.000mm

Provided Steel Area / Required Steel Area = 1.156

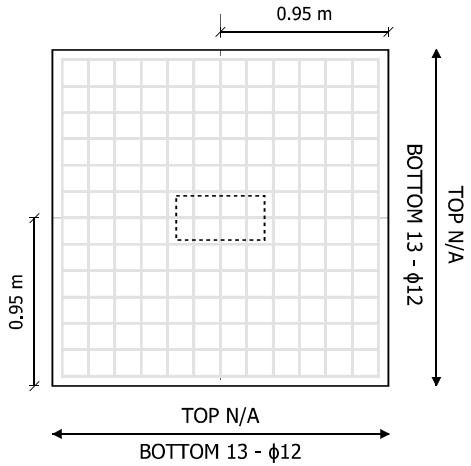
Check to see if width is sufficient to accomodate bars



Plan



Elevation



Plan

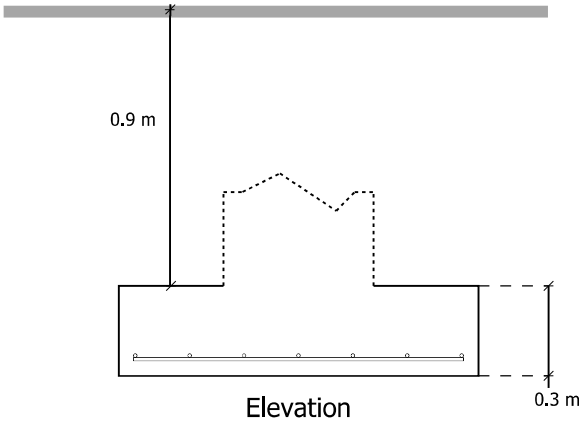
	Dimensions (m)			Reinforcement			
	L	B	D	AstX(T)	AstZ(T)	AstX(B)	AstZ(B)
Foundation	1.9 m	1.9 m	0.4 m	N/A	N/A	13 - ϕ12	13 - ϕ12
	l	b	h	Main Reinforcement		Trans Reinforcement	
Pedestal	N/A	N/A	N/A	N/A		N/A	

Print Calculation Sheet

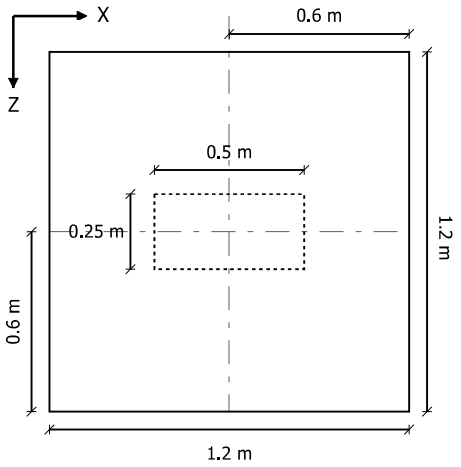
FOUNDATION F4 ALONG GRID-3 & A

Footing No.	Group ID	Foundation Geometry		
-	-	Length	Width	Thickness
47	1	1.200m	1.200m	0.300m

Footing No.	Footing Reinforcement				Pedestal Reinforcement	
-	Bottom Reinforcement(M_z)	Bottom Reinforcement(M_x)	Top Reinforcement(M_z)	Top Reinforcement(M_x)	Main Steel	Trans Steel
47	7 - ϕ12	7 - ϕ12	N/A	N/A	N/A	N/A



Elevation



Plan

Input Values

Footing Geomtery

Design Type : Set Dimension
Footing Thickness (Ft) : 300.000mm
Footing Length - X (Fl) : 1200.000mm
Footing Width - Z (Fw) : 1200.000mm
Eccentricity along X (Oxd) : 0.000mm
Eccentricity along Z (Ozd) : 0.000mm

Column Dimensions

Column Shape : Rectangular
Column Length - X (D_{col}) : 0.500m

Column Width - Z (B_{col}) : 0.250m

Pedestal

Include Pedestal? No
Pedestal Shape : N/A
Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete : 24.000kN/m3
Strength of Concrete : 35.000N/mm2
Yield Strength of Steel : 420.000N/mm2
Minimum Bar Size : #12
Maximum Bar Size : #12
Top Footing Minimum Bar Size : #12
Top Footing Maximum Bar Size : #12
Pedestal Minimum Bar Size : #16
Pedestal Maximum Bar Size : #16
Minimum Bar Spacing : 200.000mm
Maximum Bar Spacing : 200.000mm
Pedestal Clear Cover (P, CL) : 50.000mm
Bottom Footing Clear Cover (F, CL) : 50.000mm

Soil Properties

Soil Type : Cohesionless Soil
Unit Weight : 18.000kN/m3
Soil Bearing Capacity : 350.000kN/m2
Soil Bearing Capacity Type: Net Bearing Capacity
Soil Surcharge : 0.000kN/m2
Depth of Soil above Footing : 900.000mm
Type of Depth : Fixed Top
Undrained Shear Strength : 0.000kN/m2
Bearing Capacity Input Method: Fixed Bearing Capacity

Sliding and Overturning

Coefficient of Friction : 0.500
Factor of Safety Against Sliding : 1.500
Factor of Safety Against Overturning : 1.500

Global Settings

Top Reinforcement Option : Calculate only when foundation is subjected to uplift forces
Concrete Design Option : Gross Pressure
Top Reinforcement Factor : 1.000

Design Calculations

Footing Size

Initial Length (L_o) = 1.200m

Initial Width (W_o) = 1.200m

Load Combinations

Load Combination/s- Service Stress Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
10	SLS: (DL)	1.00	1.00	1.00
11	SLS: (DL+T)	1.00	1.00	1.00
12	SLS: (D+LR)	1.00	1.00	1.00
13	SLS: (D+0.75LR)	1.00	1.00	1.00
14	SLS: (D+0.7EZ)	1.00	1.00	1.00
15	SLS: (D+0.7EX)	1.00	1.00	1.00
16	SLS: (D+0.75WX1+0.75LR)	1.00	1.00	1.00
17	SLS: (D+0.75WZ1+0.75LR)	1.00	1.00	1.00
18	SLS: (0.6D+WX1)	1.00	1.00	1.00
19	SLS: (0.6D+WZ1)	1.00	1.00	1.00
20	SLS: (0.6D+0.7EX)	1.00	1.00	1.00
21	SLS: (0.6D+0.7EZ)	1.00	1.00	1.00
22	SLS: (D+WX1)	1.00	1.00	1.00
23	SLS: (D+WZ1)	1.00	1.00	1.00
39	SLS: (D+0.75WX2+0.75LR)	1.00	1.00	1.00
40	SLS: (D+0.75WZ2+0.75LR)	1.00	1.00	1.00
41	SLS: (0.6D+WX2)	1.00	1.00	1.00
42	SLS: (0.6D+WZ2)	1.00	1.00	1.00
45	SLS: (D+WX2)	1.00	1.00	1.00
46	SLS: (D+WZ2)	1.00	1.00	1.00

Load Combination/s- Strength Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
24	ULS: (1.4 DL)	1.00	1.00	1.00
25	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
26	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
27	ULS: (1.2D+0.80WX1+1.6LR)	1.00	1.00	1.00
28	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
29	ULS: (1.2D+1.6WX1+0.5LR)	1.00	1.00	1.00
30	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
31	ULS: (1.2D+EX)	1.00	1.00	1.00
32	ULS: (1.2D+EZ)	1.00	1.00	1.00
33	ULS: (0.90D+1.6WX1)	1.00	1.00	1.00
34	ULS: (0.90D+1.6WZ1)	1.00	1.00	1.00
35	ULS: (0.90D+EZ)	1.00	1.00	1.00
36	ULS: (0.90D+EX)	1.00	1.00	1.00
47	ULS: (1.2D+0.80WX2+1.6LR)	1.00	1.00	1.00
48	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
49	ULS: (1.2D+1.6WX2+0.5LR)	1.00	1.00	1.00
50	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
53	ULS: (0.90D+1.6WX2)	1.00	1.00	1.00
54	ULS: (0.90D+1.6WZ2)	1.00	1.00	1.00

Applied Loads

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
10	113.122	-12.725	-3.518	-0.255	0.986
11	113.122	-12.725	-3.518	-0.255	0.986
12	116.758	-12.822	-4.120	-0.463	0.967
13	115.849	-12.798	-3.969	-0.411	0.972
14	120.373	-12.805	-1.765	3.036	1.058
15	109.535	-7.094	-3.503	-0.260	-8.629
16	113.511	-7.540	-3.997	-0.321	-6.321
17	123.813	-14.166	-1.394	3.573	1.700
18	64.755	-0.624	-2.147	-0.033	-9.133
19	78.492	-9.459	1.323	5.158	1.562

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
20	64.287	-2.004	-2.096	-0.158	-9.023
21	75.124	-7.715	-0.357	3.138	0.664
22	110.004	-5.714	-3.555	-0.135	-8.739
23	123.741	-14.549	-0.084	5.056	1.956
39	118.550	-17.741	-3.975	-0.297	8.115
40	108.421	-13.925	-6.516	-4.393	1.492
41	71.474	-14.226	-2.118	-0.001	10.116
42	57.969	-9.139	-5.507	-5.462	1.285
45	116.723	-19.316	-3.525	-0.102	10.510
46	103.218	-14.228	-6.914	-5.564	1.679

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
24	158.370	-17.814	-4.925	-0.357	1.380
25	160.189	-17.863	-5.226	-0.461	1.371
26	160.189	-17.863	-5.226	-0.461	1.371
27	139.070	-9.817	-5.214	-0.543	-6.626
28	154.555	-18.237	0.972	8.088	2.726
29	132.575	-4.101	-4.581	-0.219	-14.385
30	154.555	-18.237	0.972	8.088	2.726
31	130.623	-7.226	-4.201	-0.313	-12.552
32	146.105	-15.385	-1.717	4.395	1.286
33	96.821	-0.235	-3.225	-0.038	-14.672
34	118.800	-14.371	2.328	8.268	2.439
35	112.168	-11.567	-0.661	4.472	0.991
36	96.686	-3.409	-3.145	-0.237	-12.848
47	144.445	-20.698	-5.190	-0.517	8.773
48	121.718	-17.724	-9.956	-8.904	2.283
49	143.326	-25.864	-4.534	-0.166	16.413
50	121.718	-17.724	-9.956	-8.904	2.283
53	107.571	-21.998	-3.177	0.014	16.126
54	85.963	-13.858	-8.600	-8.724	1.996

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o = 1.440 m^2$

Min. area required from bearing pressure, $A_{min} = P / q_{max} = 0.634 m^2$

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) = 1.200 m Governing Load Case : # 10

Width (W_2) = 1.200 m Governing Load Case : # 10

Depth (D_2) = 0.300 m Governing Load Case : # 49

Depth is governed by Ultimate Load Case

(Service check is performed with footing thickness requirements from concrete check)

Area (A_2) = 1.440 m^2

Final Soil Height = 0.900 m

Footing Self Weight = 10.368 kN

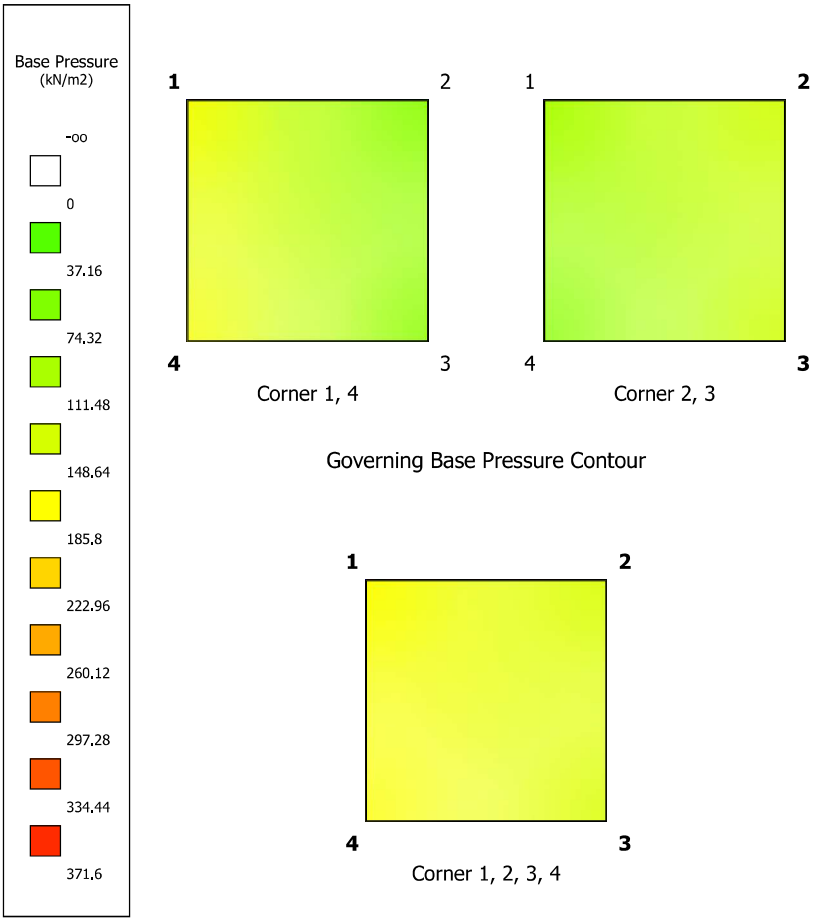
Gross Soil Bearing Capacity = 371.60 kN/m2

Soil Weight On Top Of Footing = 21.302 kN

Pressures at 4 Corners

Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

Governing Base Pressure Contour for Each Corner



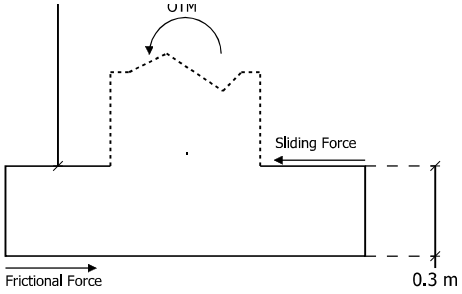
Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
45	**163.6924**	50.4637	42.4082	155.6369	0.000
22	78.1653	**126.9485**	118.6036	69.8203	0.000
22	78.1653	126.9485	**118.6036**	69.8203	0.000
45	163.6924	50.4637	42.4082	**155.6369**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
45	**163.6924**	50.4637	42.4082	155.6369
22	78.1653	**126.9485**	118.6036	69.8203
22	78.1653	126.9485	**118.6036**	69.8203
45	163.6924	50.4637	42.4082	**155.6369**

Stability Check



-	Factor of safety against sliding			Factor of safety against overturning			
Load Case No.	Along X-Direction	Along Z-Direction	Resultant	Required FOS	About X-Direction	About Z-Direction	Required FOS
10	5.689	20.578	7.755	1.500	67.003	18.088	1.500
11	5.689	20.578	7.755	1.500	67.003	18.088	1.500
12	5.788	18.014	7.793	1.500	52.963	18.499	1.500
13	5.764	18.582	7.785	1.500	55.834	18.397	1.500
14	5.937	43.081	8.317	1.500	36.759	18.619	1.500
15	9.952	20.152	12.620	1.500	65.309	13.033	1.500
16	9.628	18.163	12.030	1.500	57.900	21.458	1.500
17	5.488	55.768	7.724	1.500	29.864	15.681	1.500
18	77.254	22.452	30.491	1.500	86.727	6.467	1.500
19	5.823	41.636	8.156	1.500	12.062	15.025	1.500
20	23.937	22.888	23.395	1.500	74.300	6.836	1.500
21	6.921	149.426	9.777	1.500	21.444	21.513	1.500
22	12.397	19.928	14.887	1.500	71.496	12.101	1.500
23	5.341	921.477	7.553	1.500	18.715	14.753	1.500
39	4.234	18.897	5.843	1.500	61.146	6.707	1.500
40	5.030	10.749	6.443	1.500	13.386	14.825	1.500
41	3.625	24.351	5.071	1.500	98.750	4.303	1.500
42	4.904	8.139	5.941	1.500	7.688	13.358	1.500
45	3.841	21.048	5.344	1.500	77.541	5.461	1.500
46	4.740	9.755	6.029	1.500	10.715	13.608	1.500

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : 41
Governing Disturbing Force : -14.226kN
Governing Restoring Force : 51.572kN
Minimum Sliding Ratio for the Critical Load Case : 3.625
Critical Load Case for Overturning about X-Direction : 42
Governing Overturning Moment : -7.114kNm
Governing Resisting Moment : 54.693kNm
Minimum Overturning Ratio for the Critical Load Case : 7.688

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction : 42
Governing Disturbing Force : -5.507kN
Governing Restoring Force : 44.819kN

Minimum Sliding Ratio for the Critical Load Case : 8.139

Critical Load Case for Overturning about Z-Direction : 41

Governing Overturning Moment : 14.384kNm

Governing Resisting Moment : 61.885kNm

Minimum Overturning Ratio for the Critical Load Case : 4.303

Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

Critical Load Case for Sliding along Resultant Direction : 41

Governing Disturbing Force : 14.383kN

Governing Restoring Force : 72.934kN

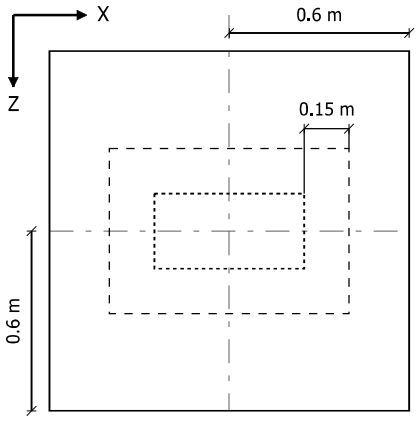
Minimum Sliding Ratio for the Critical Load Case : 5.071

Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimnesion Task Pane)

Shear Calculation

Punching Shear Check



Plan

Total Footing Depth, D = 0.300m

Calculated Effective Depth, d_{eff} = D - C_{cover} - 0.5 * d_b = 0.244m

For rectangular column, β_c = B_{col} / D_{col} = 2.000

Effective depth, d_{eff}, increased until 0.75XV_c $\geq$ Punching Shear Force

Punching Shear Force, Vu = 142.890kN, Load Case # 25

From ACI Cl.11.11.2, b_o for column= $2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ = 2.476m

Equation 11-31, V_{c1} = $0.17 \times \left(1 + \frac{2}{\beta}\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 1187.118kN

Equation 11-32, V_{c2} = $0.083 \times \left(\frac{\alpha_s \times d}{b_o} + 2\right) \times \lambda \times \left(\sqrt{f_c'} \times b_o \times d_{eff}\right)$ 1763.418kN

Equation 11-33, V_{c3} = $0.33 \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 1187.118kN

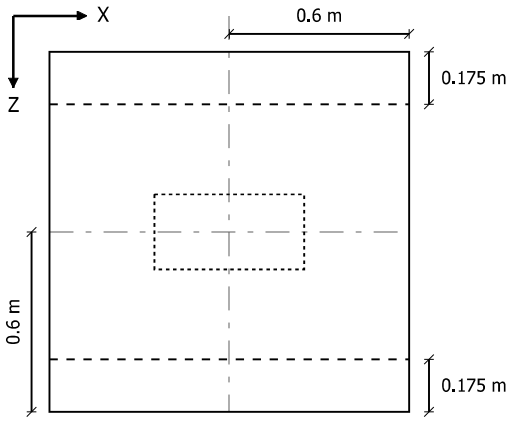
Punching shear strength, V_c = 0.75 X minimum of (V_{c1}, V_{c2}, V_{c3}) = 890.339kN

0.75 X V_c > V_u hence, OK

One-Way Shear Check

One-Way Shear Along X

(Shear Plane Parallel to Global X Axis)



Plan

From ACI Cl.11.2.1.1, V_c = $2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 287.670kN

Distance along X to design for shear, D_x = $0.5 \times (W \pm D_{col}) - d_{eff} + O_{xd}$ = 0.231m

Check that 0.75 X V_c > V_{ux} where V_{ux} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the X axis.

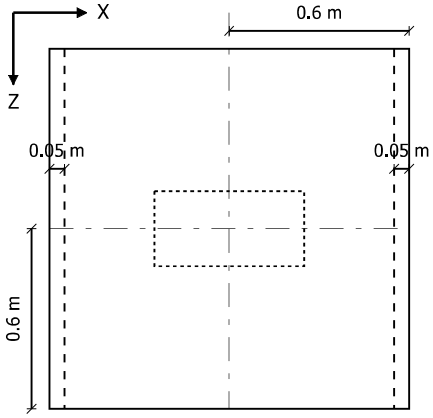
From above calculations, 0.75 X V_c = 215.753 kN

Critical load case for V_{ux} is # 30 $V_{ux} = V_{ux}|_{x=D_x}$ = 42.361 kN

0.75 X V_c > V_{ux} hence, OK

One-Way Shear Along Z

(Shear Plane Parallel to Global Z Axis)



Plan

From ACI Cl.11.2.1.1, V_c =

$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 287.670 kN

Distance along X to design for shear, D_z = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{xd}$ = 0.106 m

Check that **0.75 X V_c > V_{uz}** where V_{uz} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the Z axis.

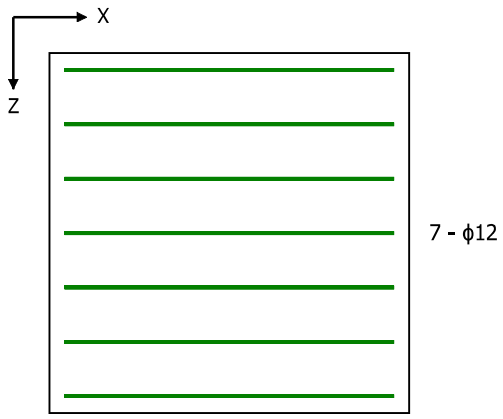
From above calculations, 0.75 X V_c = 215.753 kN

Critical load case for V_{uz} is # 49 $V_{uz} = V_{uz}|_{z=D_z}$ = 25.191 kN

0.75 X V_c > V_{uz} hence, OK

Flexure About Z-Axis

Design For Bottom Reinforcement Parallel to X Axis



Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 49

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.244	m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796	
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317	
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488	
From ACI Cl. 7.12.2, ρ_{min} =		0.00168	
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	14.118	

Calculate reinforcement ratio ρ for critical load case

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	0.350	m
Ultimate moment,	$M_u\big\vert_{z=D_z}$ =	13.900	kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	15.445	kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00052	
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00042	
Since	ρ= ρ_{min}	ρ_{min} Governs	
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	603.580	mm2

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = = 200.000mm

Selected spacing (S) = 200.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 200mm o.c.

Required development length for bars = =0.410 m

$\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ =

Available development length for bars, D_L = $0.5 \times (L - D_{col}) \cdot - C_{cover}$ = 0.300 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 7

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 791.676 mm2

d_{eff} = D - C_{cover} - 0.5 X (dia. of one bar) = 0.244 m

Reinforcement ratio, ρ = $\frac{A_{s_total}}{(d_{eff} \times W)}$ = 0.00270

From ACI Cl.7.6.1, minimum req'd clear distance between bars

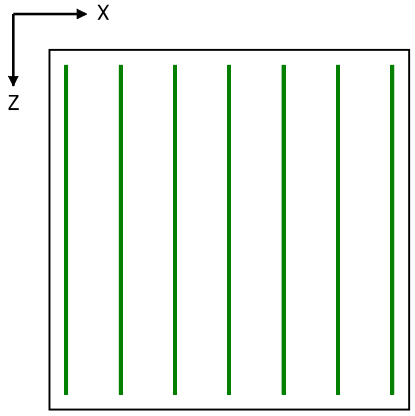
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 200.000mm

Provided Steel Area / Required Steel Area = 1.312

Check to see if width is sufficient to accomodate bars

Flexure About X-Axis

Design For Bottom Reinforcement Parallel to Z Axis



7 - ϕ12

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 30

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} = 0.232 m

Factor β_1 from ACI Cl.10.2.7.3 = 0.796

From ACI Cl. 10.3.2, ρ_{bal} = $0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$ 0.03317

From ACI Cl. 10.3.3, ρ_{max} = $0.75 \times \rho_{bal}$ = 0.02488

From ACI Cl.7.12.2, ρ_{min} = 0.00174

From Ref. 1, Eq. 3.8.4a, constant m = $\frac{F_y}{(0.85 \times F_c')}$ = 14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_z = $0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ = 0.725 m

Ultimate moment, $M_u|_{x=D_x}$ = 20.409 kNm

Nominal moment capacity, M_n = $\frac{M_u}{\Phi}$ = 22.677 kNm

(Based on effective depth) Required ρ = $\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ = 0.00084

(Based on gross depth) ρ x d_{eff} / Depth = 0.00065

Since $\rho = \rho_{min}$ ρ_{min} Governs

Area of Steel Required, A_s = $\rho \times W \times d_{eff}$ = 626.153 mm2

Selected Bar Size = #12

Minimum spacing allowed (S_{min}) = 200.000mm

Selected spacing (S) = 200.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 200mm o.c.

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ = =0.410 m

Available development length for bars, D_L = $0.5 \times (L - D_{col}) - C_{cover}$ = 0.425 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 7

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 791.676 mm2

d_{eff} = D - C_{cover} - 1.5 X (dia. of one bar) = 0.232 m

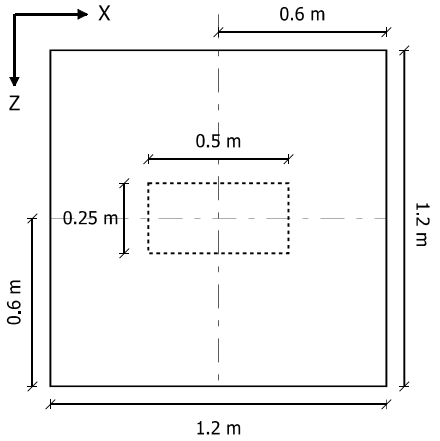
Reinforcement ratio, ρ = $\frac{A_{s_total}}{(d_{eff} \times W)}$ = 0.00284

From ACI Cl.7.6.1, minimum req'd clear distance between bars

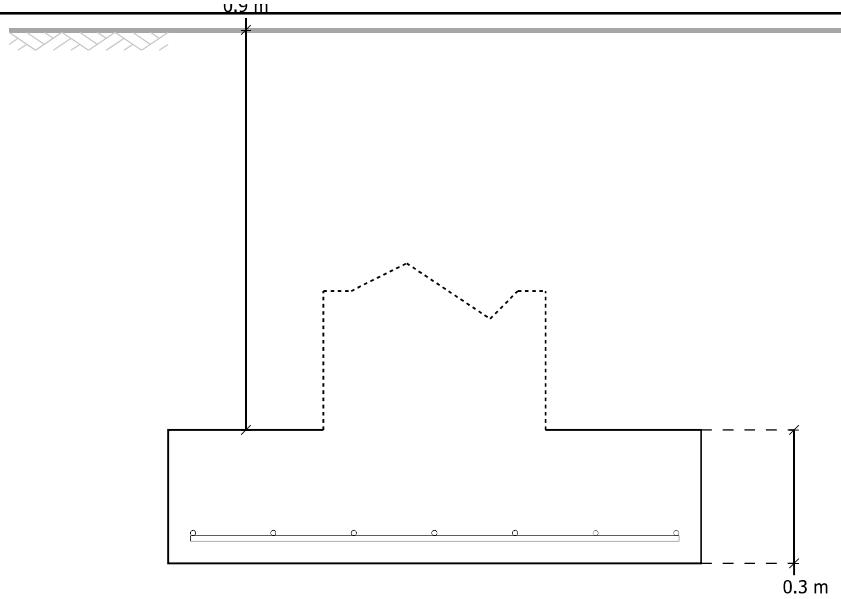
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 200.000mm

Provided Steel Area / Required Steel Area = 1.264

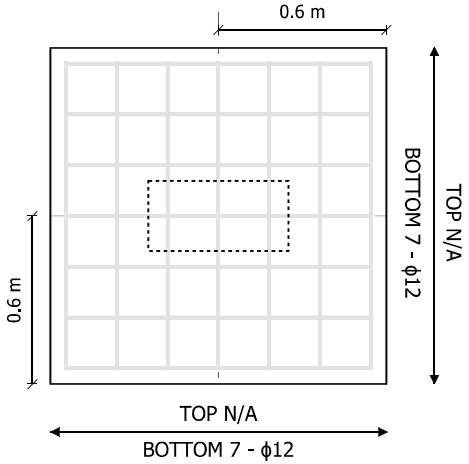
Check to see if width is sufficient to accomodate bars



Plan



Elevation



Plan

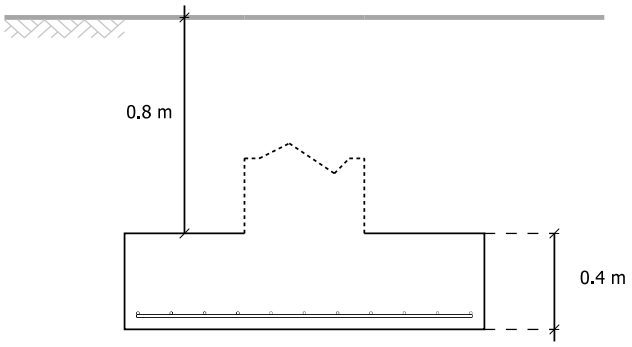
	Dimensions (m)			Reinforcement			
	L	B	D	AstX(T)	AstZ(T)	AstX(B)	AstZ(B)
Foundation	1.2 m	1.2 m	0.3 m	N/A	N/A	7 - ϕ12	7 - ϕ12
	l	b	h	Main Reinforcement		Trans Reinforcement	
Pedestal	N/A	N/A	N/A	N/A		N/A	

Print Calculation Sheet

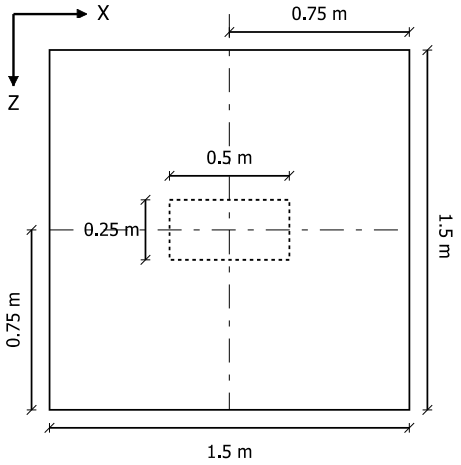
FOUNDATION-F5 ALONG GRID 2 & B

Footing No.	Group ID	Foundation Geometry		
-	-	Length	Width	Thickness
38	1	1.500m	1.500m	0.400m

Footing No.	Footing Reinforcement				Pedestal Reinforcement	
-	Bottom Reinforcement(M_z)	Bottom Reinforcement(M_x)	Top Reinforcement(M_z)	Top Reinforcement(M_x)	Main Steel	Trans Steel
38	11 - ϕ12	11 - ϕ12	N/A	N/A	N/A	N/A



Elevation



Plan

Input Values

Footing Geomtery

Design Type : Set Dimension
Footing Thickness (Ft) : 400.000mm
Footing Length - X (Fl) : 1500.000mm
Footing Width - Z (Fw) : 1500.000mm
Eccentricity along X (Oxd) : 0.000mm
Eccentricity along Z (Ozd) : 0.000mm

Column Dimensions

Column Shape : Rectangular
Column Length - X (D_{col}) : 0.500m

Column Width - Z (B_{col}) : 0.250m

Pedestal

Include Pedestal? No
Pedestal Shape : N/A
Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete : 24.000kN/m3
Strength of Concrete : 35.000N/mm2
Yield Strength of Steel : 420.000N/mm2
Minimum Bar Size : #12
Maximum Bar Size : #12
Top Footing Minimum Bar Size : #12
Top Footing Maximum Bar Size : #12
Pedestal Minimum Bar Size : #16
Pedestal Maximum Bar Size : #16
Minimum Bar Spacing : 150.000mm
Maximum Bar Spacing : 150.000mm
Pedestal Clear Cover (P, CL) : 50.000mm
Bottom Footing Clear Cover (F, CL) : 50.000mm

Soil Properties

Soil Type : Cohesionless Soil
Unit Weight : 18.000kN/m3
Soil Bearing Capacity : 350.000kN/m2
Soil Bearing Capacity Type: Net Bearing Capacity
Soil Surcharge : 0.000kN/m2
Depth of Soil above Footing : 900.000mm
Type of Depth : Fixed Top
Undrained Shear Strength : 0.000kN/m2
Bearing Capacity Input Method: Fixed Bearing Capacity

Sliding and Overturning

Coefficient of Friction : 0.500
Factor of Safety Against Sliding : 1.500
Factor of Safety Against Overturning : 1.500

Global Settings

Top Reinforcement Option : Calculate only when foundation is subjected to uplift forces
Concrete Design Option : Gross Pressure
Top Reinforcement Factor : 1.000

Design Calculations

Footing Size

Initial Length (L_o) = 1.500m

Initial Width (W_o) = 1.500m

Load Combinations

Load Combination/s- Service Stress Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
10	SLS: (DL)	1.00	1.00	1.00
11	SLS: (DL+T)	1.00	1.00	1.00
12	SLS: (D+LR)	1.00	1.00	1.00
13	SLS: (D+0.75LR)	1.00	1.00	1.00
14	SLS: (D+0.7EZ)	1.00	1.00	1.00
15	SLS: (D+0.7EX)	1.00	1.00	1.00
16	SLS: (D+0.75WX1+0.75LR)	1.00	1.00	1.00
17	SLS: (D+0.75WZ1+0.75LR)	1.00	1.00	1.00
18	SLS: (0.6D+WX1)	1.00	1.00	1.00
19	SLS: (0.6D+WZ1)	1.00	1.00	1.00
20	SLS: (0.6D+0.7EX)	1.00	1.00	1.00
21	SLS: (0.6D+0.7EZ)	1.00	1.00	1.00
22	SLS: (D+WX1)	1.00	1.00	1.00
23	SLS: (D+WZ1)	1.00	1.00	1.00
39	SLS: (D+0.75WX2+0.75LR)	1.00	1.00	1.00
40	SLS: (D+0.75WZ2+0.75LR)	1.00	1.00	1.00
41	SLS: (0.6D+WX2)	1.00	1.00	1.00
42	SLS: (0.6D+WZ2)	1.00	1.00	1.00
45	SLS: (D+WX2)	1.00	1.00	1.00
46	SLS: (D+WZ2)	1.00	1.00	1.00

Load Combination/s- Strength Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
24	ULS: (1.4 DL)	1.00	1.00	1.00
25	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
26	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
27	ULS: (1.2D+0.80WX1+1.6LR)	1.00	1.00	1.00
28	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
29	ULS: (1.2D+1.6WX1+0.5LR)	1.00	1.00	1.00
30	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
31	ULS: (1.2D+EX)	1.00	1.00	1.00
32	ULS: (1.2D+EZ)	1.00	1.00	1.00
33	ULS: (0.90D+1.6WX1)	1.00	1.00	1.00
34	ULS: (0.90D+1.6WZ1)	1.00	1.00	1.00
35	ULS: (0.90D+EZ)	1.00	1.00	1.00
36	ULS: (0.90D+EX)	1.00	1.00	1.00
47	ULS: (1.2D+0.80WX2+1.6LR)	1.00	1.00	1.00
48	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
49	ULS: (1.2D+1.6WX2+0.5LR)	1.00	1.00	1.00
50	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
53	ULS: (0.90D+1.6WX2)	1.00	1.00	1.00
54	ULS: (0.90D+1.6WZ2)	1.00	1.00	1.00

Applied Loads

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
10	284.788	-0.014	2.998	5.353	-0.354
11	284.788	-0.014	2.998	5.353	-0.354
12	312.241	-0.009	3.271	5.824	-0.378
13	305.378	-0.011	3.203	5.707	-0.372
14	284.282	-0.013	9.860	20.833	-0.358
15	284.798	7.040	2.903	5.139	-25.741
16	306.308	0.509	3.242	5.797	-2.811
17	304.673	0.134	11.604	24.720	-0.893
18	172.113	0.685	1.852	3.333	-3.464
19	169.933	0.184	13.000	28.564	-0.907

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
20	170.883	7.046	1.704	2.998	-25.599
21	170.366	-0.007	8.660	18.692	-0.216
22	286.028	0.679	3.051	5.474	-3.606
23	283.849	0.178	14.199	30.705	-1.049
39	306.301	-0.545	3.168	5.630	2.124
40	305.867	0.132	-5.224	-13.364	-0.885
41	172.104	-0.722	1.753	3.110	3.115
42	171.525	0.181	-9.437	-22.215	-0.896
45	286.019	-0.727	2.952	5.251	2.974
46	285.441	0.175	-8.238	-20.074	-1.038

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
24	398.704	-0.020	4.197	7.495	-0.496
25	412.430	-0.018	4.333	7.730	-0.508
26	412.430	-0.018	4.333	7.730	-0.508
27	386.662	0.545	4.077	7.274	-3.065
28	353.969	0.294	21.657	47.222	-1.549
29	357.456	1.094	3.819	6.853	-5.639
30	353.969	0.294	21.657	47.222	-1.549
31	341.760	10.061	3.462	6.118	-36.691
32	341.022	-0.016	13.400	28.538	-0.431
33	258.293	1.096	2.783	5.011	-5.521
34	254.806	0.295	20.621	45.381	-1.431
35	255.585	-0.011	12.501	26.932	-0.324
36	256.323	10.065	2.563	4.512	-36.585
47	386.655	-0.580	3.998	7.096	2.199
48	356.516	0.288	-14.243	-34.024	-1.530
49	357.441	-1.156	3.660	6.496	4.887
50	356.516	0.288	-14.243	-34.024	-1.530
53	258.278	-1.154	2.624	4.654	5.006
54	257.353	0.290	-15.278	-35.866	-1.412

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o = 2.250m^2$

Min. area required from bearing pressure, $A_{min} = P / q_{max} = 1.312m^2$

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) = 1.500 m Governing Load Case : # 10

Width (W_2) = 1.500 m Governing Load Case : # 10

Depth (D_2) = 0.400 m Governing Load Case : # 31

Depth is governed by Ultimate Load Case

(Service check is performed with footing thickness requirements from concrete check)

Area (A_2) = 2.250 m^2

Final Soil Height = 0.900 m

Footing Self Weight = 21.599 kN

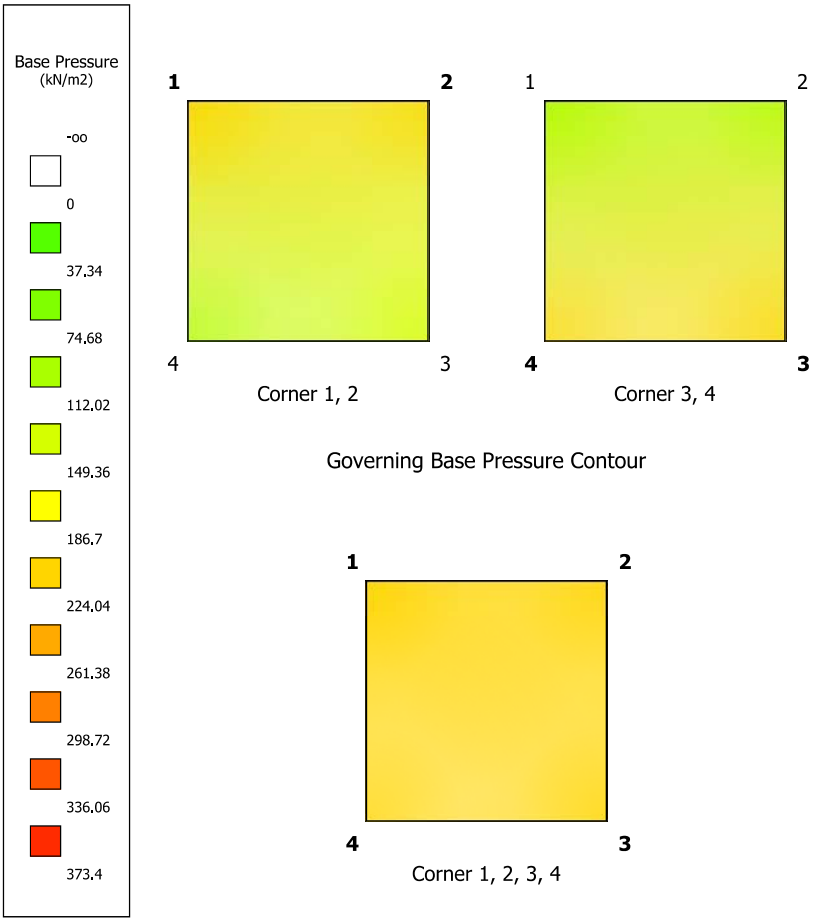
Gross Soil Bearing Capacity = 373.40 kN/m2

Soil Weight On Top Of Footing = 34.424 kN

Pressures at 4 Corners

Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

Governing Base Pressure Contour for Each Corner



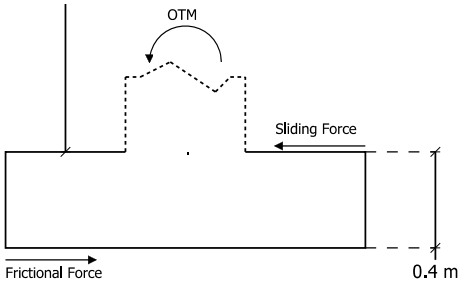
Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
46	**191.3380**	195.2765	112.1854	108.2469	0.000
46	191.3380	**195.2765**	112.1854	108.2469	0.000
23	84.3771	88.3614	**217.7312**	213.7469	0.000
23	84.3771	88.3614	217.7312	**213.7469**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
46	**191.3380**	195.2765	112.1854	108.2469
46	191.3380	**195.2765**	112.1854	108.2469
23	84.3771	88.3614	**217.7312**	213.7469
23	84.3771	88.3614	217.7312	**213.7469**

Stability Check



-	Factor of safety against sliding			Factor of safety against overturning			
Load Case No.	Along X-Direction	Along Z-Direction	Resultant	Required FOS	About X-Direction	About Z-Direction	Required FOS
10	11917.319	56.846	80.391	1.500	39.183	733.153	1.500
11	11917.319	56.846	80.391	1.500	39.183	733.153	1.500
12	19846.094	56.294	79.612	1.500	38.881	737.213	1.500
13	17155.653	56.423	79.794	1.500	38.952	736.252	1.500
14	12879.593	17.258	24.406	1.500	10.347	723.240	1.500
15	24.206	58.701	31.647	1.500	40.751	8.951	1.500
16	355.684	55.873	78.060	1.500	38.465	90.145	1.500
17	1345.801	15.542	21.978	1.500	9.252	285.634	1.500
18	166.622	61.600	81.711	1.500	42.283	45.777	1.500
19	613.539	8.690	12.289	1.500	5.053	172.730	1.500
20	16.102	66.582	22.134	1.500	46.559	5.988	1.500
21	15110.748	13.070	18.484	1.500	7.715	795.504	1.500
22	251.926	56.059	77.387	1.500	38.491	66.166	1.500
23	952.433	11.968	16.924	1.500	7.037	227.477	1.500
39	332.180	57.183	79.696	1.500	39.563	116.042	1.500
40	1375.506	34.638	48.971	1.500	17.637	289.530	1.500
41	158.048	65.080	85.104	1.500	45.195	50.263	1.500
42	629.066	12.057	17.048	1.500	6.610	176.259	1.500
45	235.107	57.939	79.558	1.500	40.061	78.581	1.500
46	974.820	20.726	29.304	1.500	11.007	231.198	1.500

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : 20
Governing Disturbing Force : 7.046kN
Governing Restoring Force : 113.453kN
Minimum Sliding Ratio for the Critical Load Case : 16.102
Critical Load Case for Overturning about X-Direction : 19
Governing Overturning Moment : 33.764kNm
Governing Resisting Moment : 170.603kNm
Minimum Overturning Ratio for the Critical Load Case : 5.053

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction : 19
Governing Disturbing Force : 13.000kN
Governing Restoring Force : 112.978kN

Minimum Sliding Ratio for the Critical Load Case : 8.690

Critical Load Case for Overturning about Z-Direction : 20

Governing Overturning Moment : -28.417kNm

Governing Resisting Moment : 170.176kNm

Minimum Overturning Ratio for the Critical Load Case : 5.988

Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

Critical Load Case for Sliding along Resultant Direction : 19

Governing Disturbing Force : 13.002kN

Governing Restoring Force : 159.775kN

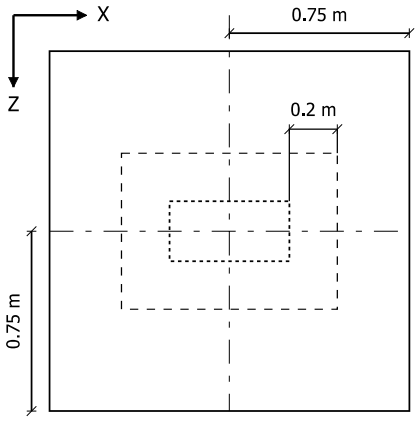
Minimum Sliding Ratio for the Critical Load Case : 12.289

Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimnesion Task Pane)

Shear Calculation

Punching Shear Check



Plan

Total Footing Depth, D = 0.400m

Calculated Effective Depth, d_{eff} = D - C_{cover} - 0.5 * d_b = 0.344m

For rectangular column, β_c = B_{col} / D_{col} = 2.000

Effective depth, d_{eff}, increased until $0.75XV_c \geq$ Punching Shear Force

Punching Shear Force, Vu = 364.074kN, Load Case # 25

$2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ = 2.876m

$0.17 \times \left(1 + \frac{2}{\beta}\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 1944.021kN

$0.083 \times \left(\frac{\alpha_s \times d}{b_o} + 2\right) \times \lambda \times \left(\sqrt{f_c'} \times b_o \times d_{eff}\right)$ 3297.265kN

$0.33 \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 1944.021kN

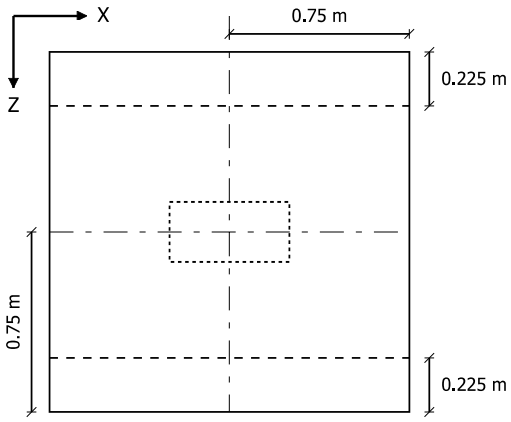
Punching shear strength, V_c = 0.75 X minimum of (V_{c1}, V_{c2}, V_{c3}) = 1458.016kN

0.75 X V_c > V_u hence, OK

One-Way Shear Check

One-Way Shear Along X

(Shear Plane Parallel to Global X Axis)



Plan

$2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 506.960kN

Distance along X to design for shear, D_x = $0.5 \times (W \pm D_{col}) - d_{eff} + O_{zd}$ = 0.281m

Check that 0.75 X V_c > V_{ux} where V_{ux} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the X axis.

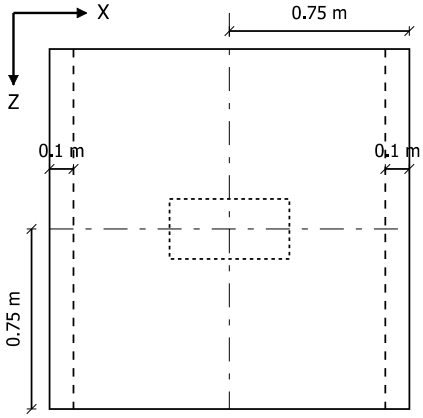
From above calculations, 0.75 X V_c = 380.220 kN

Critical load case for V_{ux} is # 28 $V_{ux} = V_{ux}|_{x=D_x}$ = 110.837 kN

0.75 X V_c > V_{ux} hence, OK

One-Way Shear Along Z

(Shear Plane Parallel to Global Z Axis)



Plan

$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 506.960 kN

Distance along X to design for shear, D_z = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{xd}$ = 0.156 m

Check that **0.75 X V_c > V_{uz}** where V_{uz} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the Z axis.

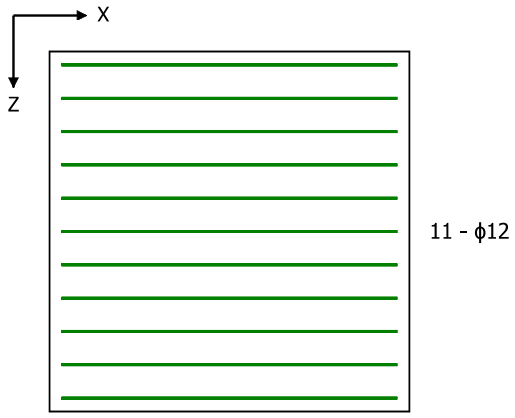
From above calculations, 0.75 X V_c = 380.220 kN

Critical load case for V_{uz} is # 31 $V_{uz} = V_{uz}|_{z=D_z}$ = 56.546 kN

0.75 X V_c > V_{uz} hence, OK

Flexure About Z-Axis

Design For Bottom Reinforcement Parallel to X Axis



Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 31

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.344 m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488
From ACI Cl. 7.12.2, ρ_{min} =		0.00168
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	1.000 m
Ultimate moment,	$M_u\big\vert_{z=D_z}$ =	43.723 kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	48.581 kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00065
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00056
Since	ρ= ρ_{min}	ρ_{min} Governs
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	1009.935 mm2

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = = 150.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

According to ACI 318 Clause No- 10.6.4 1st

Max spacing for Cracking Consideration = 250.272mm

Safe for Cracking Aspect.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

]

Required development length for bars =	$\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c'}}$ =	=0.410	m
Available development length for bars, D_L =	$0.5 \times (L - D_{col}) - C_{cover}$ =	0.450	m
Try bar size # 12	Area of one bar =	113.097	mm2
Number of bars required, N_{bar} =	$\frac{A_s}{A_{bar}}$ =	11	

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} =	N_{bar} X (Area of one bar) =	1244.062	mm2
d_{eff} =	D - C_{cover} - 0.5 X (dia. of one bar) =	0.344	m
Reinforcement ratio, ρ =	$\frac{A_{s_total}}{(d_{eff} \times W)}$ =	0.00241	

From ACI Cl.7.6.1, minimum req'd clear distance between bars

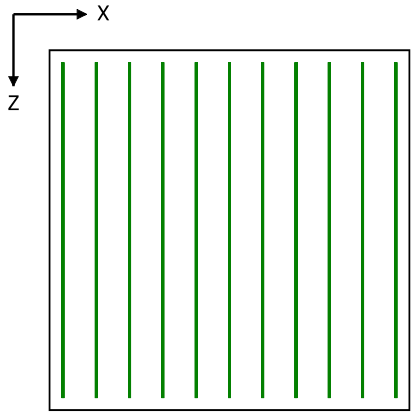
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 150.000mm

Provided Steel Area / Required Steel Area = 1.232

Check to see if width is sufficient to accomodate bars

Flexure About X-Axis

Design For Bottom Reinforcement Parallel to Z Axis



11 - ϕ12

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 28

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.332	m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796	
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317	
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488	
From ACI Cl.7.12.2, ρ_{min} =		0.00173	
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{(0.85 \times F_c')}$ =	14.118	

]

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_z = $0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ = 0.875 m

Ultimate moment, $M_u|_{x=D_x}$ = 74.423 kNm

Nominal moment capacity, M_n = $\frac{M_u}{\Phi}$ = 82.692 kNm

(Based on effective depth) Required ρ = $\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ = 0.00120

(Based on gross depth) ρ x d_{eff} / Depth = 0.00100

Since ρ= ρ_{min} ρ_{min} Governs

Area of Steel Required, A_s = $\rho \times W \times d_{eff}$ = 1038.196 mm2

Selected Bar Size = #12

Minimum spacing allowed (S_{min}) = 150.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

According to ACI 318 Clause No- 10.6.4 3rd

Max spacing for Cracking Consideration = 250.272mm

Safe for Cracking Aspect.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ = =0.410 m

Available development length for bars, D_L = $0.5 \times (L - D_{col}) - C_{cover}$ = 0.575 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 11

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1244.062 mm2

d_{eff} = D - C_{cover} - 1.5 X (dia. of one bar) = 0.332 m

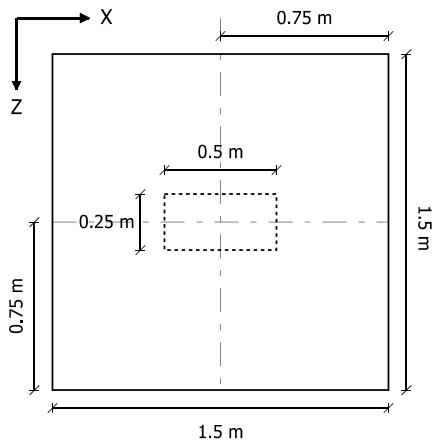
Reinforcement ratio, ρ = $\frac{A_{s_total}}{(d_{eff} \times W)}$ = 0.00250

From ACI Cl.7.6.1, minimum req'd clear distance between bars

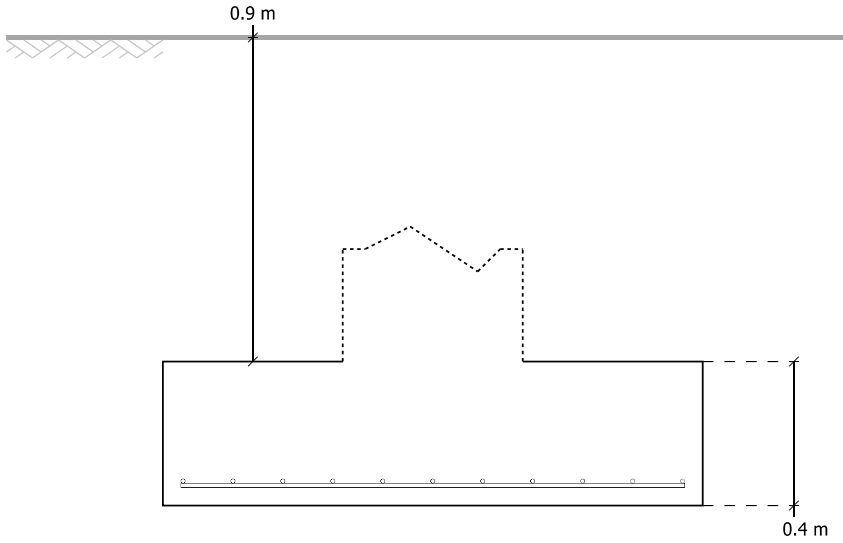
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 150.000mm

Provided Steel Area / Required Steel Area = 1.198

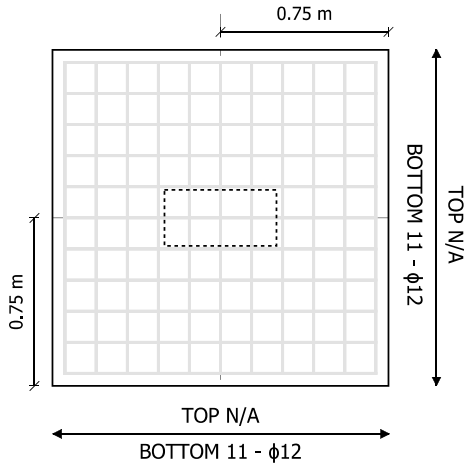
Check to see if width is sufficient to accomodate bars



Plan



Elevation



Plan

	Dimensions (m)			Reinforcement			
	L	B	D	AstX(T)	AstZ(T)	AstX(B)	AstZ(B)
Foundation	1.5 m	1.5 m	0.4 m	N/A	N/A	11 - ϕ12	11 - ϕ12
	l	b	h	Main Reinforcement		Trans Reinforcement	
Pedestal	N/A	N/A	N/A	N/A		N/A	

Print Calculation Sheet